Emotion Recognition and Solution Recommendation via Tone and Facial Analysis

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CS/2019/045

Under the supervision of

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CSCI 43018 – Research Project

Thesis

Academic year

2022/2023

Submitted in partial fulfilment of the requirements for the

Bachelor of Science Honours in Computer Science Degree

Faculty of Computing and Technology
University of Kelaniya

Date:19/08/2025

DECLARATION

I, **K.S.Malki Dilhara**, hereby declare that the thesis **entitled Emotion Detection and Solution Suggestion through Tone and Face Analysis**, submitted to the Faculty of Computing and Technology, University of Kelaniya, in fulfilment of the requirements for the award of the degree of **Bachelor of Science Honours in Computer Science**, is the result of my own independent work carried out under the guidance and supervision of **Dr Rasika Rjaphaksha**.

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Acknowledgement

I was able to successfully complete this research work due to the enthusiastic support of people who have provided me with immense help, guidance and encouragement during my educational journey. At this time, I express my love and deep gratitude to all those who contributed to the successful completion of this work.

First of all, I would like to express my deep gratitude to Dr. Rasika Rajapaksa, who served as my research advisor, for his invaluable guidance, creative ideas and continuous support. His experience and wisdom contributed greatly to the direction and depth of this research. It was a real honor for me to study under his guidance.

Also, I would like to express my sincere gratitude to the faculty, including the Research Expert Committee, for their valuable, forward-looking suggestions and extensive analysis. Their contribution played a fundamental role in improving the quality of this research work.

Additionally, the continuous support provided to me by the academic and non-academic staff of the Faculty of Computer and Technology, University of Kelaniya, helped to create a conducive environment for learning and research.

Finally, I express my deep gratitude to my beloved parents and supportive friends who have given me financial and mental strength throughout this journey. Their patience, understanding and constant enthusiasm were the primary energy that enabled me to carry out the research properly from start to finish.

Abstract

In today's fast-paced and highly demanding world, many individuals face emotional and psychological challenges stemming from occupational stress, unstable relationships, and the complexities of daily life. However, access to mental health support is often limited by factors such as high costs, lack of time, and the fear of social stigma. This thesis presents the development of a modern, emotion-aware software system named *Crew AI*, designed to provide free, on-demand personal support through intelligent emotional recognition and psychological assistance. The system detects users' emotional states by analyzing both facial expressions and vocal cues using state-of-the-art deep learning techniques. A dedicated Convolutional Neural Network (CNN) is employed for visual emotion recognition, while a hybrid CNN-LSTM model is used to interpret vocal tone and prosody, ensuring robust performance even in detecting subtle or ambiguous emotions.

What distinguishes Crew AI is its use of a multi-agent system architecture, where autonomous agents collaborate to interpret the user’s current emotional state, identify possible psychological triggers, and generate context-sensitive solutions. Unlike traditional systems that rely solely on static, pre-programmed responses, Crew AI dynamically tailors its advice using a scientific understanding of human psychology. These recommendations span immediate emotional relief strategies to long-term behavioral interventions, all while being sensitive to the user’s cultural, social, and individual context.

The system architecture combines Python-TensorFlow-based backend that performs inference of the models, Flask API layer to interact with the services, and a responsive frontend using React and TypeScript. The accuracy, adaptability and usability of the system have been demonstrated by the extent of verification using cross-validation methods and field testing. Crew AI is a new combination of artificial intelligence, the science of mental health, and human compassion with the idea of an accessible and hyper-personalized intelligent assistant in emotional well-being support.

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1.Introduction

In today's world, pressure and stress on the professional front, insecure relationships, and other socio-economic factors create significant mental turmoil. Many individuals prefer not to discuss their emotional challenges with others due to social stigma, fear of judgment, or lack of trust. This silence exacerbates mental health issues, leaving individuals vulnerable and isolated.

Emotions have a profound impact on human behavior, decision-making, and daily behavior. The decisions we make in various areas such as work, relationships, projects, and health are often based on internal moods such as happiness, sadness, anger, or anxiety. Mental health expert Annie Miller points out that "understanding and paying attention to how our minds and bodies respond to different situations can help us manage stress more easily." This makes it clear that it is especially important to always recognize and manage negative emotions. Otherwise, they can seriously damage our mental and physical health.

The key research issue targeted is the requirement for an advanced system to recognize and classify varied emotional states in human beings from an amalgamation of visual and auditory clues. The effort aims at enhancing our knowledge of human behavior as well as promoting enhanced mental help and human interactions. The work proposes a multimodal architecture utilizing both visual and audio data to detect human emotional states. The study centers on the examination of emotional states from human face images as well as voice samples through the application of deep learning architectures to identify latent relations. Audio architecture derives features including MFCC, Chroma, as well as Mel spectrograms and injects them into multimodal audio architecture comprising a CNN and an LSTM in a composite manner. Visual architecture deploys a Custom CNN architecture for the evaluation of human face expression as an architecture well-poised to extract and retain spatial features from an image effectively. The integrated architectures represent a multimodal system for emotion identification through the application of late fusion methods.

This work addresses an important gap in literature. Though interest has been on the rise in emotion recognition systems, multimodal models specifically designed for human emotions, by merging both visual and audio data, remain under development. The bulk of the work utilizes the single-modal approach where the audio analyses commonly depend on individual features like MFCC. Audio data in the work is depicted through MFCC, Chroma, Mel spectrograms, and spectral audio features for the construction of an audio multimodal model through the integration of LSTM and CNN architectures. The multimodal model fuses the visual and the audio models through the implementation of late fusion techniques whose performance is juxtaposed with established models for the establishment of the effectiveness of the model in the detection of human emotions with the focus on the accuracy and validation of the visual detection of the Custom CNN.

Also identifying negative mental states such as stress, anxiety, and depression in real time is not only important for improving personal happiness, but also for productivity and safety in the workplace. By paying attention to such moods early, appropriate interventions can prevent them from developing into more serious mental problems. However, most existing meditation recognition methods are based solely on user feedback or are limited to a single input medium, so they lack the depth to detect real moods. The results from traditional methods are largely inadequate to provide personalized and realistic responses.

Considering these shortcomings, this research proposes a comprehensive multimedia meditation recognition system that can be implemented using modern deep learning and artificial intelligence. It is designed to detect and classify people's moods in real time by simultaneously analyzing speech tone and facial micro-expressions. This dual-modal approach captures both verbal and non-verbal cues, and more accurately identifies meditations.

Another unique module of this system is the Serenity module, which immediately suggests suggestions based on the detected mood. Through this, if someone is suffering from stress, it can suggest supportive media such as breathing exercises, mindfulness exercises, or calming music/meditation. In severe cases, professional counseling signals can also be provided. This fills a gap between recognizing meditation and providing appropriate support for them.

2.Background Literature

Emotion recognition using computer models has gained immense importance, especially in the fields of mental health measurement, human-computer interaction, and affective computing. Various input channels for emotion recognition have been investigated, with facial expressions and speech signals both considered reliable and informative ways to detect emotions.

Facial Emotion Recognition refers to the computer-aided detection of human emotions from images of face by using algorithms. Facial Emotion Recognition is very significant in application fields such as human-computer interaction, virtual reality, mental state estimation, and driver attention-based vehicles. FER systems attempt to detect states of happiness, sadness, anger, surprise, fear, disgust, and neutral from vision information.

The classification process is normally of three stages: face detection, feature extraction, and emotion classification. Depending on the data type, these systems can be for still image-based or real-time recognition from video inputs.

Facial emotion recognition has evolved from traditional machine learning techniques relying on handcrafted features to sophisticated deep learning approaches that automatically learn rich representations from data. Initially, algorithms such as Support Vector Machines (SVM) and k Nearest Neighbors (k-NN) and engineered features such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG) were used. While these methods provided a creative foundation, their limitations in being widely applicable to diverse and constrained domains required more robust models.

Deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized the field of facial emotion recognition by providing the ability to learn from raw images without the need for experimental features. This has led to increased accuracy and robustness, and modern representation models such as VGG16, ResNet50, and EfficientNetB0 have been widely used in research fields.

The PhD thesis that is conducted by ITALTEL focuses on the development of a Facial Expression Recognition System (FERS) for an innovative telemedicine platform called Doctor LINK. This research focuses on the development and implementation of a Deep Learning-based facial expression recognition system for the Doctor LINK [1] telemedicine platform. A person's facial expressions reflect their mental state and emotions, and research has shown that emotions are expressed in about 7% of written communication, 38% of speech, and 55% of facial expressions. The aim of this design is to add an accurate FER module to Doctor LINK that can be integrated into real-world applications such as remote patient assessments. This is an innovative telemedicine platform developed by ITALTEL, which uses medical sensors and real-time video communications to monitor the health status of patients. The system can monitor clinical parameters such as blood pressure, ECG, POW, temperature, oxygen saturation, respiratory rate, and spirometry, and the deep learning-based FER module is integrated with a microservice architecture and GDPR-compliant security

The CNN models VGG-19 and ResNet-152 were used in this study. The VGG-19 model achieved 98% accuracy on the CK+ database and was successfully integrated for real-time FER in the Doctor LINK application. ResNet-152 achieved 90% accuracy in initial tests, and VGG19 showed superior performance in the overall results. The FER process is divided into four stages: face detection, pre-processing, feature extraction, and classification, and the Viola-Jones algorithm was used for face recognition. The datasets used include CK+, FER2013, and JAFFE, and the large variety of data required, and the significant computational resources required to train models such as VGG-19 are a challenge. Future studies aim to improve the ability to recognize emotions more accurately and diversely by combining multimodal techniques.

In emerging fields such as computer vision and affective computing, multimodal emotion recognition is an important area of research. It has been developed with the aim of overcoming the limitations of monomodal systems. Bharat Gupta and Manas Gupta [2](2024) proposes a comprehensive framework that attempts to improve the accuracy of emotion recognition by combining visual, audio, and textual media. Their model achieves an overall accuracy of 81% on a test set of 300 short video clips. This demonstrates the power of integrating facial clarity, speech patterns, and textual content. This approach is in line with the current trends in the fields of human-computer interaction and social robotics for improving the reliability and generalizability of multimedia systems.

Focusing on facial sensitivity recognition, Bharat used several datasets, including FER2013 dataset. They achieved a validation accuracy of 79.6% using a VGGNet trained on 60 sensitivities. Their pipeline uses positive preprocessing steps that include data cleaning and data augmentation (e.g., rotation, left-right flipping, meshing). This helps provide a balanced representation across the 7 sensitivity categories. This accuracy surpasses previous records in FER2013, demonstrating the power of deep CNN models to capture sensitive information from facial images. Their work can be considered a robust search compared to recent CNN models such as MobileNetV2. In the field of audio sensitivity recognition, Gupta have used datasets such as CREMA-D, SAVEE, RAVDESS, and TESS. They extract features such as MFCCs, Mel Spectrograms, and Chroma Vectors using the Librosa library, and feed them into a multilayer perceptron (MLP) model. They achieve a validation accuracy of 60% after 50 training sessions. This demonstrates the complexity of audio sensitivity recognition and the limitations of traditional MLP models. Although my research mainly uses CNN models, their feature engineering methods can provide useful guidance for innovations such as convolutional models or integrated systems.

Skowroński, Gałuszka, and Probierz (2024) [3] further strengthen this by introducing a multimedia framework designed for social robots. It analyzes speech, text, facial expressions, and biomedical data to perform emotional state recognition.Of particular importance in this research is the use of deep learning models such as CNNs and LSTMs for speech and text emotion recognition. They achieve 84.6% accuracy on the Polish nEMO dataset and 87% to 96% accuracy for Polish-language text classification. The methodology they use Mel spectrograms and tokenized text features are a composite strategy that empowers face-sensitivity CNNs based on MobileNetV2.Furthermore, in this research, the authors emphasize the importance of linguistic and cultural factors. They show that models trained on English datasets (RAVDESS and CREMA-D) lose more than 30% of their accuracy when used on Polish data.

This motivation is clear, as language-specific datasets and appropriate model architecture are needed a point that is consistent with the central issue of my research, namely the generalizability across different user populations. By addressing both technical and cultural challenges, Skowroński and colleagues' research provides a solid foundation for designing robust, multilingual multimedia systems. It also serves as a benchmark for extending my research beyond visual input to the broader integration of emotion-based computational systems.

Saravia et al. (2018) [4] introduced the CARER dataset for textual emotion recognition and showed that recurrent models such as LSTM are efficient in capturing social and temporal relationships [Saravia et al., 2018]. Based on this, Zhou et al. (2019) [5] showed that Bidirectional LSTM can achieve strong performance in emotion classification tasks using both past and future [Zhou et al., 2019]. These studies show how recurrent network techniques can be applied to textual emotion recognition. My research is for video and audio media, but their methodology provides a strong foundation for integrating textual media into multimedia systems. A notable contribution to multimedia systems was the development of fusion strategies. For example, Zadeh et al. (2017) [6] introduced the Tensor Fusion Network to capture intermedia interactions, while Li et al. (2019) and Mai et al. (2019) investigated the fusion of multimedia streams using decision-level and threshold-based methods. These proposals show that decision-level fusion produces reasonably good results. The same approach is also applicable to the fusion of audio and video streams based on my CNN. Finally, researchers have also highlighted the limitations of multimedia sensing systems. The IEMOCAP database proposed by Busso et al. (2008) remains a benchmark today, but it is limited in size. In this regard, Poria et al. (2020) have presented open challenges such as database imbalance, generalization, and integration of body motion recognition. These limitations also apply to my work. However, the roadmaps proposed in the literature provide a strong direction for the development of more complete multimedia sensing systems in the future.

Shanti et al. (2025) [7] presents an integrated framework that combines facial emotion recognition (FER), spoken emotion recognition (SER), standardized clinical scales (e.g., HAM-D, YMRS), and a doctor chat interface to generate comprehensive depression assessments. Their model achieves high accuracy rates 91.3% for FER using the FER2013 dataset and 82% for SER using datasets such as CREMA-D, RAVDESS, TESS, and SAVEE demonstrating the effectiveness of emotion analysis as a proxy for mental health assessment. The FER component uses a CNN architecture enhanced with multi-task cascaded convolutional neural networks (MTCNN) for face recognition and preprocessing, yielding a robust benchmark that closely aligns with my own work on visual-based emotion recognition using CNN variants such as MobileNetV2. For SER, the study uses Mel Frequency Cepstral Coefficients (MFCCs) extracted through the Librosa library and implements a multi-model approach that improves performance by up to 90% for commonly confused emotion categories, providing a powerful framework for auditory emotion modeling. Moreover, the integration of emotion analysis modules with a secure chat based clinical interface highlights the translational potential of such systems in real-world healthcare settings. This combination of FER, SER, and standardized psychological scales provides a unique methodological and application-driven perspective that complements the more architecture-focused work in my dissertation and expands its potential scope in practical affective computing. Therefore, Shanti et al.'s study represents a significant step forward in the application of emotion recognition models in interdisciplinary domains, particularly at the intersection of AI, healthcare, and human-computer interaction.

Mobbs, Makris, and Argyriou (2025) [8] present a comprehensive survey that synthesizes state of-the-art (SOTA) approaches to both emotion recognition (ER) and emotion generation (EG), covering facial, speech, and text methods. Their review highlights the growing shift toward multimodal emotion analysis, where the integration of visual, auditory, and linguistic signals improves the robustness and contextual accuracy of emotion-aware systems an approach that closely parallels the goals of my dissertation. Of particular relevance is the analysis of facial expression recognition (FER) techniques, which include convolutional neural network (CNN)based models such as Poster++ and EmoFAN, which achieve up to 92% accuracy on the RAFDB dataset and up to 75% on Affect Net. These benchmarks contextualize the performance of lightweight architectures like MobileNetV2, which I follow in my research to balance computational efficiency with predictive power. The authors list key datasets such as FER2013 for face data and Affect Net and CREMA-D for speech that reinforce their prevalence in ER benchmarking and justify their choice in my experimental design. Moreover, Mobbs et al. discuss important challenges in affective computing, including dataset bias, real-time limitations, and ethical risks associated with deep learning technologies, which directly inform the limitations of my study and areas of future work. Their comprehensive overview not only supports the technical direction of my work but also anchors it in the broader landscape of ethical, scalable, and multimodal emotion-aware systems.

Multimodal emotion recognition systems are increasingly being developed not only for affective computing tasks but also for real-world mental health applications, with personalized intervention strategies gaining prominence. Pathirana and his team (2024) [9] present6 a reinforcement learning (RL)-based framework that integrates facial expressions, vocal tones, and text data to predict emotional states and provide culturally specific cognitive behavioral therapy (CBT) recommendations. Using CNN-based models such as VGG16 for facial recognition on the FER2013 dataset, their system achieves a test accuracy of 76.21%, which fine-tunes to 72.58% on a Sri Lankan dataset highlighting the importance of regional and cultural adaptation in model performance. Similarly, their hybrid CNN-LSTM model for speech emotion recognition achieves 81.1% on combined datasets like CREMA-D, TESS, and RAVDESS, with performance tapering to 73.26% after localization. These results reflect the trade-offs between generalizability and cultural specificity, a challenge central to current debates in emotion recognition. Most notably, the authors implement a reinforcement learning agent that dynamically suggests CBT activities based on emotional state predictions, significantly reducing depression and anxiety levels in participants, as measured by DASS-21 scores. This integration of emotion recognition and therapeutic intervention distinguishes the study from traditional ER approaches, offering a roadmap for applying affective computing systems in personalized mental health care. The study's methodological alignment with CNN-based vision and audio architectures, combined with its innovative application layer, complements the technical and applied dimensions of my research, particularly in the development of generalized, real-world-ready emotion recognition systems.

A framework based on reinforcement learning (RL) by Amod Pathirana and their team et al. They combine facial expressions, vocalizations, and text data to make emotional state predictions and provide suggestions for culturally appropriate literary genres (CBT). Using CNN-based models such as VGG16 on the FER2013 dataset, their system achieves a test accuracy of 76.21% for face recognition. The fact that this accuracy drops to 72.58% after local smoothing for a Sri Lankan dataset highlights the importance of developing models that are based on local and cultural capabilities. Their convolutional CNN-LSTM model trained on composite datasets such as CREMA-D, TESS, and RAVDESS achieves an accuracy of 81.1%, which decreases to 73.26% after localization. These results demonstrate a trade-off between generalization and cultural specificity, a fundamental challenge in current emotion recognition research discussions. Specifically, they employed a reinforcement learning agent that provides

CBT activity suggestions based on emotional state predictions. This significantly reduced pain and stress among participants as measured by the DASS-21 score. The integration of emotion recognition and therapeutic interventions sets this research apart from traditional ER (Emotion Recognition) approaches. It thus demonstrates a new way to use emotion-based systems for personalized mental health care. The methodology of this research is compatible with CNNbased visual and audio models, and the level of innovative applications they employ provides a foundation that is compatible with the technical and practical human aspects of my own research. In particular, this is a valuable benchmark for my goal of developing a generalizable, real-life-ready sensitivity recognition system that can be applied to all user populations.

Radoi et al. (2021) [10] introduced a novel end-to-end architecture, Temporally Aggregated Audio-Visual Network (TA-AVN), that asynchronously aggregates audio and visual information using randomly selected analysis windows across temporal segments. This multimodal approach achieved state-of-the-art results of 84.0% accuracy on the CREMA-D dataset and 78.7% on RAVDESS, outperforming many traditional monomodal systems. The framework uses Convolutional Neural Networks (CNNs) for both facial feature extraction and audio analysis - the visual stream includes MTCNN-based face detection and alignment, which parallels the CNN-based strategies adopted in my thesis using MobileNetV2. On the auditory side, Log-Mel is processed through a spectral CNN-based audio pipeline, demonstrating the value of synchronous aural-visual learning. One of the study's most compelling contributions is its natural data augmentation strategy: by providing small temporal offsets and random samples in analysis windows, the model improves generalization without extensive annotated data—an insight that is particularly relevant given the limited size of most emotion datasets. With an overall processing latency of 25.9 ms per audio-visual sequence, TA-AVN also offers viable architecture for real-time deployment, a growing demand in HCI applications. While the study is limited to a few datasets and cultural contexts, its architectural innovations, performance benchmarks, and temporal integration strategies provide a valuable comparative framework for my own research on multimedia and real-time facial expression recognition systems.

Teye et al. (2022) [11] developed a multimedia framework for emotion recognition systems that considers both technical performance and moral representation. It combines speech and image data to identify seven core emotions anger, disgust, fear, happiness, sadness, persistence, and neutral as well as “sarcasm.” Their model achieved excellent classification performance, reporting accuracy ranging from 85% to 96% for individual emotions, and an overall average accuracy of 95.7%. This is due to the novel AFME (Audio-Frame Mean Expression) algorithm they introduced. Using audio features such as Log-Mel spectrograms and MFCCs, and a 3-layer CNN model trained on a dataset of 1000 culturally diverse Ghanaian images from FER2013, their system can be considered an example of robust CNN-based multimedia emotion recognition. This is particularly consistent with the design of models such as MobileNetV2 for visual emotion recognition in my research. More importantly, this study addresses a major weakness of current datasets the underrepresentation of pre-existing Asian and black African populations and builds on this by incorporating Ghanaian face data to design models that are culturally appropriate. This cultural specificity not only enhances the accuracy of emotion recognition but also provides a basis for the application of AI systems to diverse populations, including fairness and politeness. Another important point is the secondary role played by the AFME algorithm it not only improves the accuracy of the sensitivity prediction but also performs very well in detecting interesting sensory cues such as "sarcasm." Although their research was limited by geographical limitations, it provides a great deal of insight into the field of emotion-based computing, including bias reduction, cultural coverage, and ethical responsibilities. These are considered to be crucial aspects of designing human-human interaction systems for CNN-based multimedia sensitivity analysis, which is the focus of my research.

Facial Emotion Recognition (FER) is a fundamental challenge in the fields of affective computing and human-computer interaction. Modern research has increasingly shifted from handcrafted feature engineering to end-to-end deep learning models. A comprehensive review of FER methods by Vats et al. (2023) [12] divides FER methods into older methods such as Local Binary Patterns (LBP), Support Vector Machines (SVM) and deep learning methods, especially Convolutional Neural Networks (CNNs).In this review, the accuracies of traditional methods range from 50% to 97%, while studies by researchers such as Minaee and colleagues have reported that CNN-based deep learning models have achieved accuracies of 99.3% on the FERG dataset and 98% on CK+. These results demonstrate high performance that can be achieved with CNN-based models. Accordingly, efficient but powerful models such as

MobileNetV2 are also suitable for my own research. In addition to the methodological insight, Vats and colleagues present a clear analysis with detailed and comparative details of the sequential accuracies on several widely used datasets such as FER2013, CK+, MMI, JAFFE, and RAF-DB. They also take into account existing challenges such as occlusion, illumination variance, and real-world variability. Although not proposing new methodologies, this review provides a valuable perspective to understand the evolution of the FER field, the adoption of CNN-based systems, and the areas where innovation is still needed. This kind of analysis, which is a synthesis of traditional methods and deep learning methods, provides a strong background understanding for my thesis on the development of FER methods.

Elsheikh et al. (2024) [13] introduced a new CNN architecture called anti-aliased deep convolutional network (AA-DCN) as a recent contribution to this field. This is fundamental to improving the FER accuracy by solving the aliasing problem caused by the down-sampling process of traditional CNN models. Their architecture integrates MaxBlurPool layers. This acts as a low-pass filter before the pooling operations, helping to preserve the necessary spatial features. This innovation yielded state-of-the-art results, achieving 99.26% accuracy on the CK+ dataset, 98% on JAFFE, and 82% on RAF-DB. It also outperforms traditional CNN models such as VGG16, ResNet50, and EfficientNetB0.These results provide a strong validation for the robust use of anti-aliasing in FER and provide a starting point for comparing the performance of lightweight models such as MobileNetV2 in my thesis. Another important point is that they focus on data preprocessing and augmentation using techniques such as rotation, scaling, and flipping to improve generalization across different datasets. The RAF-DB dataset is a more complex dataset. It is affected by real-life noise, occlusion, and lighting variation. On this dataset, the AA-DCN model outperforms traditional CNN models by 6%. This shows that it is also suitable for real-life applications. Although their research does not address multimodal integration, its technical depth, structural innovation, and robust experimental validation can be considered a foundational benchmark for FER-centric systems. This work is particularly relevant to my thesis goals of improving the generalization and robustness of FER models using CNN-based structures.

The development of the facial recognition system presented in this research was inspired by previous work presented in CS230 course projects at Stanford University [14]. In particular, the Facial Expression Recognition with Deep Learning project, presented in the winter 2020 term, implemented the FER method using an ensemble of pretrained CNN models such as ResNet50, SeNet50, and VGG16. These models were combined with transfer learning, class weighting, and support datasets (CK+, JAFFE), and the system achieved a validation accuracy of 75.8%. In addition, the spring 2021 term project DeepLie, Detect Lies with Facial Expression introduced a custom CNN model. It has 8 convolutional layers and 4 fully connected layers and is designed to recognize fraudulent facial patterns using GRU embeddings. Although these projects have not been formally reviewed in journals, they can be considered as efficient, practical, and commonly written formal methods based on the FER2013 dataset. These methods have provided important inspiration for the structural design and data preparation strategies of the research presented so far. Facial Expression Recognition (FER) is still the foundation for computationally based emotion analysis and integrating it with speech emotion recognition (SER) is essential for designing robust multimedia systems. Parvez et al. (2025) have made a special contribution to this field by developing a CNN-based speech emotion recognition model that achieves 95.56% accuracy on a collection of CREMA-D, RAVDESS, TESS, and SAVEE datasets. Their framework classifies seven basic emotions using 1D CNNs using audio features such as Mel-Frequency Cepstral Coefficients (MFCCs), Zero Crossing Rate (ZCR), and Root Mean Square Energy (RMSE). A highlight of this study is the focus on cross-dataset generalization. This allows the model to adapt to different speakers, environments, and
pronunciation features. This is a great advantage for real-life applications. The robustness of the model is improved by using data augmentation techniques such as pitch shifting, time stretching, and sound injection, which is consistent with the goal of creating speech recognition systems that are suitable for environments that are not the target of your research. Furthermore, the social significance of this study is also shown. That is, through applications in the fields of facilitating communication for deaf and hard of hearing people and use in mental health prevention. Although this model only uses sound, it provides a strong sound-based technical foundation for a multimedia speech recognition system. It is also an invaluable addition to your CNN-based FER research. It also provides a guide for integrating sound cues into a practical multimedia system.

A recent paper titled "Emotion Recognition Using Speech: A Deep Learning-Based Speech Recognizer Model" [15] in the International Journal of Recent Research and Review (Vol. XVIII, Issue 1, March 2025) explores CNN-LSTM-based architecture to classify emotional states using speech signals. This research paper describes a model that uses deep learning techniques to recognize emotions from speech (Speech Emotion Recognition - SER). SER is a subject area that has attracted significant attention in the fields of human-computer interaction, artificial intelligence, and emotional computing. In addition to conveying information through words, speech expresses a wide range of human emotions, such as happiness, sadness, anger, fear, and neutrality. The ability to recognize these emotional states from speech has become extremely important for improving human-computer interaction, creating emotionally intelligent systems, and improving user experiences.

Traditional SER systems relied on hand-crafted features such as pitch, energy, and formants, as well as classical machine learning algorithms such as support vector machines or decision trees. While these systems showed some efficiency, they often had difficulty generalizing across different speakers, languages, and acoustic environments. Furthermore, hand-crafted features were limited in capturing the complex temporal and frequency patterns of emotional speech. With the rise of deep learning, especially with Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) (such as Long Short-Term Memory - LSTM networks), the performance of SER systems has improved significantly. This paper proposes a hybrid CNNLSTM model for emotion recognition from speech. The CNN layers act as feature extractors, capturing the local time-frequency patterns of the input Mel Frequency Cepstral Coefficients (MFCC).

These features are then fed to LSTM layers, which model temporal dynamics such as changes in the pitch and rhythm of speech. This approach allows the system to learn both spatial and sequential representations, which is crucial for emotion recognition. Mel Frequency Cepstral Coefficients (MFCCs) were chosen as the primary feature set in feature extraction. MFCCs are widely used in speech processing due to their ability to mimic human auditory perception. In this study, 40 MFCCs were extracted from each audio clip. Two well-known datasets, RAVDESS and CREMA, were used for this research. RAVDESS contains high-quality recordings of professional actors expressing eight different emotions (neutral, happy, sad, angry, afraid, surprised, disgust, and neutral), while CREMA-D contains recordings of different speakers with emotional ratings obtained from human commentators.

Training and evaluation were performed using deep learning libraries such as Keras. Categorical Cross-entropy was used as the loss function and Adam optimizer was used for optimization. Standard classification metrics including accuracy, precision, recall, and F1-score were used to evaluate the model. After training on the RAVDESS and CREMA-D datasets, the CNN-LSTM model achieved the following average performance metrics: Overall Accuracy: 82.4%, Precision: 81.3%, and Recall: 80.7%. F1-Score: 80.9%

According to the Confusion Matrix, emotions such as "neutral", "happy", and "angry" were identified with high accuracy (often exceeding 85%). Higher confusion rates were experienced for emotions such as "fear" and "disgust". For example, "fear" was often misclassified as "sad" or "angry", possibly due to their overlapping acoustic patterns.

In a comparative analysis with existing methods, traditional models such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) have reported accuracy rates of 65% to 72% on similar datasets. Deep learning models using only CNN or LSTM components have achieved values of 74% to 78%. The hybrid CNN-LSTM model presented in this study showed superior performance by 4-10% over both approaches, highlighting the value of the combined approach. Impact of feature engineering: MFCCs provided the best compromise between model complexity and classification accuracy, which also echoes the findings of previous studies. They efficiently capture changes in pitch, tone, and rhythm, which are critical indicators of emotional state. Although the model produced robust results in controlled experiments, several limitations were identified. The data sets used consisted of acted out emotional stories, which may not reflect the spontaneity or subtlety of real-life emotional expression. The model can learn speaker-specific features, whereas a speaker-independent model would require large and diverse datasets. The model's performance in noisy environments has not been extensively tested. The model may be affected by having fewer samples for some emotion classes (e.g., "disgust" or "fear") than others. The study focused only on English-language datasets. The "black box" nature of deep learning models is a concern, and ethical issues such as data privacy and misuse of emotional data need to be addressed. In summary, this paper presents a hybrid deep learning model that combines CNNs and LSTMs for emotion recognition from speech. Using MFCCs as features, the model achieves high accuracy on the RAVDESS and CREMA-D datasets, demonstrating superior performance over traditional and single deep learning approaches. The model has the potential to be integrated into real-time applications such as virtual assistants, therapy bots, and emotion-aware conversation systems.

3.Research Gaps

While there are several emotion detection systems that have been developed using facial expressions or tone analysis, most existing approaches are limited to using a single input modality. This limits the accuracy and capability of emotion assessments. Furthermore, most traditional recommendation systems, especially those used in mental health tools, rely on static databases and are unable to handle the individual preferences and contexts of their users. These systems face challenges in providing excellent and evidence-based support, especially since they are unable to evolve with current inputs. Also, there are few cases where dynamic agentic systems, which can make rational decisions among psychological, contextual, and cultural factors, are integrated into the design of emotional support. This project directly addresses those limitations by combining multimodal emotion detection of faces and tones with an agent based intelligent recommendation system, helping to fill the void in live, adaptive psychotherapy solutions.

3.1. Addressing Research Gaps

While many existing emotion recognition systems focus solely on detecting emotions, they often lack the ability to provide meaningful, personalized responses or recommendations based on the user’s psychological context. Additionally, most current systems either rely on static, predefined responses or limited rule-based decision trees, which fail to adapt to diverse user needs. This project addresses these gaps by integrating a dynamic, agent-based recommendation system (Crew AI) that not only interprets emotions but also offers scientifically validated, context-aware solutions. Unlike traditional database-driven approaches, this agentic model adapts in real time, ensuring relevance, psychological grounding, and greater accessibility for users seeking emotional support.

In addition, when exploring multi-agent systems, recent advances in team AI systems have been researched to integrate emotion recognition capabilities to improve decision-making and human-AI collaboration, especially in high-risk and intelligence-intensive environments such as aviation, space travel, and healthcare. Many studies have shown that incorporating emotion sensitive logic processes into agent systems can significantly improve team safety, mental health, and task efficiency. For example, Zhang et al. (2022) presented a conversational AI agent called Crew Bot that recognizes emotions such as shock and discomfort in spacefaring astronauts and responds to them with motivational language or humorous speech. Halverson et al. (2023) also developed a conversational-based Crew AI system that uses real-time emotion input to adjust task complexity and speech style while under constant stress. Such systems rely on agent-based behavior, such as autonomous decision-making, goal prioritization, and multimedia perception. This is often achieved through natural language processing, affective computing, or reinforcement learning. Although these models offer robust emotion-response mechanisms, they often lack a standard approach for inter-agent coordination or modular workflow design. This gap highlights the need for constructible and explicit agent workflow systems, as demonstrated in new multi-agent frameworks such as AutoGen (Wu et al., 2023) and CrewAI (Moura, 2023), which use YAML-based configurations.

3.2. Research Problem

Although the prevalence of emotion detection technologies is increasing, many existing systems are limited to identifying emotions. They do not provide actionable, personal support. People who experience emotional difficulties such as depression and anxiety but do not have access to professional psychological services due to financial or social constraints are deprived of critical and useful guidance. In addition, most traditional recommendation systems rely on static databases or rule-based outputs and lack adaptability, psychometrics, and contextual understanding. The goal of this project is to provide an intelligent, accessible, and evidence-based solution, one that not only recognizes emotions but also provides context-sensitive recommendations. By combining emotion detection and psychologically validated, dynamically generated assistance tools using a multi-agent system, this research seeks to fill a critical gap in both emotion AI and psychological assistance technology.

4.Aims & Objectives

4.1. Aim

To develop a multimodal emotion-aware recommendation system that detects user emotions through facial and vocal analysis and provides personalized, evidence-based solutions using an agentic decision-making framework.

4.2. Objectives

- Design and implement a multimodal emotion recognition module using facial expression and voice tone analysis.
- Build an agent-based recommendation engine (using Crew AI) that dynamically generates support strategies based on the detected emotion and user-provided context.
- Compare and evaluate traditional database-driven recommendation methods with the agentic system in terms of adaptability, relevance, and user-friendliness.
- Ensure accessibility by enabling the system to offer free, real-time mental health guidance for users who may not have access to professional psychological services.
- Validate effectiveness through qualitative and cross-validation techniques to assess accuracy and practical applicability of the support suggestions.

5. Methodology

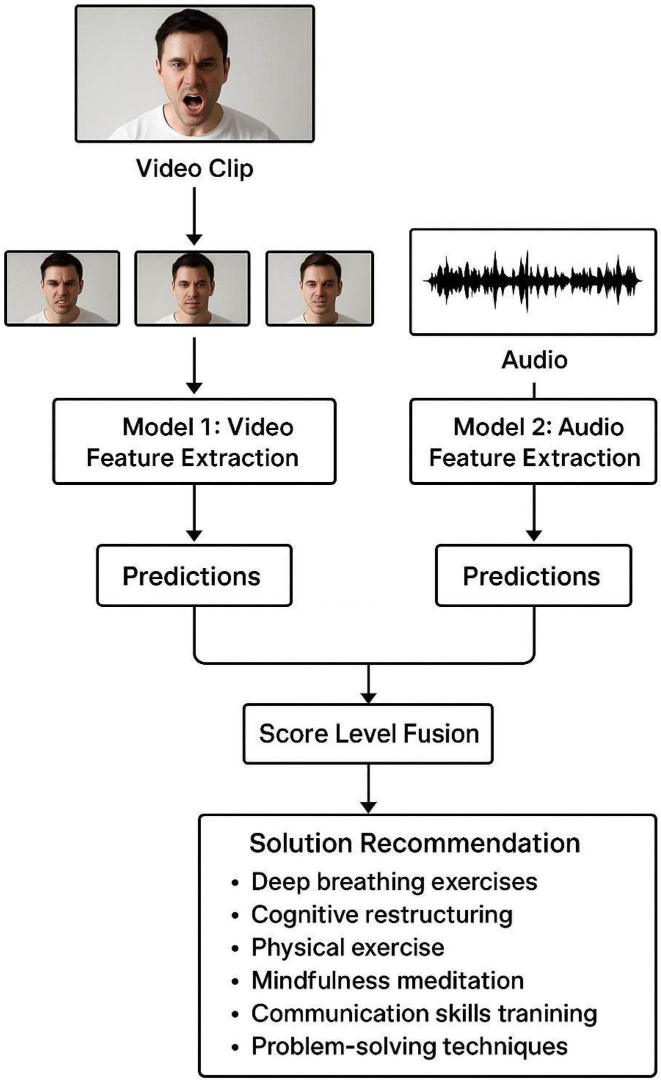


Figure 1:Overall system's methodology

5.1. Overview

In this research, two specialized models for short video clip analysis are integrated, based on the visual media such as face clarity and voice tone. The framework is implemented by drawing on qualified methods for image and audio. The model allows the labelling of unlabeled data via fusion across modalities and aims to develop lightweight models with performance comparable to memory-intensive ones. The model's block diagram is referenced below.

5.2. Audio Approach

Human voice is used for impulse recognition. This involves collecting and pre-processing audio data. Then, features such as MFCC, Mel spectrograms, Chroma features, and Spectral Contrast are extracted. After that, CNN and LSTM networks are trained to perform impulse classification. Both CNN and LSTM models were trained with different learning rates and network layers to find the best configuration. By combining the best performing models and building a multimedia system, the accuracy of impulse recognition from audio input can be improved.

5.3. Image Model Approach

The computational complexity was reduced by strengthening the model using data augmentation and oversampling techniques. After testing the various models, a specific CNN model was used to improve performance.

5.4. Multimodal Approach

In this approach, audio and image data are combined using late fusion. Late fusion refers to the technique of combining the outputs of individual models, each trained separately on either audio or image data, at a later stage of the decision-making process. The objective of this fusion is to create a unified system that leverages both audio and image modalities for impulse recognition. By combining the distinct features extracted from both sources, the system aims to enhance the accuracy and robustness of impulse recognition, making the model more resilient to noise and variations in either modality. This approach allows the model to benefit from complementary information provided by both audio and visual data, leading to improved overall performance.

5.5. Loss Function

In this research, the categorical cross-entropy loss function was used to train models. Categorical cross-entropy is a loss function commonly used for classification problems, especially when the target variable is categorical. It is used to measure whether a model performs accurately, especially when the model output is given as a probability value between 0 and 1. In these cases, categorical cross-entropy perfectly measures the difference between the predicted probabilities and the actual labels. That is, it adequately expresses the difference by comparing the predicted probabilities associated with each class with the actual labels.

5.6. Image Approach

The face approach identifies the feelings of human beings on the basis of their face characteristics. For the start, a varied series of images of human faces is downloaded from Kaggle, i.e., the FER2013 dataset.

5.6.1 Tools & Technology

5.6.1.1 Python

This code is written in Python, a language widely used for certain applications, especially Machine Learning and Data Science. This is due to the simplicity of the Python language and its extensive library support.

5.6.1.2 Google Colab

This script is implemented on Google Colab. This is a cloud-based Jupyter Notebook environment, which is proven by mounting the Drive for saving and accessing files on Google Drive. Google Colab provides free GPU access, which allows for high computational speed.

5.6.1.3 Pandas

This library is used to load and manipulate the FER2013 dataset from a CSV file into a Data Frame. This is very useful for managing structured data easily.

5.6.1.4 NumPy

NumPy is used for numerical operations. For example, converting pixel data into an array and reshaping it to fit as input to a neural network.

5.6.1.5 Sequential Model

The Sequential model is used to sequentially build and introduce layers for the CNN structure.

5.6.1.6 Layers

Conv2D (convolution layers) – to extract features from the image,

MaxPooling2D (pooling layers) – to reduce the feature matrix,

Batch Normalization (normalization) – to increase the training speed and reduce overfitting,

Dropout (random stop) – to randomly disable some units from the layers to prevent overfitting,

Dense (model layers) – used to perform the final classification.

5.6.1.7 Optimizers

The Adam optimizer is used to minimize the loss function.

5.6.1.8 Imagedatagenerator

The ImageDataGenerator is used to provide structural diversity to the image data and reduce overfitting. This uses transformations such as rotation, shifting, zooming, and flipping.

5.6.1.9 To Categorical

Translate the labels for sensitivity into one-hot encoded form. This is required for multi-class classification.

5.6.1.10 Early Stopping

A callback that stops training if the validation accuracy stops increasing, which helps prevent overfitting and preserve the best model weights.

5.6.1.11 Activation Functions

Activation functions add nonlinearity to the model, which allows it to learn complex patterns and relationships in the data. Without nonlinearity, the model behaves like a linear regressor, regardless of the number of layers in the neural network. In this model, the ReLU (Rectified Linear Unit) and SoftMax activation functions are used at different stages of the network, each with a specific purpose.

5.6.1.12 ReLU (Rectified Linear Unit)

The ReLU activation function is used in most of the hidden layers of the model. Mathematically, this is defined as:

$$f(x) = \max(0, x)$$

Where x is the input to the activation function. That is, if the input is a positive number, the output is the same as the input, otherwise the output is zero.

5.6.1.12.1 Purpose

ReLU reduces the Vanishing Gradient Problem (the inability of the network to learn properly due to very small densities). ReLU provides large densities for positive inputs, so the model learns quickly and efficiently. By eliminating negative values, it introduces sparsity into the model's activations, which makes training efficient.

5.6.1.13 SoftMax

The SoftMax function is used in the output layer of the model for multi-class classification. It converts the raw output values (also known as logits) into probabilities by assigning a probability score to each class. The SoftMax function ensures that the sum of the predicted probabilities for all classes equals 1, making it suitable for classification tasks where only one class can be predicted at a time.

5.6.1.14 Multiple Seed Evaluation

While most experimental setups were evaluated using a single training run, additional care was taken for high-performing models by rerunning them under multiple random seeds. This step aimed to confirm that the observed accuracy was not a one-off result due to a favorable initialization or split. By repeating training with different fixed seeds, the consistency of performance could be verified, lending greater credibility to the model's generalization ability. Only models that consistently achieved strong metrics across runs were considered reliable, and the best or average accuracy from those runs was reported accordingly.

5.6.1.14 Model Evolution & Cross Validation

The k-fold cross-validation method was used to verify the reliability and generalization of the emotion detection models. In particular, the 5-fold cross-validation method was used for the facial expression (CNN-based) model. [23] This helped to test the performance of the model on different data sets by applying the training set and the validation set interchangeably. This helped to minimize overfitting and obtain reliable measures such as accuracy and F1-score. Accordingly, it was confirmed that the models performed more reliably on different emotional inputs.

5.7 Data collection

Here, an image-based methodology is followed to explore facial emotion identification in human beings by using the well-known FER2013 dataset obtained from Kaggle. for validation, and 3,589 images for testing. Each image is labeled with one of seven emotion classes, which include happiness, sadness, surprise, anger, disgust, fear, and neutral, with the numbers of images varied per emotion category due to inherent class imbalance present in the dataset.

Figure 2:Various kinds of samples in FER 2013 dataset

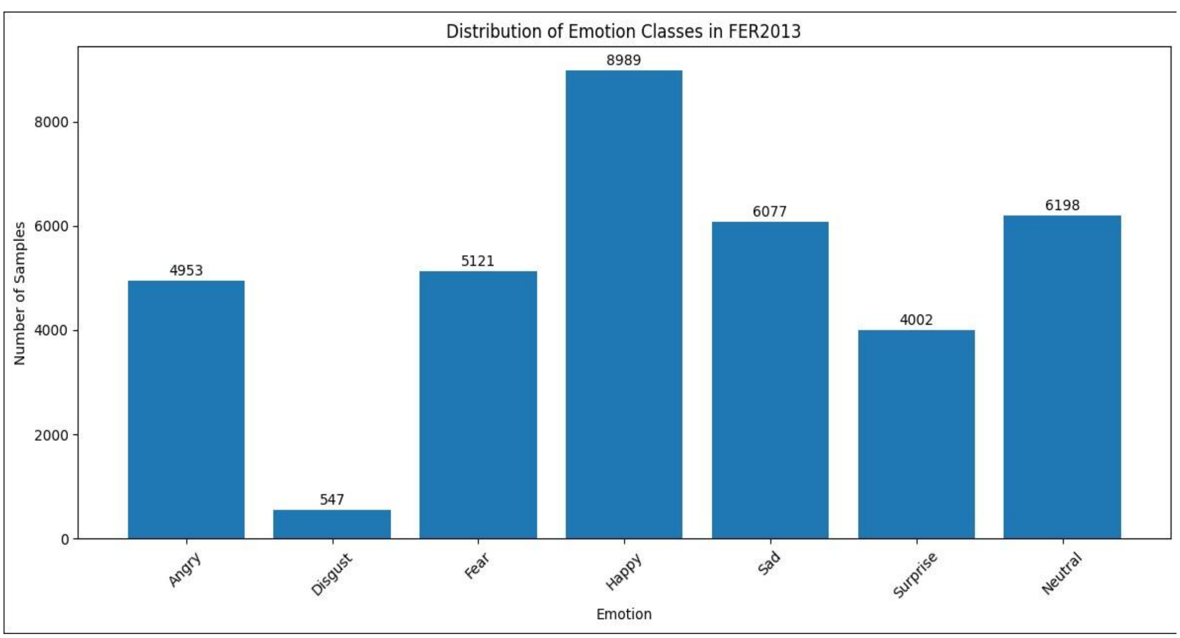


Figure 3:Original data distribution in FER2013

For the current study, all seven emotions are taken into account, covering the entire range of data provided to facilitate thorough analysis. For ease of handling the data as well as training the model, the images for these emotions are systematically processed and organized with the intention of having streamlined access as well as ordering during the time of developing and testing the model for emotion identification.

5.8 Data Preprocessing

Preprocessing is a significant step within the dataset preparation process for both machine learning and deep learning, particularly in tasks of facial expression recognition (FER). The FER2013 dataset, used most commonly for this application, is made up of 35,887 grayscale images labeled into seven emotional groups: angry, disgust, fear, happy, sad, surprise, and neutral. The dataset is split into subsets used for training, validation, and testing and has a strong impact on the performance of FER models due to its structure. Due to its common application, the comprehension and implementation of good preprocessing methods on the FER2013 dataset are key to achieving model accuracy and robustness in facial expression detection.

Preprocessing constitutes a series of tasks intended to promote the quality of the data. These comprise the resizing of the images, the elimination of noise, and the creation of additional data to enhance variability. Seeing as the dataset is class imbalanced, that is, some emotional categories have fewer instances preprocessing comes in handy to balance the dataset and help the model acquire representations that generalize well across all the emotion classes. Methods such as mean imputation when handling missing values and extensive data augmentation methods help acquire more diverse and resilient training data while curbing the chance of overfitting and enhancing the model's generalization to an extent.

Cultural diversity is another factor to consider when using FER2013 since the dataset has largely images from Western and African populations. The limitation prompts the question of how well-trained models would perform in the broader, multicultural environments the system would eventually be deployed into. For this reason, researchers should factor cultural differences in facial expression when preprocessing the input datasets and training the models to create FER systems that are more inclusive. In conclusion, preprocessing is a vital component of both deep

Learning and Machine Learning Pipelines in FER applications. Thoughtfully designed preprocessing not only enhances input quality but also boosts the effectiveness of trained models, contributing to more accurate and fair facial emotion recognition across diverse use cases.

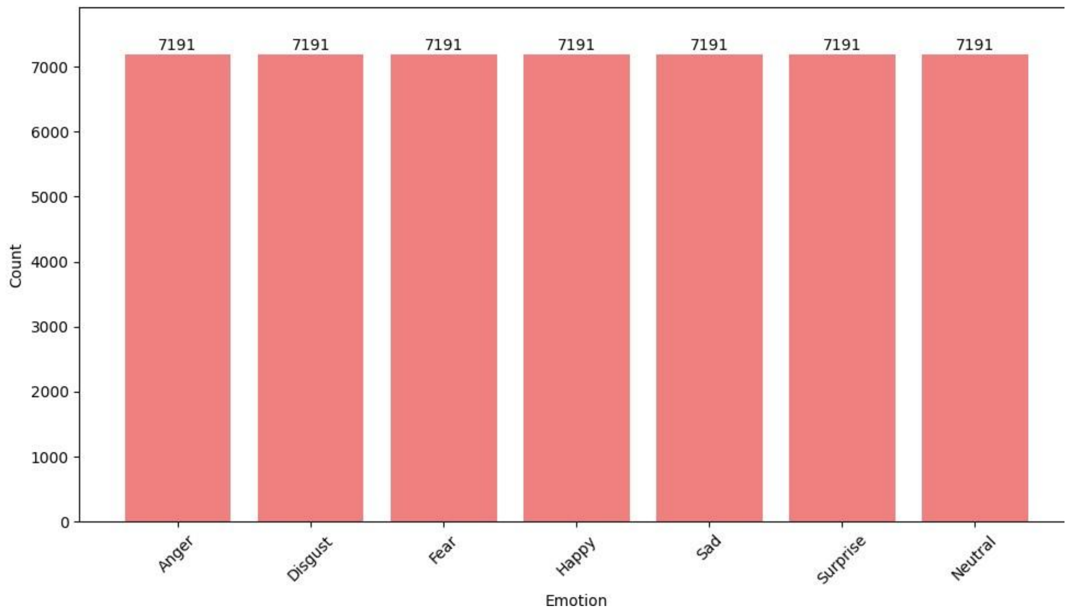


Figure 4: Balanced dataset for training process by applying augmentation and oversampling in FER2013

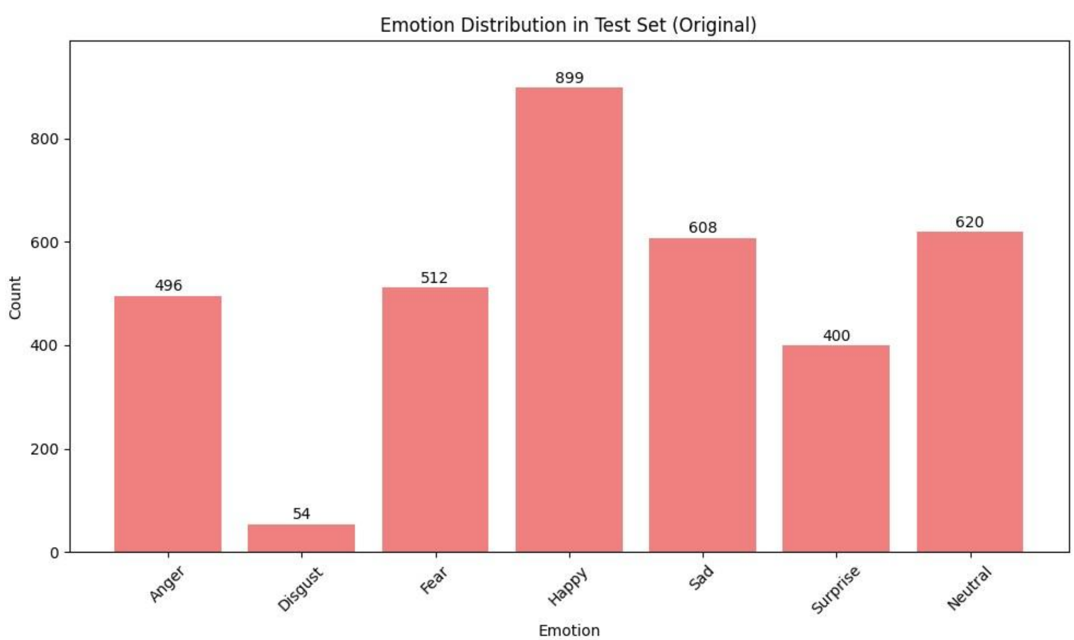


Figure 5:Original dataset for Testing images inFER2013

5.9 Normalization

Normalization is a very important preprocessing step in preparing the FER2013 dataset for facial emotion recognition tasks. This process involves scaling the pixel intensity values of the grayscale images from their original range of 0 to 255 down to a normalized range of 0 to 1. This is done by dividing each pixel value by 255.0, the maximum intensity that can be produced by an 8-bit grayscale image. This helps to ensure that all input data is of the same scale.

There are several reasons why normalization is important for the FER2013 dataset. First, it normalizes the input data. This makes the training process of deep learning models more stable and faster. Second, it prevents numerical instability and helps in faster convergence by formal algorithms such as gradient-based optimizers (e.g. Adam). Finally, the large frequency range of individual features also helps prevent the model from developing a bias.

FER2013 images are grayscale, with pixel values ranging from 0 to 255. Accordingly, Normalization within this range is a simple and efficient method. This is widely used in scientific research and practical applications using the FER2013 dataset. It directly addresses the need to normalize data representation in deep learning systems. Overall, normalization smoothes the behavior of training models on the FER2013 dataset and improves efficiency. This increases the ability to obtain accurate and reliable emotion recognition results.

```
X=X/255
```

```
[ ] X
```

```
array([[0.2745098 , 0.31372549, 0.32156863, ..., 0.41568627, 0.42745098,
        0.32156863],
       [0.59215686, 0.58823529, 0.57647059, ..., 0.75686275, 0.71764706,
        0.72156863],
       [0.90588235, 0.83137255, 0.61176471, ..., 0.34509804, 0.43137255,
        0.59607843],
       ...,
       [0.06666667, 0.06666667, 0.0627451 , ..., 0.60392157, 0.52156863,
        0.44313725],
       [0.11764706, 0.10980392, 0.10980392, ..., 0.1372549 , 0.11764706,
        0.10980392],
       [0.0745098 , 0.05098039, 0.05490196, ..., 0.74117647, 0.78039216,
        0.78823529]])
```

Figure 6:How normalization is applied to preprocessing

5.10 Data augmentation

Data augmentation is a very important preprocessing technique when analyzing the FER2013 database for facial emotion recognition. [24] It improves the robustness of deep learning models by improving the feature extraction process. This technique is important in view of the challenges of FER2013, namely, the generally small number of images (35,887) compared to large datasets such as ImageNet, and the "in-the-wild" nature of the data, which results in unstructured lighting, facial poses, and background variations. Augmentation is implemented as a solution to the limited training samples, which is always a major problem in such FER tasks. By artificially increasing the size and diversity of the training dataset through various methods, the model can be trained to recognize a wider range of intentional facial features.

In this research, as implemented in previous updates, data diversification is implemented on the same training set. This involves using Keras tools such as `ImageDataGenerator` to represent light, location, and other environmental variables to learn invariant features for the CNN model. The diversification strategies used include random circular rotation (up to 15°, then up to 20°), width and height shifts (up to 10%, then up to 20%), zooming (up to 10%, then up to 20%), and horizontal flips. In the latest versions, we have also added smooth transformations such as shear transformation (up to 20%), and luminance range (from 0.8 to 1.2), which specifically address the natural light and location problems of FER2013. All of these distortion methods are implemented in real-time (on-the-fly) during the training epoch, so memory usage is not increased, and computational efficiency is ensured.

The main advantage of data diversification for FER2013 is to avoid the risk of overfitting. Such a risk can exist due to the quantitative limitation of the dataset and the complexity of the design such as the 8-layer CNN model. By showing the model different versions of the same image, it learns in a more generalized way, instead of just memorizing the training data. Such a method has been strongly validated by practical experiments, which also show inflation on FER datasets with high accuracy. Techniques such as GANs (Generative Adversarial Networks) have also been demonstrated in recent literature, which expand the dataset by generating representative face images. However, our current model only uses physical and lighting-related variations. This structural and systematic diversification method has been a consistent method in all my experiments.

Therefore, data diversification is a very essential pre-processing step when faced with uncontrolled environmental conditions in FER2013 and contributes greatly to obtaining accurate and reliable intentional inferences.

Figure 7:Example of one image with multiple augmentations in the training dataset

5.11 Data Splitting

Data splitting is a very important step in the preprocessing process for facial expression recognition tasks especially for the FER2013 dataset. In this process, the dataset is divided into separate subsets for training and testing. This helps in accurately evaluating the performance of the model and is also important to avoid problems such as overfitting. In here a common practice is to split the data into 80% for training and 20% for testing. This split allows the model to be trained on a large amount of data, while still retaining enough data to test its generalization capability.

5.12 Class Imbalance Considerations

The FER2013 dataset exhibits a severe class imbalance. The number of samples in different motivational categories is inconsistent. For example, there are only 436 samples in the "Disgust" class, while there are 7,215 samples in the "Happy" class. This imbalance is especially important to consider when partitioning the data. This is because a naive partition may result in the least represented classes being replicated in the testing set. Techniques such as Random Over Sampling can be used to address this problem. Such techniques maintain a balance between classes in the training set, allowing the model to learn efficiently from both the majority and minority classes.

5.12.1 Standardization and Noise Reduction

A common preprocessing technique is to standardize the dimensions of the face images in the database. This allows the model to be trained on inputs of the same size. Filtering techniques can also be used to reduce noise in the images, thereby improving the quality of the training data. For example, research has shown that applying histogram equalization improves the performance of the model by adjusting the contrast of the images. This helps to identify important features more clearly.

5.12.2 Down sampling and Ensemble Learning

Another powerful technique is down sampling, which is the process of down sampling images to a lower resolution. Previous research has shown that ensemble models, which combine multiple models trained on high- and low-resolution images, perform better than single models. Their research showed that combining predictions from different resolutions can reduce the top 5 error rate. This suggests that multiple down sampled images can also act as a data augmentation method. This type of ensemble learning can improve the robustness of the model, as it provides multiple perspectives on the data.

5.13 Model Architecture

5.13.1 CNN

CNNs (Convolutional Neural Networks) are models inspired by the human vision system that are capable of automatically extracting hierarchical features from images. Traditional machine learning methods require manual feature engineering, which CNN models do not. Instead, CNN models automatically learn relevant features from raw pixel data, using a series of layered operations. This capability makes CNN models well suited for image-based tasks such as face emotion recognition (FER). [22] This allows them to recognize patterns such as lines, textures, and facial landmarks related to emotions. For example, CNNs can easily identify facial patterns related to emotions such as happiness, sadness, and anger.

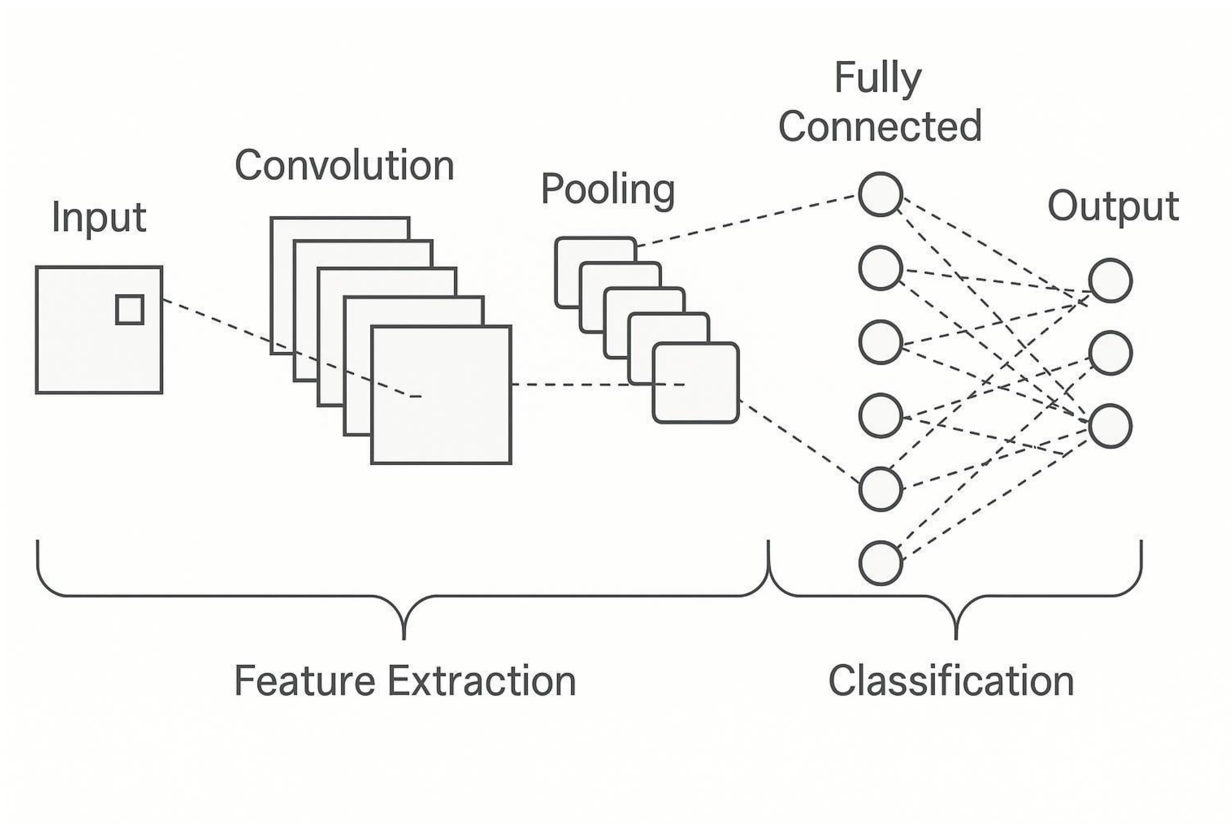


Figure 8:CNN layered architecture how works

To perform emotion recognition using these facial images, I designed and implemented a specialized Convolutional Neural Network (CNN) architecture. This decision was made to design a model specifically tailored to the features of the facial broadcast data, and to provide the ability to increase accuracy by providing appropriate control over the joints and hyperparameters. CNN models are particularly well-suited for visual recognition. They are able to automatically learn the visual patterns of the lens and sequentially extract spatial features from raw image input. This makes them ideal for analyzing complex patterns in the eyes, mouth, eyebrows, and other important areas of the face all of which are particularly important for emotion recognition.

5.13.2 Input Layer

The network uses 64×64-pixel grayscale face images as input. These images are obtained by resizing the original 48×48-pixel images from the FER-2013 database using bicubic interpolation. Since facial recognition relies heavily on structural changes in the face, using grayscale images helps to preserve the necessary features while reducing mathematical complexity. The input format given to the network is (64, 64, 1) which represents only a single-color channel.

5.13.3 Convolutional Blocks (Feature Extraction Layers)

The model contains five convolutional blocks, each carefully structured to learn increasingly abstract and complex features: Two Conv2D layers per block, each with a kernel size of 3×3, use 'same' padding to preserve spatial resolution and ReLU activation to introduce non-linearity.

The number of filters increases progressively:

- Block 1: 64 filters
- Block 2: 128 filters
- Block 3: 256 filters
- Block 4 and 5: 512 filters each

This hierarchical expansion allows the model to first capture simple patterns (e.g., edges) and later learn more complex and high-level features (e.g., shape of the mouth or furrowed eyebrows). Following each convolution, I added a Batch Normalization layer to stabilize learning by normalizing the output and help reduce internal covariate shift to improve training speed. A MaxPooling2D layer (2×2) is applied after each pair of convolutional layers to down sample feature maps, effectively reducing dimensionality while retaining the most salient features.

Dropout layers with a rate of 0.3 follow the pooling layers to randomly deactivate neurons during training. This is the regularization method which avoids excessive adaptation of the model to individual neurons and makes its generalization more valid.

5.13.4 Flatten Layer

After all convolutional blocks, the output feature maps are flattened into a one-dimensional vector. This step transforms the 3D spatial structure into a format suitable for processing by dense layers.

5.13.5 Fully Connected Layers (Classification Head)

The flattened vector is passed through a series of three fully connected layers:

- Dense (1024)
- Dense (512)
- Dense (256)

Each layer uses ReLU activation, followed by Batch Normalization and Dropout (rate = 0.4). These layers enable the network to learn complex, nonlinear relationships among the abstract features extracted from the convolutional layers. Dropout continues to act as a safeguard against overfitting.

5.13.6 Output Layer

The final layer is a Dense layer with 7 neurons, each corresponding to one of the target emotion classes like Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral. A SoftMax activation function is used to generate a probability distribution over the classes,

5.13.7 Results and Evaluation

The FER model was trained on a training set over 100 epochs then evaluated after being unseen in a test set to measure how well it performed. This critique sheds light on the performance of the model in terms of its ability to discriminate facial emotion using the input that was given. The performance of the model was evaluated with the help of such crucial measures, as precision, recall, f1-score, and the confusion matrix, as these provide a more convincing picture of its work in each of the emotion categories

5.14 Audio Approach

5.14.1 Overview

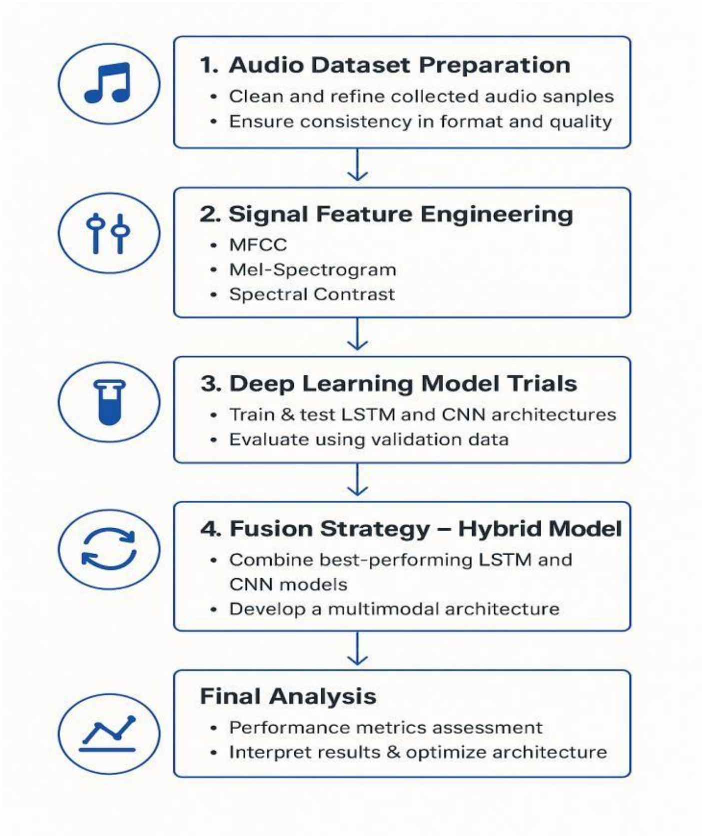


Figure 9: Audio approach overview

The provided figure shows the complete methodology for a deep learning system based on sound. First, the sound dataset is prepared, where the basic sound patterns are cleaned and standardized to ensure regularity and quality. This is followed by signal feature engineering, which is an essential step that involves extracting key sound features such as Mel-frequency cepstral coefficients (MFCC), Mel-Spectrograms, and Spectral Contrast. These features are crucial for efficiently representing sound information.

The extracted features are fed into deep learning model testing, where various LSTM and CNN structures are trained and rigorously evaluated using validation data to select the best models for the task. Then, a compositional approach is implemented to assemble the high-performance LSTM and CNN models to create a multimedia model that works as a whole. This increases the robustness and accuracy of the system. The methodology ends with a final analysis, where the performance measurements are fully evaluated, the results are analyzed, and the entire system structure is adopted for the desired purposes.

5.14.2 Tools and Technologies

5.14.2.1 Python: Used to code the entire audio processing pipeline, from data collection and preprocessing to model development and evaluation.

5.14.2.2 Colab: This tool provides an interactive opportunity to quickly code, analyze data, and visualize model performance. It facilitates frequent code testing and bug detection.

5.14.2.3 GPUs: These devices are used to accelerate the training of deep learning models such as CNN and LSTM. This helps in efficiently managing large datasets and complex calculations.

5.14.2.4 TensorFlow and Keras: Used to generate and train CNN and LSTM models. TensorFlow manages low-level operations, while Keras facilitates the creation and training of models through high-level abstraction**.**

5.14.2.5 Librosa: This is the package needed to extract important audio features such as MFCC, Mel-spectrograms, and Chroma features.

5.14.2.6 NumPy and Pandas: These packages are used to manage numerical data and feature arrays, preprocess, and train models.

5.14.2.7 SciPy: This library contains additional signal processing tools needed for processing and analyzing audio data.

5.15 Data collection

The effectiveness of emotion recognition systems largely depends on the quality and diversity of the datasets used for training and evaluation. For this study, the datasets used were selected with a direct design to ensure robust model development and unbiased performance evaluations. The CREMA-D and RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) datasets served as the main sources for model training and validation. These datasets provide a large range of emotion expressions covering a variety of speakers, emotion situations, and linguistic content, and are essential for the development of general emotion recognition capabilities.

To independently and unbiasedly evaluate the performance of the final model and its generalization ability to unseen data, a dataset generated using the SAVEE (Surrey Audio-Visual Expressed Emotion) database was used for testing purposes only. By rigorously separating training/validation data and test data into excellent datasets, my goal was to provide a rigorous and realistic evaluation of the performance of the developed system in recognizing previously unseen speakers and emotion expressions.

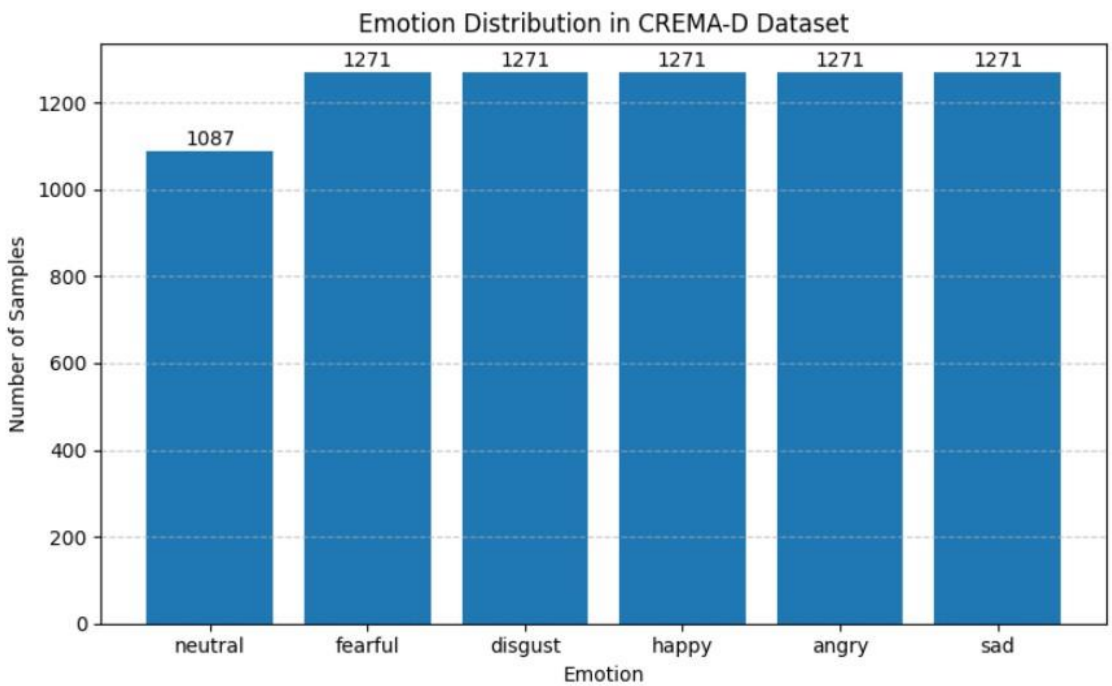


Figure 10:Crema-d data distribution

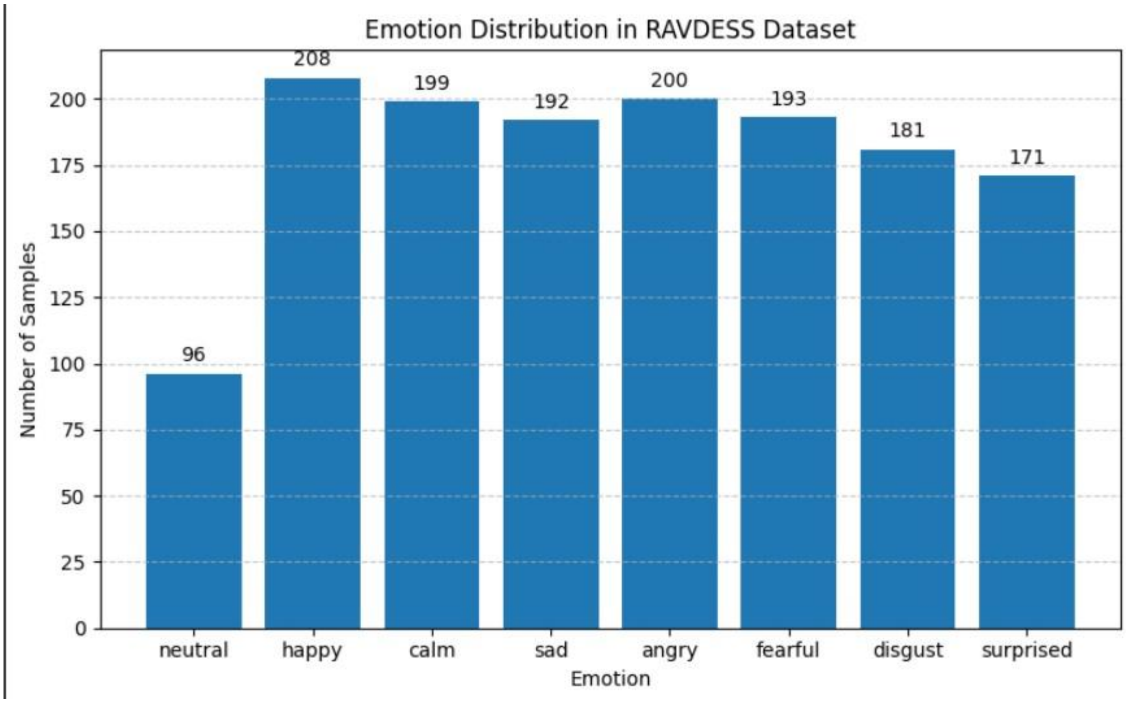


Figure 11:RAVDESS data distribution

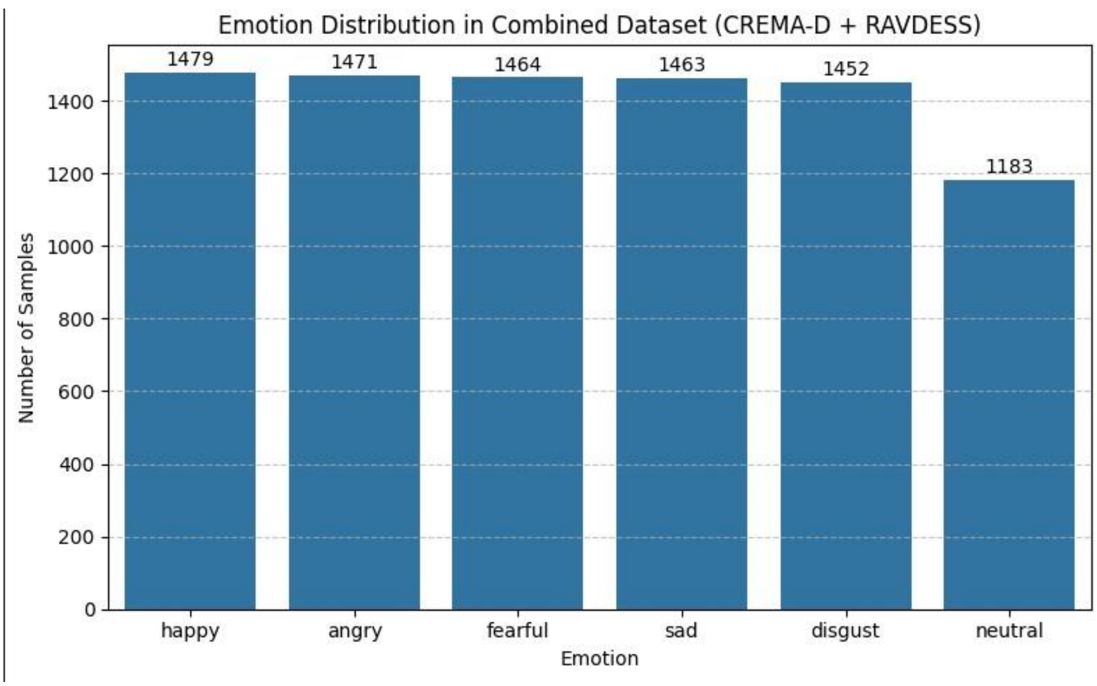


Figure 12:Total audio data distribution

5.16 Data Preprocessing

Data preprocessing is a critical step in preparing audio data for the model training phase. This ensures that the input is clean, standardized, and suitable for learning. The various processes performed during the preprocessing phase include:

5.16.1 Audio Normalization: Normalizes all audio files across the dataset to a single standard. By normalizing the sound levels of different details, it reduces the variance caused by different environmental and equipment differences. This is done to train the models on sounds of constant intensity, increasing the ability to learn matching features.

15.16.2 Audio Segmentation: Pre-groups clips into 3-second segments and further processes them to keep them of similar length. This can include clipping or padding the clips, thereby ensuring uniformity in feature extraction and model training.

15.16.3 Mono Conversion: If the audio file is in stereo, it is converted to mono. This reduces complexity and focuses feature extraction on just one sound channel, which is sufficient for impulse classification.

5.17 Data Augmentation Strategy

Some oversampling was used in the training set as a strategy to “target data augmentation,” given the extreme skew in the classes in the combined dataset. The activity was directed toward the minority emotions classes, in particular, and the aim was to augment the population of the emotions artificially. They reduced their representation to a set limit; that is, 800 examples per category. For each, underrepresented emotion, original audio files were randomly selected (with replacement) and transformed through a series of three distinct augmentation techniques implemented using the Librosa library. These techniques included: 1) Noise Injection, where random white noise was added to the waveform to improve the model's robustness against environmental noise; 2) Time Stretching, which varied the audio's speed within a random range of 0.85x to 1.15x to simulate different speaking rates; and 3) Pitch Shifting, which adjusted the vocal pitch by a random factor between -3 and +3 semitones to account for tonal variations. The parameters for each transformation were randomized for every new sample generated, ensuring a high degree of variability. This strategic creation of new, varied training instances was essential

for mitigating potential biases towards the majority classes and improving the model's generalization
capabilities across all emotion categories.

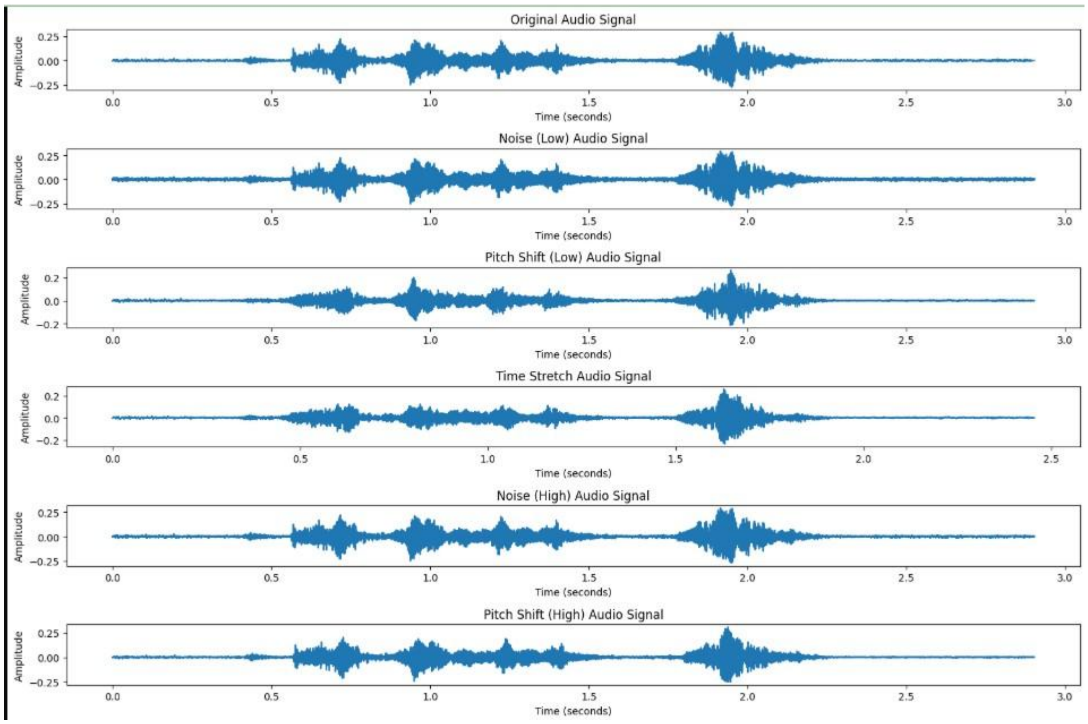


Figure 13:Augmented Audio signal samples

5.18 Feature Extracting

5.18.1 Mel-frequency cepstral coefficients (MFCCs)

Mel-frequency cepstral coefficients (MFCCs) are a widely used and highly efficient feature representation in the field of speech processing, especially for impulse recognition. They mathematically represent the short-term power range of a sound symbol into a nonlinear Melfrequency range, which is more in line with the way the human auditory system processes sound.

This representation is important because it describes the filtering characteristics of the airway and the vocal folds for sound generation. Both of these vary significantly depending on the impulse or “impulse state” of an individual. For example, the spectral pattern of a fearful voice, with high fundamental frequency and jitter, is significantly different from that of a happy voice, with high amplitude and constant intensity.

This sensitivity to detecting physical changes that occur with different impulse states makes MFCCs very efficient at detecting differences between different impulse states. This special ability is essential for emotion recognition, as the full spectral content of vocal expressions such as voice or laughter is strongly influenced by the underlying emotion state. MFCCs serve as a feature that accurately captures this reality.

5.18.2 Mel-Spectrogram

A time-frequency representation is defined as Mel-Spectrogram representation of sound. As the sound human experience, it is simulated and developed such that signal is mapped mathematically into the nonlinear Mel range. This is either through the use of Mel-filter banks and commonly logarithmic compression on Short-Time Fourier Transform (STFT) power spectrum. This representation uniquely shows the energy distribution of psychologically relevant signal ranges over time. This is crucial for speech impulse recognition, as it accurately captures subtle but important differences in vocal muscle settings and sound generation sources that vary psychologically during impulse expression nfor example, music, timbre, and rhythm. Accordingly, Mel-Spectrograms serve as an effective and rich input feature for deep learning models to identify different impulse states from speech.

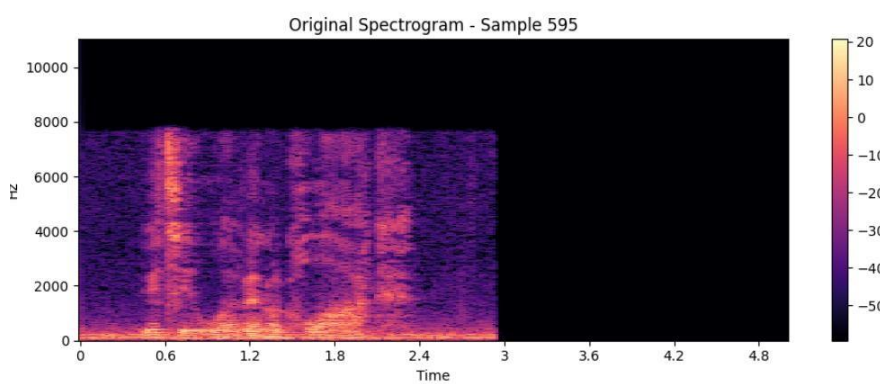


Figure 14:Mel spectrum representation for sad

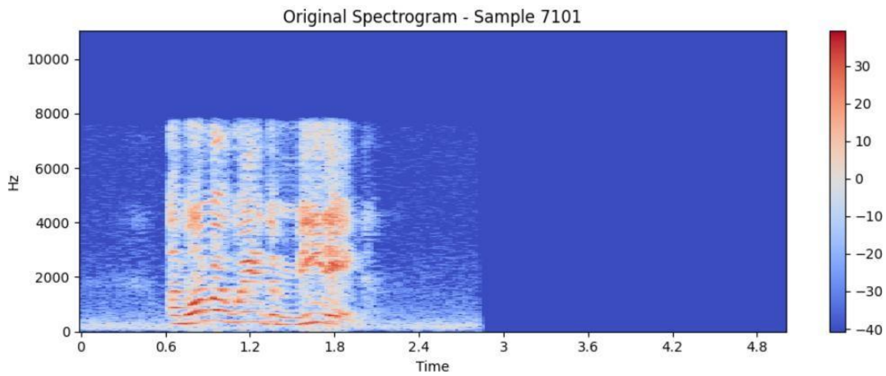


Figure 15: Mel spectrum representation for Angry

5.18.3 Spectral Contrast

Spectral Contrast is an audio feature designed to quantify the relative amplitude difference between the peaks and valleys in the short-term spectrum of a sound signal. It segments the frequency spectrum into distinct sub-bands and, for each band, computes the difference between the average energy of the spectral peaks and the average energy of the spectral valleys. Perhaps this aspect is more efficient in the acoustic nature of the presence or the lack of pronounced spectral peaks as opposed to the noisiness of the spectrum or the absence of significant spectral peaks. In speech emotion, contrasts along the spectrum can offer significant contributions to voice quality or articulation differences cited in connection to emotional states e.g., an angry voice may be sharp and reflect well-defined spectral crests compared to the sad or the muffled voices and the sharp difference may therefore be a discriminate feature in terms of identifying emotional differences.

5.19 Model architecture

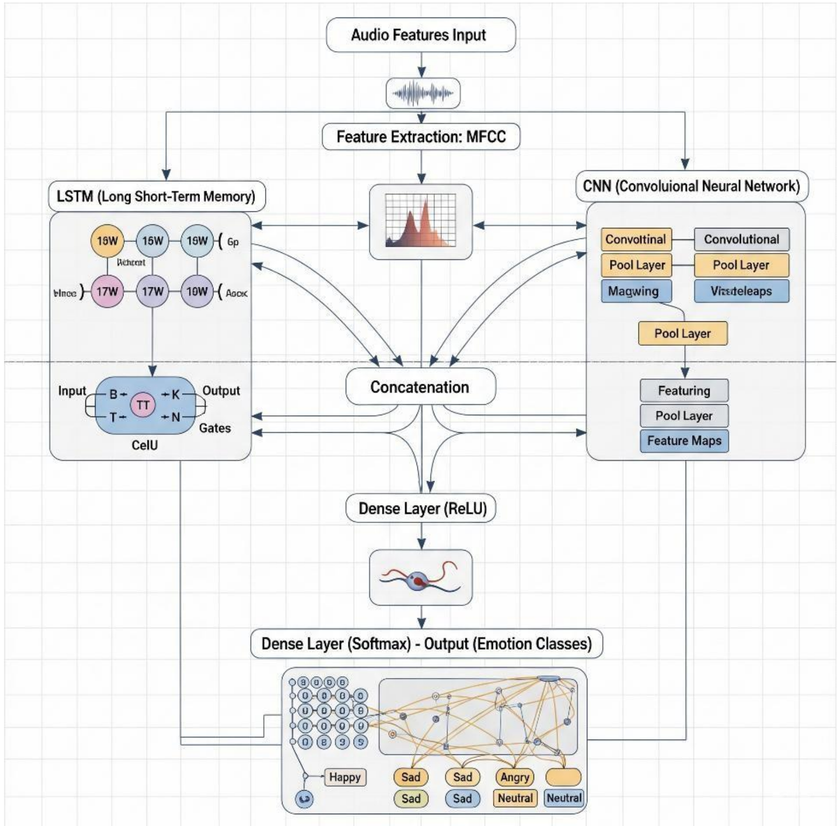


Figure 16:Audio Multimodal Architecture

5.19.1 CNN

This study implements a convolutional neural network (CNN) [26]as a key feature extraction component in a hybrid architecture for speech emotion recognition. Specifically, the CNN branch is designed to process a 2D Mel-Frequency Cepstral Coefficients (MFCC) feature map. By treating the MFCC feature map as an image, CNN uses its ability to detect and isolate meaningful Spectro-temporal patterns such as pitch, strength, and shifts in form. These patterns play a very important role in the distinction between emotional states during the speech. By ensembling CNN to perform feature extraction the model will learn more complex representations of emotional cues at higher levels aiding the classification tasks to identify the type of emotion.

5.19.2 Mel-Frequency Cepstral Coefficients (MFCCs) as Input to CNN Model

In the proposed architecture, the input to the Convolutional Neural Network (CNN) section is a convolutional Mel-Frequency Cepstral Coefficients (MFCC) feature map. The dimensions of the MFCC feature map are specified by the parameters of the feature extraction process. Specifically:

- The time axis of a 3-second audio segment is represented by a height of 130, which captures the temporal structure of the audio signal over a range of 130-time steps.
- The spectral shape of the sound is captured by 40 MFCC features, which act as the width of the feature map. These features represent the main spectral components of the sound, capturing the basic spectral properties.
- A single channel is used, which allows CNN to treat the MFCC feature map as a grayscale image. This allows the features to be extracted by the convolutional layers of the CNN.
- Accordingly, the final model input shape for each audio input is (130, 40, 1), where 130 is the time steps, 40 is the MFCC coefficients, and the channel dimension represents the single channel.

5.19.3 CNN Structure

The convolutional neural network (CNN) architecture employed in this study consists of two sequential convolutional blocks, each designed to extract hierarchical features from the input MFCC feature map.

- Block 1: The first convolutional block is responsible for learning low-level features from the MFCC input. It focuses on capturing basic spectral patterns and pitch variations inherent in the audio signal.
- Block 2: The second convolutional block processes the output from the first block, learning more complex and abstract feature combinations. This enables the model to capture higher-level patterns and temporal structures, thus improving its ability to detect subtle speech quality variations.

The final output of the CNN branch is a flattened feature vector that serves as a compact, high-level abstraction of the key features identified by the network.

5.19.4 Convolutional Layers

The convolutional layers form the core analytical components of the CNN architecture. These layers apply filters to the input MFCC feature map, capturing both spatial and temporal feature sequences. The first convolutional layer identifies basic, low-level patterns, while the second layer builds upon these features to extract more complex and abstract representations.

5.19.5 Max Pooling Layers

After each convolutional operation, a max pooling layer is applied. Max pooling reduces the spatial dimensions of the feature maps by selecting the maximum value within a small window (typically 2x2). This operation helps reduce computational complexity and enhances the model's ability to generalize by making it more invariant to small shifts in pitch or energy.

5.19.6 Activation Functions: ReLU and SoftMax

Activation functions introduce non-linearity to the network, enabling it to model complex patterns in the data. This network utilizes two activation functions: ReLU and SoftMax.

5.19.7 ReLU (Rectified Linear Unit)

ReLU is applied to all hidden layers. It is defined mathematically

$$ReLU(x) = max\ (0, x)$$

Where x is the input. If x is positive, the output is x; otherwise, the output is 0. This activation function is widely used for its simplicity, computational efficiency, and effectiveness in mitigating the vanishing gradient problem.

5.19.8 SoftMax

The SoftMax function is used in the output layer to transform the raw output logits into a probability distribution. Given an output vector $Z = [z1, z2, \ldots, zk]$, the SoftMax function computes the probability. This makes such output values to be between 0 and 1 and also allows the total probability of any classification to sum to 1 and therefore suitable in multi-class categorization.

5.19.9 Adam Optimizer Algorithm

The Adam (Adaptive Moment Estimation) optimizer is used to minimize the model's loss function. It computes adaptive learning rates for each parameter by using moving averages of both the gradient (first moment) and the squared gradient (second moment). Adam combines the benefits of Momentum and RMSprop, providing an efficient method for optimizing complex models. In this study, Adam is employed with a learning rate of 0.001.

5.19.10 LSTM Branch of the Model

The LSTM (Long Short-Term Memory) [18] layers are aimed at learning the temporal relations among the features sequence processed through CNN branch. The time-based dependencies are crucial to speech emotion recognition since often emotions in speech are bounded by their temporal ordering of acoustic features.

The LSTM portion of the model has used three stacked Bidirectional LSTM layers. We take the number of LSTM layers as 1 and the number of units per LSTM layer as 128 with Bidirectional wrapper to process the sequence both backward and forward. This can be quite effective in speech emotion recognition, to enable the model to learn dependencies based on both past and future context, which can often be essential to determine emotional aspects of speech.

5.19.10.1 Reshaping for LSTM Input

The output of the CNN branch is reshaped to match the expected input format for the LSTM layers. Specifically, the feature map from the CNN branch is reshaped to a 3D tensor, where the first dimension represents the time steps, the second represents the feature dimension, and the third dimension is the sequence length. This reshaped output is then fed into the LSTM layers.

5.19.10.2 Bidirectional LSTM Layers

First Bidirectional LSTM Layer: This layer processes the input sequence from both directions (forward and backward) and returns a sequence of outputs, preserving the temporal dependencies for further processing by subsequent LSTM layers.

Second Bidirectional LSTM Layer: This layer continues to process the sequence in both directions, further extracting temporal dependencies from the data.

Third Bidirectional LSTM Layer: Similar to the first two layers, the third LSTM layer processes the sequence and provides a sequence output for further attention processing. This layer captures additional higher-level temporal patterns and is crucial for the temporal context of speech.

5.19.10.2 Dropout Layers

After each LSTM layer, a dropout layer is applied to regularize the model and reduce overfitting. A dropout rate of 0.3 is used, meaning 30% of the units are randomly deactivated during training. This technique prevents the model from relying too heavily on any one-time step or feature.

5.19.10.3 Temporal Attention Mechanism

A temporal attention mechanism is applied to the output of the third LSTM layer to focus on the most relevant time steps for emotion recognition. The attention mechanism computes attention scores for each time step, and these scores are passed through a SoftMax activation function to normalize them into a probability distribution. This allows the model to focus more on the time steps that are more indicative of the emotional content of the speech.

The weighted sum of the LSTM outputs, weighted by the attention scores, is computed to create a context vector. This context vector highlights the most important time steps and helps the model concentrate on the relevant portions of the speech sequence.

5.19.10.4 Global Average Pooling

After computing the context vector, a Global Average Pooling operation is applied. This operation reduces the sequence length by averaging the features across all time steps, resulting in a single feature vector that represents the temporal dependencies of the speech sequence.

5.19.10.5 Concatenating CNN and LSTM Features

The outputs of the CNN and LSTM branches are concatenated to form a combined feature vector. This vector contains both spatial features (captured by the CNN) and temporal dependencies (captured by the LSTM and attention mechanism). This combined feature vector is then passed through a fully connected layer to make the final prediction.

The use of Bidirectional LSTMs, in combination with the attention mechanism, allows the model to effectively capture both the spatial patterns (via CNN) and temporal dynamics (via LSTM) necessary for accurate speech emotion recognition.

5.19.11 Results and Evaluation

The model is trained using a batch size of 32 for up to 50 epochs. The categorical cross-entropy loss function is employed to quantify the error between the predicted probabilities and the true labels. To enhance training efficiency and prevent overfitting, the following callbacks are utilized:

- Early Stopping: Monitors the validation loss and halts training when the loss stops improving.
- Model Checkpoint: Saves the model's weight whenever the validation performance improves.
- ReduceLROnPlateau: Reduces the learning rate if the validation performance plateaus.

Once training is complete, the model is evaluated on an untouched test set. Key evaluation metrics, including accuracy, precision, recall, and F1-score, are computed. Additionally, a confusion matrix is generated to provide a comprehensive performance evaluation across all categories.

5.20 Combined Approach with Both video and audio

In this study, a multimodal approach that combines video and audio data was used to improve the efficiency of the Facial Emotion Recognition (FER) task. Late Fusion was chosen as the method for combining these two media because it has been shown to be efficient in successfully combining complementary information from different sources at the decision level.

5.20.1Tools & Technology

5.20.1.1 Late fusion

A late fusion occurs where the results of models trained independently on sets of media are combined after converting the inputs to each model. Two models were in this case trained in isolation using either the audio data (containing speech features) or using the video data (containing facial features).

5.20.1.2 Data collection

The database used in this study is based on the Savee (Surrey Audio-Visual Expressed Emotion) database. It contains both audio clips and image data.

5.20.1.3 Audio data

The Savee database contains 480 audio clips of 3 seconds in length. This covers emotions such as anger, disgust, fear, happiness, sadness, and neutral. The audio data is used to train the SER model. For preprocessing, Mel-Frequency Cepstral Coefficients (MFCCs) are obtained and applied as features to the SER model.

5.20.1.4 Image data

The corresponding facial expressions for the 480 audio clips containing similar emotions were manually generated. This is the facial expression captured from the video and matches the emotion of the corresponding audio clip. All images were scaled to 64x64 pixels and normalized. For the simulations, the audio clips and images were paired with each other and the emotion labels angry, disgust, fear, happy, sad, neutral, and surprise were applied. During the testing phase, only the active emotions angry, disgust, fear, happy, and sad were focused on.

5.20.1.5 Pre-trained Models

In this approach, both the FER and SER models are pre-trained and saved as .h5 files. These models are used without further training during the testing phase.

5.20.1.6 FER Model

The Facial Emotion Recognition (FER) model is a Convolutional Neural Network (CNN) trained to predict emotions based on facial expressions. It is loaded with a pre-trained .h5 file.

5.20.1.7 SER Model

The Speech Emotion Recognition (SER) model is trained to predict emotions from speech features. The MFCCs extracted from the audio are used as input to this model, which is also loaded from a pre-trained .h5 file.

Both models are used independently to make predictions, and then their outputs are combined using the late fusion approach.

5.20.1.8 Decision-Making

Late fusion integrates the FER and SER model forecasts after processing of their modalities. The benefit of this method is that it is computationally less complicated, one does not need to jointly train or extract features of both modalities simultaneously. Rather, the audio and video models will be used together, pooling predictions made by individual models.

5.20.1.9 Model Predictions

The FER model processes the image data (facial expression) and predicts the emotion. The SER model processes the audio data (speech clip) and predicts the emotion.

5.20.1.10 Fusion Weights

Fusion weights are computed based on the pre-trained model accuracies. These weights indicate how much each model should contribute to the final decision. Weighs are calculated using the following formulas.

$$\text{Weight}_{\text{FER}} = \frac{\text{FER Accuracy}}{\text{FER Accuracy} + \text{SER Accuracy}}$$

$$\text{Weight}_{\text{SER}} = \frac{\text{SER Accuracy}}{\text{FER Accuracy} + \text{SER Accuracy}}$$

Figure 17:How to calculate fusion weight

5.20.1.11 Combining Predictions

The output of FER model (video) and SER model (audio) are selected and then their outputs are fused using calculated fusion weights. This leads to a unifying prediction vector. The last step to identifying the emotion is taking the most probable emotion after combining predictions.

5.20.1.12 Final Decision

The emotion corresponding to the highest probability after combining the predictions is selected as the final prediction for the test sample.

5.21 Solution Recommendation Using Agentic Architecture (Crew AI) Introduction

In the proposed emotion-aware support system, after identifying a user's emotion and its cause, the system should provide supportive, appropriate, and psychologically based recommendations. Traditional approaches often rely on pre-existing databases, where preexisting suggestions are associated with specific emotions. However, this approach lacks multilevel, scientific validation, and user-specific modeling. To overcome these limitations, this research implements a multi-level agent system based on a large language model (LLM) called

Crew AI. Crew AI can automatically collaborate with agents to perform process-related tasks, and generate personalized, psychologically valid, natural language recommendations for the user.

5.21.1 Why Crew AI?

To make the recommendation process more intelligent and responsive, this study replaces traditional, static solution-mapping methods with a dynamic, agent-based reasoning system. Rather than relying on a fixed database of emotional triggers and predefined suggestions, the system uses Crew AI an open-source framework designed to simulate collaboration between multiple reasoning agents. [16] Each agent takes on a specialized psychological or cognitive role, such as understanding emotional input, validating strategies through scientific evidence, or generating user-friendly guidance. Together, these agents form a coordinated pipeline that interprets user emotion and reason, researches the latest strategies from psychological literature, and delivers personalized, scientifically informed support in natural, accessible language. This approach allows the system to remain up to date without needing constant manual database updates and better mimics how a trained human support team might respond to emotional needs.

Feature	**Traditional DB Method**	**Crew AI Agentic Approach**
Solution Generation	Keyword/Tag Matching	Real-time reasoning based on emotion + reason
Personalization	Minimal	Highly personalized using context, tone, and user needs
Scientific Validity	Manually curated, often outdated	Validated using research-backed methods (meta-analysis, CBT)
Emotional Tone	Generic or robotic	Warm, trauma-informed, conversational
Adaptability	Static	Easily updatable and extensible via agents and LLMs

User-Friendliness	Often overly clinical or rigid	Simple, everyday language with intelligent insights

Table 1:Difference between traditional databases and Crew ai agentic system

Crew AI-based solutions are designed as a small team of mental health agents, each with a specific role in the sentiment analysis process. This collaborative structure mimics a real-life interdisciplinary healthcare team. [20]

5.21.2 Agentic Workflow

Emotion Intake Agent

- **Role**: Emotion Assessment Specialist
- **Function**: Validates the detected emotion (e.g., sadness, joy, anxiety) using psychological models such as the Circumplex Model of Affect or DSM-5 criteria.
- **Outcome**: A clinically grounded emotion profile to start the recommendation process.

Evidence Based Strategy Analyzer

- **Role**: Clinical Researcher
- **Function**: Searches for proven, evidence-backed short-term coping strategies based on recent meta-analyses, Cochrane Reviews, and APA guidelines.
- **Outcome**: A portfolio of validated short-term and long-term interventions for the detected emotion.

Personalized Solution Generator

- **Role**: Therapy Designer
- **Function**: Customizes recommendations using user-specific context (reason, culture, stress level, support system) and adapts them for accessibility and simplicity.
- **Outcome**: Practical and immediate strategies the user can follow (e.g., journaling, “no contact”, reflective writing, etc.).

Trauma Informed Presenter

- **Role**: Empathetic Communicator
- **Function**: Presents suggestions using traumatically informed language, validating the user’s emotions while ensuring emotional safety and user agency.
- **Outcome**: A user-friendly, gentle interaction that builds trust.

Contextual Analysis Researcher

- **Role**: Root Cause Analyst
- **Function**: Analyzes the user’s stated reason (e.g., “I failed an exam”, “They left me”) using biopsychosocial models, attachment theory, and neuropsychological reasoning.
- **Outcome**: A deeper understanding of the cause behind the emotion and its long-term implications.

Final Presenter Agent

- **Role**: Support Summary Agent
- **Function**: Combines all prior agent outputs into a readable, encouraging support plan using natural language.
- **Outcome**: A human-like support message, including immediate actions, long-term advice, encouragement, and referral guidance if necessary.

The following figure illustrates the Crew AI architecture used in the system:

Note: Each agent is designed to be stateless and reusable, making it easy to scale the system to include new mental modules or multi-modal analysis (e.g. text + facial signals).

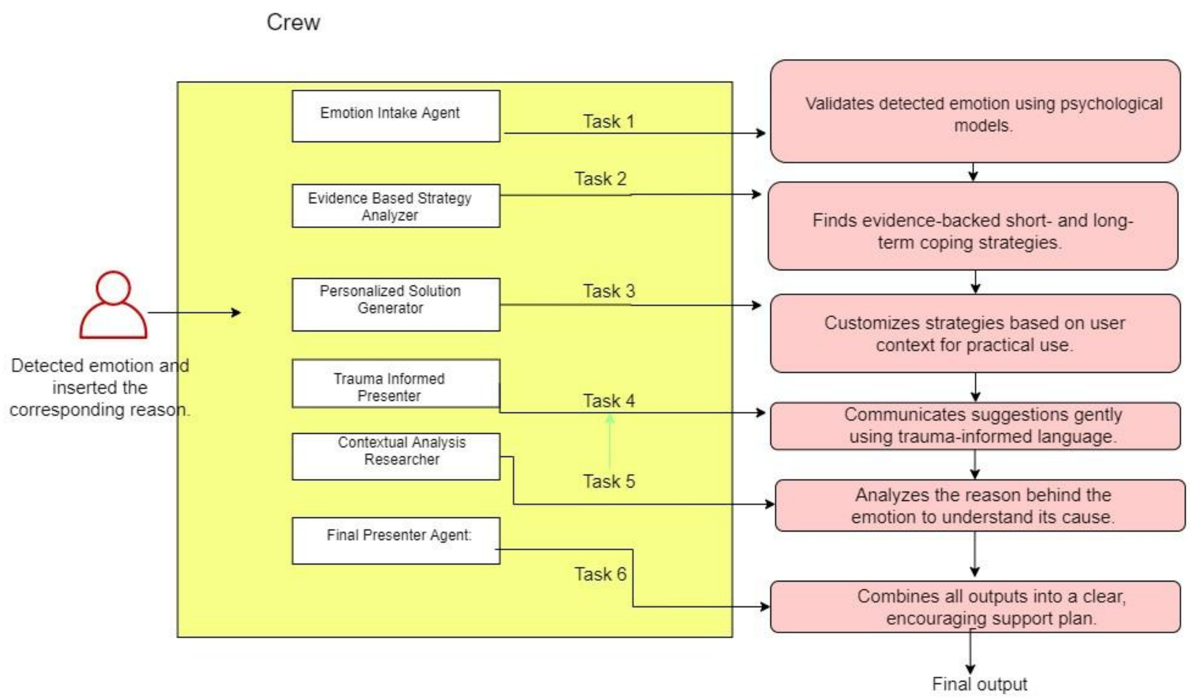


Figure 18: Architectural diagram for crew behavior

In summary, Crew AI is an innovative, scalable, and evidence-based way to implement a mental solution recommendation system. By delegating intellectual subtasks to expert agents, the system bridges the gap between mental error expertise and AI-powered responsiveness. It acts as a human-like, medically validated framework that can provide mental coordination through technology.

5.22 System Overview

The developed system is a web-based emotional wellness platform designed to detect and respond to human emotions in real time. It combines a React and TypeScript frontend with a Flask backend, creating a smooth interaction between the user interface and the AI-driven emotion recognition engine. The platform focuses on analyzing facial expressions and voice patterns to identify emotional states and then provides tailored wellness suggestions based on the detected mood. [19]

On the front end, the system uses React together with TypeScript to build a responsive and user-friendly interface. To ensure a user-friendly experience, the interface was first prototyped using Figma. This allowed for careful planning of the user journey and design elements before moving into development. The final frontend was then implemented in React and TypeScript based on the Figma designs, ensuring consistency and usability. This choice allows for a modular design and ensures type safety, which helps maintain the application as it grows. The dashboard offers live emotion analysis, showing the user's current emotional state along with a confidence level. Users can choose between face detection, voice detection, or a combination of both. Depending on the detected emotion, the interface dynamically updates to show relevant wellness tips, music therapy suggestions, and general guidance to support the user.

The backend is built with Flask, [25] which serves as the processing layer that connects the user interface to the AI models. It handles incoming image or audio data, preprocesses it, and runs inference using trained deep learning models. Flask then sends the predicted emotion and confidence score back to the frontend in real time. Its lightweight nature makes it well-suited for integrating machine learning workflows and providing fast, reliable API communication.

The workflow of the system is simple yet efficient. When the user starts the camera or microphone, the front end captures the input and securely sends it to the backend. The Flask server processes the data through the appropriate model, determines the emotion, and responds

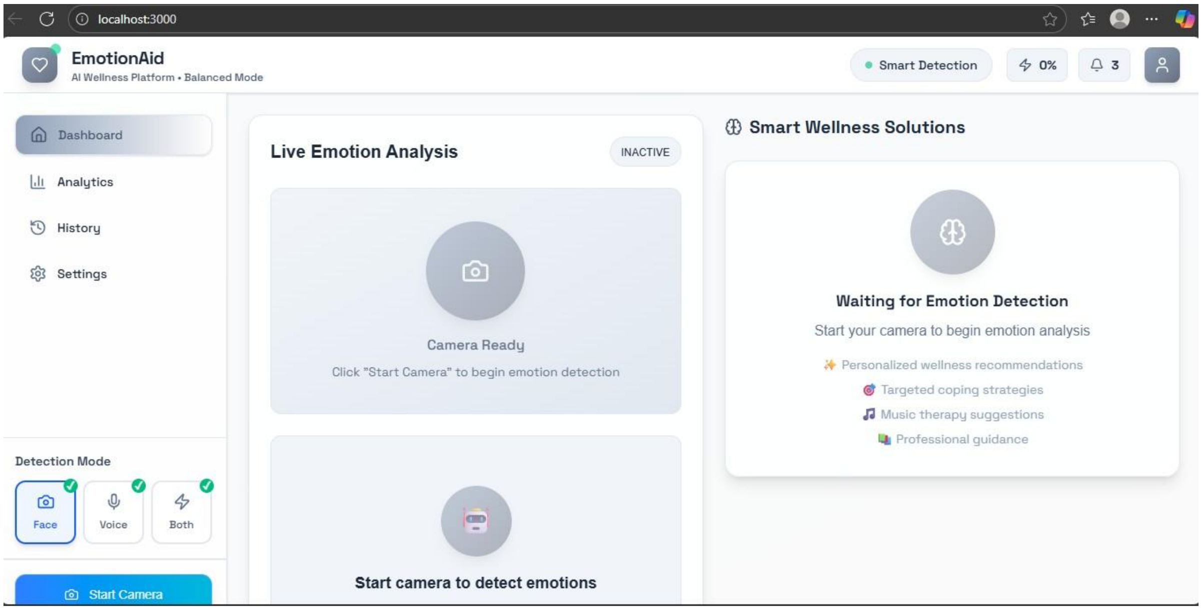


Figure 19: Main Interface of the final Emotion recognition system

with the results. The interface then updates instantly, giving the user immediate feedback and personalized wellness recommendations based on their current emotional state.

In designing the user experience, the system incorporates a color-coded interface to visually represent different emotional states. Six distinct colors were chosen to correspond with specific emotions, with careful consideration given to psychological associations of color. For instance, when the system detects sadness, the interface adopts a calming blue tone, while anger is represented with a light red, and happiness with a bright yellow. These color choices were not arbitrary; they were selected to match the emotional context and to enhance the user's interaction with the platform. The decision to use blue for sadness carries particular significance, as blue is known to be an eye-friendly color and was chosen under the assumption that a large number of users might access the platform while experiencing sad feelings. [21]

To further support emotional regulation, the system integrates auditory feedback by automatically playing background sounds designed to help users cope with the detected emotion. For privacy and comfort, the video capture process is limited to six seconds per detection session, after which the camera closes automatically to avoid prolonged exposure that might make users feel self-conscious. In contrast, the voice detection feature offers more flexibility, allowing users to engage at their own pace without time constraints. This combination of visual cues, sound-based therapy, and thoughtful user experience design aims to create a supportive and non-intrusive emotional wellness environment.

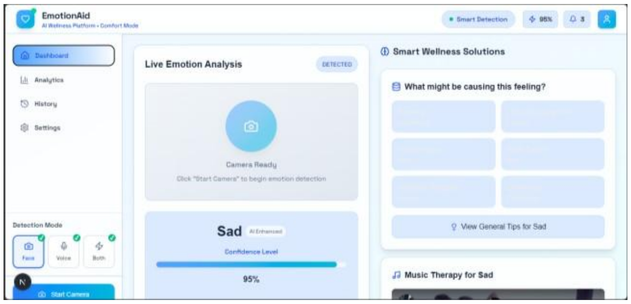


Figure 20:Blue Interface sad emotion

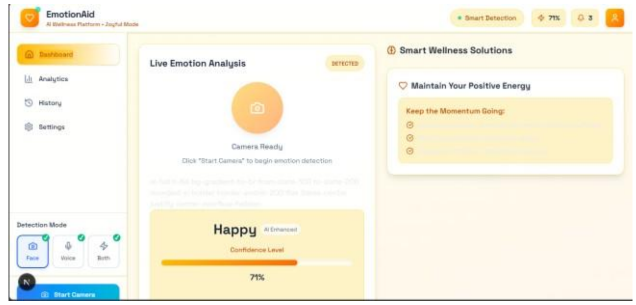


Figure 21: Yellow Interface for happy emotion

In addition to its core functionality, the system is designed to be intuitive and accessible to a wide range of users with different levels of technical knowledge. The interface emphasizes simplicity, ensuring that even those with minimal technical experience can navigate the platform easily. Furthermore, the backend architecture allows the system to adapt and learn quickly from diverse input patterns, enabling it to improve emotion detection accuracy over time. This combination of ease of use and adaptability ensures that the platform can serve a broad user base effectively, providing seamless experience for individuals from various backgrounds and levels of digital intelligence.

6 Experiments and Results

6.1 Image Approach

For the visual method, I conducted various experiments to identify the effect of different features on human emotion recognition. I also tested with pre-trained models such as, MobileNetV2 EfficientNetB0, ResNet50, VGG16 etc. I selected and tested SVM and Decision Tree as other machine learning models. Then I selected and tested the CNN model. I also tested changing the number of layers in CNN and changing the feature extraction methods. Initially, I tested on the entire database, but later I realized that some prediction results were inconsistent due to the data sets. Therefore, I found many different cases where some emotions were incorrectly detected from this database. Going through them, I learned a lot of new things each time. And in the end, I was able to get good results.

In initial experiments pretrained models are used for emotion detections. Experiments are done using Mobile Net ResNet50, VGG16, and a simple CNN.

6.1.1 Experimental Setup for FER-2013 Emotion Detection

I conducted a series of controlled experiments to select the most efficient CNN (Convolutional Neural Network) model for facial emotion recognition on the FER-2013 dataset. Here, the performance of the process was revealed by varying the depth of the CNN model, and by systematically varying the number of Convolutional and Fully Connected layers. In order to obtain better quality features, efforts were made to reduce the inter-class bias using the Oversampling method, and to focus more attention on important features by applying Squeeze and-Excitation (SE) blocks and attention mechanisms. Data preparation tasks included removing watermarks from the images, increasing image clarity, and using progressive pooling methods such as Max pooling, Average pooling, and Global pooling. All these strategies combined helped to increase the model's robustness, reduce noise, and better represent facial emotion-related features.

6.1.1.1 Transfer Learning with ResNet50 for Facial Expression Recognition

In a separate effort to improve the performance of this FER, a transfer learning method was implemented using the ResNet50 model pre-trained on ImageNet. To fit this model, the grayscale images of FER2013 were pre-processed with CLAHE and bilateral filtering and then converted to a 3-channel model. Initially, only a custom classification head was trained, confirming the base layers of the ResNet50 model. In this setup, the validation accuracy reached about 34–35 percent, but the test accuracy was very low. To better utilize the model's capabilities, all layers were re-enabled, and the fine-tuning process was initiated at a reduced learning rate. This resulted in training accuracy exceeding 96%, and validation accuracy approaching 80%, while testing accuracy dropped to 45.81%. This showed that while high-capacity models are capable of learning patterns quickly, they can quickly be overfit on small and heterogeneous datasets like FER2013 especially when new differences are not detected in
the testing data.

To address this, a second, more controlled training set was designed. Again, CLAHE and Bilateral filtering was used to preprocess, and the heterogeneity between classes was controlled using random oversampling. A custom classification head with dropout and regularization was appended to the frozen base, followed by selective fine-tuning of the top 30 layers. However, even with this improved setup, the model plateaued at ~33% validation accuracy, and test accuracy remained low (33.47%). While emotions like Happy and Neutral were reasonably well recognized, there were still significant challenges in recognizing emotions like Fear, Disgust, and Sad. These results suggest that large pre-trained models like ResNet50 may not be suitable for applications like FER. To properly generalize datasets with highly complex emotion variations, such as low-resolution images, simpler models, more domain-specific pretraining, or the inclusion of multimodal inputs may be required.

	precision	recall	f1-score	support
0	0.26	0.22	0.24	945
1	0.45	0.20	0.27	92
2	0.24	0.16	0.19	964
3	0.46	0.42	0.44	1759
4	0.27	0.30	0.29	1185
5	0.45	0.45	0.45	651
6	0.29	0.42	0.34	1211

Figure 22:Classification report for Resnet50 model

Figure 23:Data set balancing for Resnet50 model

6.1.1.2 Transfer Learning using MobileNetV2

In this experiment, we investigate emotion recognition on the FER2013 dataset using MobileNetV2 through transfer learning. The original grayscale images (48×48) were converted to three-channel RGB to match the input requirements of MobileNetV2. The labels were one hot encoded according to seven emotion categories. A stratified train-validation split (80:20) was implemented to maintain class balance. For data augmentation, the input data was arranged in various ways using techniques such as rotation, magnification, inversion, photo complexity changes, and spatial shift. This was implemented using the ImageDataGenerator in Keras. The MobileNetV2 architecture was initially implemented with pre-trained weights (except for the classification head) on ImageNet. A new classification head was added with global average pooling, two dense layers with ReLU activation and dropout, and a SoftMax output layer. The base model was initially sampled, and only the top layers were trained using categorical cross entropy with Adam optimizer and label smoothing.

After sampling (convergence), the base model was unfrozen and trained at a lower learning rate for fine tuning. The training was optimized using callbacks such as learning rate reduction and early stopping. The best validation accuracy obtained during training was recorded as 26% as a symbol. The final test accuracy was also recorded as 24%. The confusion matrix plot showed balanced classification for most emotion categories.

6.1.1.3 EfficientNetB0 with CLAHE Preprocessing

A unique preprocessing pipeline was used, where each 48×48 grayscale image was contrast enhanced using Contrast Limited Adaptive Histogram Equalization (CLAHE). This was done to improve the model's ability to distinguish minor facial features such as smiles, anger, and tears under varying lighting and photo complexity. The main structure was built using the EfficientNetB0 backbone (without pre-trained weights), followed by global average pooling. Two dense layers (with dropout rates of 0.4 and 0.3) along with Dropout and Batch Normalization were incorporated to reduce overfitting and improve learning stability. The final layer was a SoftMax classifier with 7 output units for 7 emotion categories. The data were normalized to the range [0, 1] and one-hot encoded. The data was stratified and split into 80% for training and 20% for testing, with a further 10% of the training data set reserved for validation. The model was trained for up to 50 epochs, utilizing callbacks such as early stopping, learning rate reduction on plateau, and checkpoint saving. However**,** the model showed relatively low generalization ability, attributed to the absence of pre-trained weights and the limitation of image resolution. The best validation accuracy achieved during training was approximately 49%, and the final test accuracy was 48.04%.

6.1.1.4 Hybrid Preprocessing with Attention-Efficient Net

This was a combination of a custom deep network architecture based on EfficientNetB0 with improved image preprocessing techniques and an attention mechanism. The preprocessing stage used techniques such as CLAHE (Contrast-Limited Adaptive Histogram Equalization), bilateral filtering, and gamma correction, which were derived from peer-reviewed studies especially for improving the visual quality of low-contrast facial images. After preprocessing, the data was balanced using RandomOverSampler to address class imbalance. The input was kept as

grayscale (48×48) and converted to 3 channels for Efficient Net requirements. The base network was EfficientNetB0 (without pre-trained weights). After that, feature maps were enhanced using an attention block, and a dense head was added with Gaussian noise and dropout regularization. This helped improve the robustness and generalization of the model, especially under noise and imbalance. Aggressive data augmentation, learning rate scheduling, and early stopping callbacks were used in the training pipeline. Mixed precision training improved memory usage and training speed. Finally, the model achieved a test accuracy of 65.63%, demonstrating a strong balance between structural complexity and generalization, using input of 48×48 resolution.

6.1.2 Experiments with machine learning algorithms

In this study, I applied several machine learning models to facial expression recognition, considering the practicality and accessibility of running on a personal computer. Compared with large-scale deep learning networks that require high-end GPUs and large computing resources, traditional and lightweight machine learning methods provide a more efficient solution in limited hardware environments. These models provide competitive baseline performance, fast modeling, short training time, and easy annotation capabilities. Here, I tested the code only after proving that they were highly accurate and suitable, avoiding unnecessary wastage of colab's computing power units.

6.1.2 .1 A Lightweight Machine Learning Framework for Facial Emotion Recognition Using Principal Components and XGBoost

A proportional dataset was utilized, comprising deep feature representations extracted from facial images alongside their corresponding emotion labels. The original CSV file was preprocessed to separate the feature vectors from the target labels. Label encoding was employed to convert the categorical emotion labels into a numerical format, enabling compatibility with the classification algorithm.

To prepare the data for modeling, standard normalization was applied using StandardScaler to ensure that each feature had zero meaning and unit variance. This step promotes uniformity in feature contribution during training. Dimensionality reduction was then performed using Principal Component Analysis (PCA), retaining 300 principal components. This not only enhanced computational efficiency but also helped reduce noise and mitigate the risk of overfitting by removing redundant or less informative features.

The dataset was divided into training and testing subsets using stratified sampling with an 80:20 ratio. Stratification preserved the original class distribution of emotions in both subsets. The 80% training portion was used for model training and cross-validation, while the remaining 20% was reserved strictly for final model evaluation on unseen data.

For the classification task, the XGBoost algorithm was selected due to its proven effectiveness and robustness in handling structured data. The classifier was configured with 500 estimators, a maximum depth of 12 for the decision trees, and a learning rate of 0.02. To regulate model complexity and prevent overfitting, regularization parameters (reg_alpha = 0.5, reg_lambda = 1) were used. Additionally, row (subsample = 0.8) and column (colsample_bytree = 0.8) subsampling strategies were employed to enhance model generalization.

Model validation was conducted using 5-fold stratified cross-validation on the training set. Each fold trained on 80% of the data and validated on the remaining 20%. Performance metrics recorded for each fold included accuracy, mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and the coefficient of determination (R^2). Detailed classification reports and confusion matrices were also generated to assess per-class performance.

The confusion matrices from each fold were aggregated to produce an average confusion matrix, offering insight into the classifier's consistency across different emotion classes. The matrix revealed balanced classification performance without significant class-specific misclassifications, indicating model robustness.

Finally, the trained model was evaluated on the reserved 20% test set. During cross-validation, the XG Boost classifier achieved an average accuracy of 52.78% with a low standard deviation of ±0.19%, indicating stable performance across folds. The average mean absolute error (MAE) was 1.1961, and the mean squared error (MSE) was 3.9037, corresponding to a root mean squared error (RMSE) of 1.9757. The R^2 score averaged 0.0241, suggesting modest explanatory power in this high-dimensional feature space. Despite the moderate accuracy, the consistency

Classification Report (Test Set):

	precision	recall	f1-score
angry	0.44	0.40	0.42
disgust	0.84	0.91	0.88
fear	0.44	0.35	0.39
happy	0.36	0.37	0.36
neutral	0.38	0.41	0.40
sad	0.46	0.45	0.46
surprise	0.72	0.79	0.76
accuracy			0.53
macro avg	0.52	0.53	0.52
weighted avg	0.52	0.53	0.52

Figure 24:Classification report with XgBoost algorithm

and reliability of the model across different validation sets support its robustness and make it a viable candidate for further tuning and ensemble integration.

6.1.2.2 Deep Visual Features Using LightGBM with Stratified Cross-Validation

In this experiment, I used the csv file that I already created. The features and labels were separated, and `Label Encoder` was used to numerically encode the categorical emotion classes. When preparing the features for model training, standard normalization was performed using `StandardScaler` to obtain a mean of zero and unit variance for each feature. This is important to maintain the numerical stability and learning efficiency that are essential for tree-based models such as LightGBM. The dataset was split into training and testing sets in a ratio of 80:20 using a ratio compression. The class distribution was fully preserved by the ratio compression. The LightGBM classifier was chosen for emotion recognition due to its efficiency in managing large amounts of structured data. The model was configured with 400 estimators, a maximum tree depth of 16, and a learning rate of 0.05. Parameters such as `class weight='balanced'`, `subsample=0.9`, and `colsample_bytree=0.9` were used to reduce class disparity and overclassification. To evaluate the performance and robustness of the model, a 5-fold stratified Cross validation was performed on the training set. At each fold, the model was trained on 80% of the data and validated on the remaining 20%. Performance metrics including accuracy, mean absolute error (MAE), mean squared error (MSE), root mean square error (RMSE), and R^2 score were recorded for each fold. Correlation matrices and classification reports were generated to analyze class-level performance. LightGBM classifier showed superior performance over previous models. It achieved an average inter-validation accuracy of 62.28%, showing consistent performance across folds with a standard deviation of about ±0.0037. The average MAE value was 0.9298, RMSE was 1.7393, and R^2 score was 0.2436, showing a significant relationship between the results and the true labels. After the inter-validation was completed, the model was re-trained on 80% of the full training data and used to evaluate the remaining 20% of the test set. The final fitting matrix and classification report showed consistent performance even on unseen data, confirming the model's generalization ability. These results show that LightGBM is a powerful and scalable classifier using deep feature representations for the task of impulse detection.

6.1.2.3 Feature Importance Analysis Using Random Forest

In this experiment, I used a Random Forest Classifier to identify emotions, implementing a broad machine learning methodology. This model was chosen because of its robustness and ability to handle the complex, high-dimensional features of the FER2013 dataset. The Random Forest model builds multiple decision trees on random subsets of the data and features, and the final prediction is determined by majority vote. This significantly improves accuracy and generalization. A key advantage of this model is the ability to measure feature importance, which provides valuable insight into which facial features are most powerful for classifying specific emotions.

The experiment began with a rigorous data preprocessing phase. First, `Label Encoder` was used to convert the classified emotion labels into a numerical format, which is a prerequisite for many machine learning algorithms. Then, standardization was performed using `StandardScaler` to ensure a mean of zero and unit variance for all features, a critical step in increasing the numerical stability of the model.

The main component of the test was a rigorous, two-stage model training and validation protocol. The dataset was first split into an 80% training set and a 20% test set. This split was done with stratification to maintain the original sentiment class distribution in both subsets. The first stage involved a nested cross-validation process on the 80% training data. An outer loop used a 5-fold Stratified Fold cross-validation to obtain an unbiased estimate of the model's performance. In each outer fold, `GridSearchCV` was run with `cv=3` to systematically adjust hyperparameters such as `n_estimators` and `max_depth`. The ``class weight='balanced’ parameter was a key feature to minimize the effects of class heterogeneity.

Throughout this process, key performance metrics including accuracy, precision, recall, and f1score were recorded. These metrics provided a quantitative measure of the model's predictive ability. After this cross-validation phase, a final model was retrained on the full 80% training set using the optimized hyperparameters. The final performance of this model was evaluated on the previous 20% test set, providing a measure of its generalization capabilities.

The Random Forest classifier achieved a test accuracy of 79.04%, confirming the robustness of the optimized model.

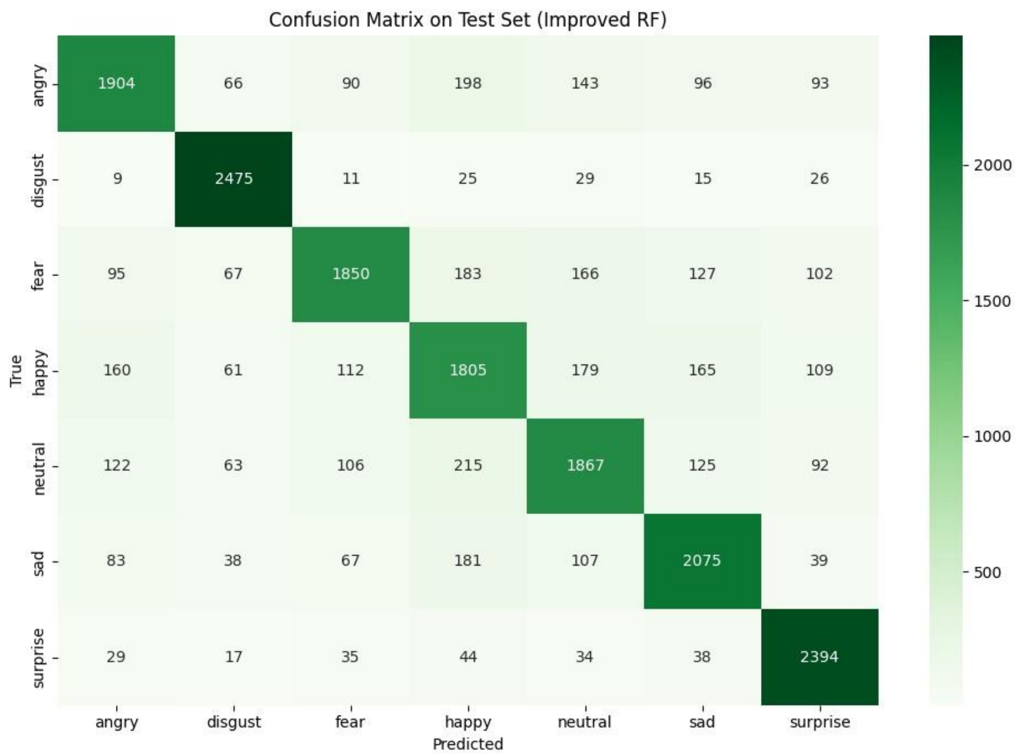


Figure 25:How test samples are distributed with Random Forest classifier

Overall, seven models were tested, as shown in Table below the models utilized either VGG16based deep features or custom extracted features, with variations in preprocessing (standardization, PCA), class balancing, and hyperparameter tuning.

Classifier	**Features**	**Preprocessing**	**Tuning**	**Accuracy**
Random Forest	VGG16	PCA (100) + Scaling	GridSearchCV	0.7881
XGBoost	VGG16	None	Manual	0.6634
XGBoost	VGG16	PCA (300) + Scaling	Manual	0.7897
Random Forest	Custom Features	Scaling	GridSearchCV	0.7888
Random Forest	Custom Features	Scaling	Extended GS	0.7904
LightGBM	Custom Features	Scaling	Manual	0.6423

XGBoost	Balanced VGG16	PCA (300) + Scaling	Manual	0.5278

Table 2:Test Accuracy comparison table for ML Algorithms in FER

And these results were further illustrated through a plot

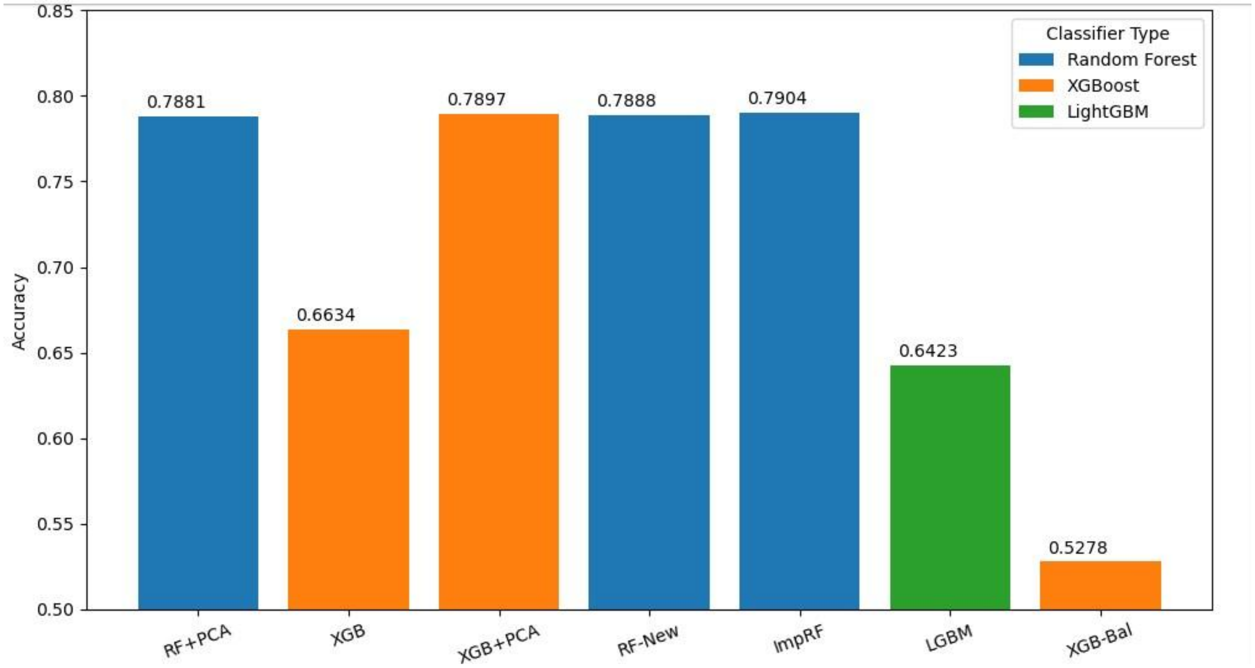


Figure 26:Visualization of ML algorithm accuracies in a bar chart

As shown in above plot, the improved Random Forest and XGBoost with PCA models achieved the highest test accuracy, around 79%. However, it was later identified that class balancing, through augmentation or oversampling, had been applied prior to the train-test split. This approach potentially introduced data leakage, as similar patterns may exist in both the training and test sets, leading to inflated evaluation metrics. Therefore, this limitation should be carefully considered when interpreting the reported accuracies.

6.1.3 Experiments Related to CNN model

6.1.3.1 CNN mechanism using Progressive Dropout and Batch Normalization

In this effect, a fully motivated convolutional neural network (CNN) principle was created without using pretrained models. Here, the aim was to improve generalization by applying a progressive dropout strategy and batch normalization. 4 convolution layers with 3×3 kernel (64, 128, 256, 512 filters respectively) were added, and batch normalization, ReLU activation, max pooling and dropout (from 0.2 to 0.5) were applied after each layer. The generated feature layer was flattened, passed through 512 units fully connected dense layer with ReLU activation, and finally classified into 7 face-like resolutions by a SoftMax classification layer. The FER2013 dataset was preprocessed as a NumPy array, formatted as a 48×48×1 shape and normalized to the range \ [0, 1]. One-hot encoding of the label was applied for the observations. The model was compiled with Adam optimizer (learning rate = 0.001) and trained using categorical cross entropy loss. Early stopping, learning rate reduction, and model checkpoint callbacks were applied, and up to 50 epochs were trained with batch size 64, early stopping stopped at epoch 35 and replaced the weights from the best performing epoch 20. Finally, this model yielded 91.56% training accuracy, 63.01% validation accuracy, and 61.13% testing accuracy (test loss = 1.20). Although it showed good convergence on the training data, the stability of the validation results showed mild overfitting. However, this was a strong initial benchmark for a CNN model created without transferring learning or attention mechanisms.

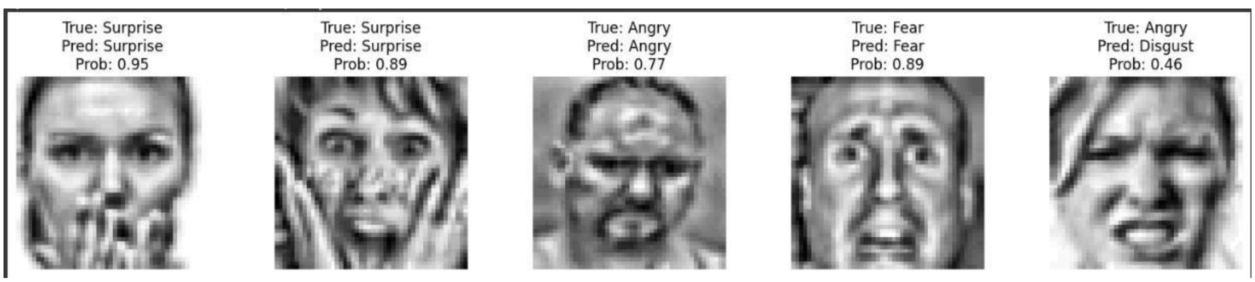


Figure 27: Examples of correctly and incorrectly classified facial expressions with confidence scores

6.1.3.2 Baseline CNN with RGB Inputs and Class Balancing

As an additional preprocessing step, the grayscale images were converted to RGB format. The FER2013 CSV file was analyzed and reshaped to a 48×48×3 format to fit the CNN input requirements by channel repetition. A stratified train-validation partition was performed to preserve the class label distribution as it existed. Class imbalance was adjusted by calculating class weights. To improve the data diversity, several real-time augmentations — small rotations, zooms, shifts, horizontal flips, and photo complexity changes — were applied during training using ImageDataGenerator.

The CNN model consisted of triple convolutional blocks with thirty (32), sixty (64), and one hundred twenty-eight (128) filters in increasing filter size, followed by max pooling layers.

Then, regularization was performed using two dense layers (256 and 128 units) and dropout layers (0.5 and 0.3, respectively). The network was compiled and trained using Adam optimizer and categorical cross-entropy loss. Early stopping, learning rate reduction on plateau, and CSV logging were used as callbacks to ensure training accuracy and reproducibility. After training, the model was evaluated on the validation set. The accuracy and loss curves were plotted, and the convergence was analyzed. The performance of the emotion categories was evaluated using the classification report and confusion matrix.

6.1.3.3 CLAHE and Bilateral Filter-Based Preprocessing with Compact CNN Architecture

This experiment focused on enhancing the FER pipeline by incorporating a hybrid preprocessing strategy, combining CLAHE-based contrast enhancement with bilateral filtering for noise reduction. These steps were designed based on techniques from recent literature to improve the quality of grayscale facial inputs before training. The CNN architecture used was relatively compact, consisting of three convolutional blocks followed by a dense classification head, and regularized with high dropout to prevent overfitting. The model was trained using balanced data obtained through random oversampling and further supported by moderate data augmentation. After 65 epochs, the model achieved a test accuracy of 62.45%, showing improved generalization compared to earlier baselines. The training and validation trends remained mostly stable, suggesting that the introduced preprocessing pipeline effectively contributed to learning discriminative features even within a simpler architecture.

Then we shifted toward a more conservative preprocessing strategy by softening the parameters of CLAHE and the bilateral filter. The goal was to test whether a gentler enhancement of facial features could help reduce overfitting while still maintaining important expression details. Alongside this, the CNN architecture was slightly expanded by adding a third convolutional block, and the dropout rate was reduced to allow the network to learn more expressive features. We also lowered the learning rate to encourage more stable training. Importantly, class balancing was applied only after splitting the data to avoid any leakage from the test set. Although the model showed some improvement during training, the validation accuracy fluctuated, and the final test accuracy settled at 53.36%. These results suggest that while the approach showed promise, it might be more vulnerable to instability and class confusion, especially without stronger regularization or more robust augmentation**.**

In the following experiment, I was rather determined to find a more attentive and methodologically sound means of refining the preprocessing pipeline. Instead of pertaining the enhancements on the entire dataset, the training and test sets were first partitioned prior to an update of the transformation in order to prevent the leakage of data. It was also possible to relax the preprocessing by using conservative CLAHE parameters as well as mild bilateral filtering to get closer to a reasonable balance between enhancement and naturalness. Subsequently, the training data had its data balanced, using oversampling. The CNN architecture employed during this step was of a small size having two convolutional blocks with a dense layer. with moderate dropout prior to regularization. Robustness was achieved using Data augmentation. As the validation performance was stable and even demonstrated good generalization (validation accuracy exceeded 80% by the last few epochs), the actual test set accuracy achieved 61.05%, which is to say that although the network learned to fit a validation distribution well enough, its generalization on the unseen test set can still be improved. This experiment, however, further reaffirmed the use of careful data handling and controlled. preprocessing in Facial Expression Recognizing (FER) applications.

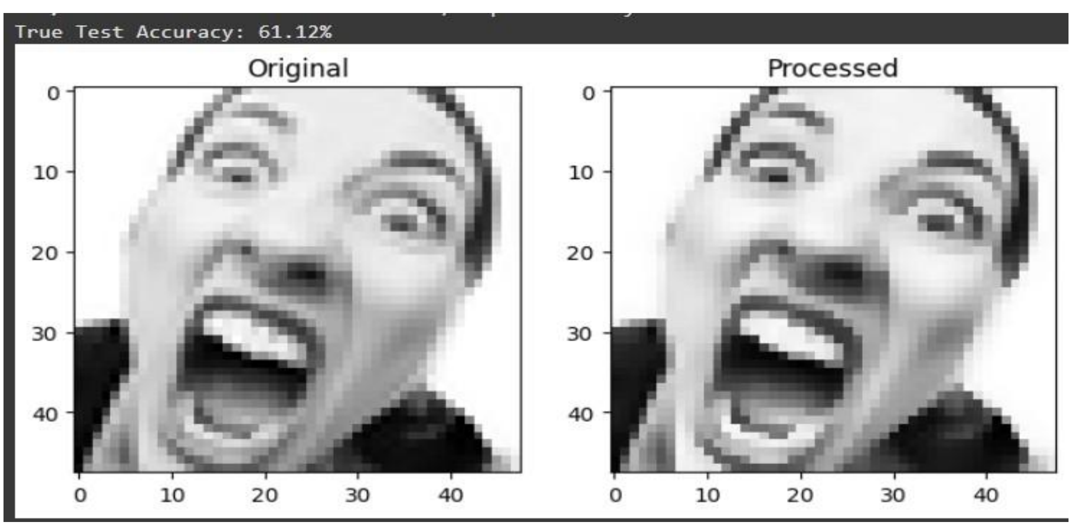


Figure 28: Original image sample vs bilateral filter applied processed image sample

6.1.3.4 Lightweight Regularized CNN with Data Augmentation

The aim of this experiment was to evaluate the performance of a simple CNN structure using L2 regularization and extensive data augmentation. The model was designed with 3 convolutional layers (32, 64, and 128 filters, respectively), and batch normalization, ReLU activation, and max pooling were applied after each layer. Overfitting was attempted by gradually increasing the dropout rate from 0.3 to 0.5. The dense layer (256 units) was also subjected to L2 regularization ($\lambda = 0.0001$) and finally classified into 7 impulses by a SoftMax output layer. A reduced learning rate (0.0005) was used with the Adam optimizer. Data propagation strategies such as rotation, zoom, horizontal flips, brightness variation, and translation were applied to simulate architectural conditions.

However, even after regularization and augmentation, the model failed to learn meaningful patterns from the FER2013 dataset. Training was stopped by early stopping at epoch 9 because the validation accuracy had plateaued. The best validation accuracy obtained was 14.21%, and the final test accuracy was 14.32% and test loss = 2.38. These results confirm that even when training extremely simple CNN structures for facial expression recognition, even regularization and augmentation cannot provide robust propagation.

6.1.3.5 Balanced CNN with Mild Regularization and Augmentation

In this experiment, a balanced CNN structure was investigated by combining data distribution with moderate regularization and two-phase training. The model was built with 4 convolutional layers (with the number of filters increasing from 64 to 512), and batch normalization, ReLU activation, and max pooling were applied after each layer. A fixed L2 weight decay ($\lambda = 0.00005$) was applied to all convolutional and dense layers, and moderate dropout rates (0.25 – 0.4) were also applied. The model was initially pre-trained for 5 epochs without data distribution, and then the goal was to provide a stable weight initialization. Then, data augmentation was applied to the last 45 epochs, which included minor shifts, zooms, horizontal flips, and brightness changes an attempt to prepare the model for architectural conditions. However, despite the use of structured regularization and structured propagation, the model failed to achieve good performance on the entire data. Although training and validation accuracy initially increased, they dropped sharply after augmentation began indicating difficulty in adapting to augmented data. Finally, the test accuracy was 16.13% and the test loss was 2.14. According to the classification report, many predictions were incorrect, with a bias toward common categories such as Fear, and misclassification of categories such as Happy and Disgust. These results show that while regularization and phased training can stabilize convergence, excessive regularization or conflicts between secondary training design and propagation can limit learning ability — especially in facial expression recognition tasks.

6.1.3.6 Three-Phase Training with Gradual Regularization and Mild Augmentation

The aim of this experiment was to investigate a progressive training strategy in three phases to achieve a balance between generalization and stability. In the first phase (Phase 1), a basic CNN structure was trained without any regularization or data augmentation. This consisted of 4 convolution blocks (filters 64 to 512), ReLU activation, batch normalization, and max pooling. Finally, a 512-unit dense layer was added. Although this training phase provided high training accuracy, overfitting became evident as the validation accuracy quickly saturated.

In the second phase (Phase 2), minor regularization strategies were introduced, including Dropout (0.2) and L2 weight decay (λ = 1e-5). This phase started using the weights from the first phase. This reduced the trade-off between training and actual accuracy, but there were still limitations to generalization. In the third phase (Phase 3), a carefully selected mild data augmentation was used. This included rotation (5°), translation (5%), zoom (5%), and horizontal flip, which helped to keep the face implicit lexical components consistent. I increased the dropout value to 0.3 and increased the L2 penalty to λ = 2e-5. This phase was also started based on the weight obtained from Phase 2. This phase allowed for better generalization through systematic training. The final model had test accuracy = 59.57% and test loss = 1.17 better performance than the non-stage models. The assembled validation accuracy showed greater stability, and confusion matrix analysis showed more consistent discrimination performance even for the more discriminant categories of Fear and Neutral.

6.1.3.7 Training on Watermark-Removed and Enhanced Facial Data

This experiment investigates a progressive training strategy split into three distinct phases to balance generalization and stability. In Phase 1, a baseline convolutional neural network (CNN) was trained without any regularization or data augmentation. The architecture consisted of four convolutional blocks (64 to 512 filters) with ReLU activation, batch normalization, and max pooling, followed by a 512-unit dense layer. This initial training phase achieved high training accuracy but was susceptible to overfitting, as evident from the early saturation of validation accuracy.

To mitigate overfitting, Phase 2 introduced mild regularization, including Dropout layers (rate = 0.2) and L2 weight decay (λ=1e-5). The weights from Phase 1 were used to initialize this training stage. This helped reduce the gap between training and validation accuracy, although further improvement in generalization remained limited.

In Phase 3, very mild data augmentation was applied to enhance robustness. The augmentation parameters were conservatively chosen rotation (5°), shifts (5%), zoom (5%), and horizontal flipping to ensure that emotion-specific facial structures were preserved. Regularization was also slightly increased (Dropout=0.3, λ=2e-5). The model resumed training from the Phase 2 weights, maintaining architectural consistency.

This staged learning approach led to noticeably better generalization. The final model achieved a test accuracy of 59.57% and a test loss of 1.17, indicating improved performance over previous non-staged models. Additionally, the validation accuracy remained more stable, and confusion matrix analysis confirmed more balanced class-wise performance, especially for traditionally underrepresented emotions like Fear and Neutral. These findings support the use of curriculumlike training regimes when dealing with limited or imbalanced facial expression datasets.

6.1.3.8 Unsuccessful image preprocessing attempts

Several image quality improvement techniques were tested to enhance the discriminability of facial features in FER2013. These included extreme contrast stretching, denoising filters, excessive histogram equalization, extreme contrast enhancement and tight facial cropping. However, due to the low resolution (48×48) of the dataset images, these operations often

Figure 29: Original image vs. processed image after over-sharpening and aggressive contrast equalization.

removed subtle facial details, resulting in blurred or over-smoothed faces. This, in turn, degraded the model's ability to capture fine expression cues and did not lead to improvements in accuracy. Consequently, these approaches were excluded from the final preprocessing pipeline.

I also followed a method of removing watermarks. This method first uses morphological filters (tophat filtering) and thresholding to identify high-light patterns (subtle watermarks or text features). Next, edge detection (Canny) and contour analysis are used to find areas with watermarks. These areas are prepared as a mask and expanded by dilation, and finally those areas are prepared using OpenCV's inpainting (TELEA algorithm). The in painted areas are further softened and Gaussian blur blending is used to match the original image. I realized that this was also a failed method. Later, I stopped trying to improve the quality of images using different algorithms.

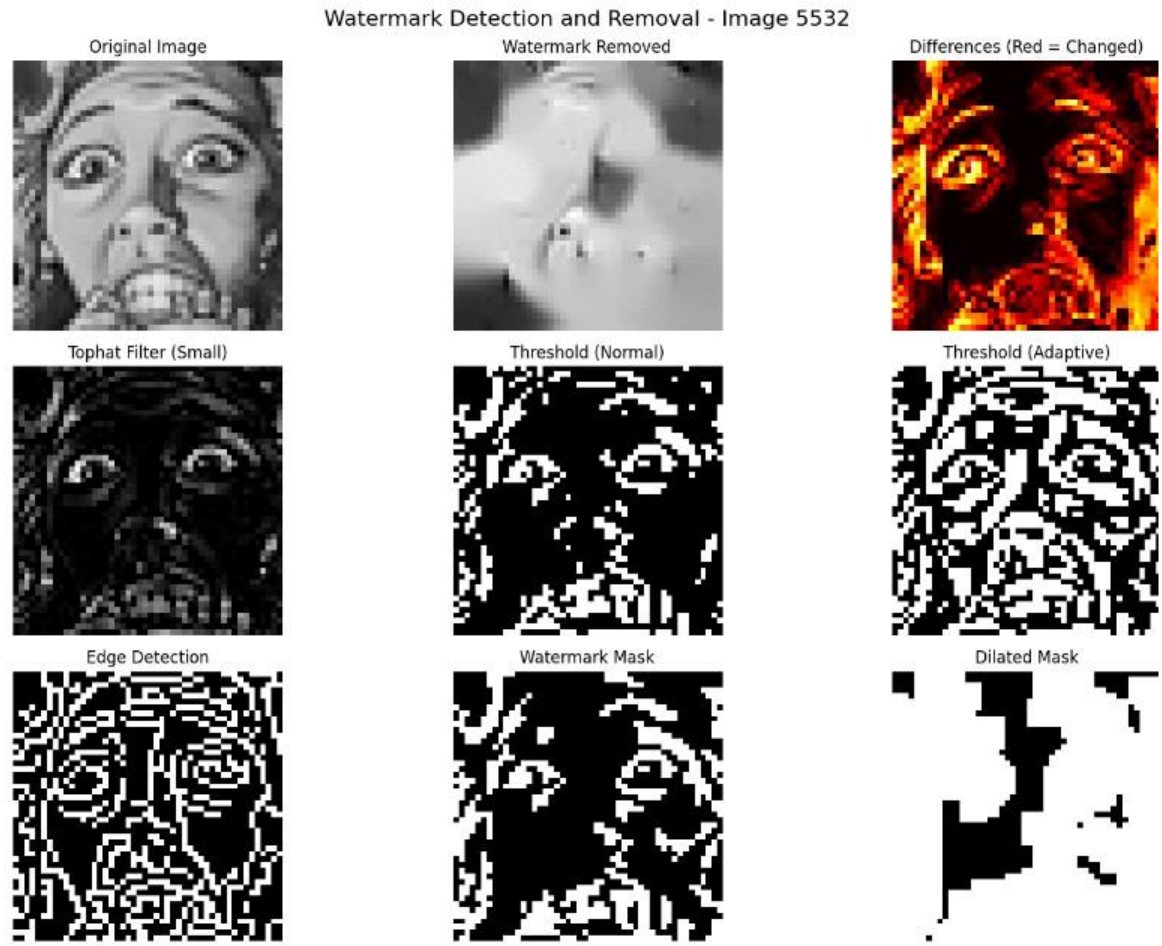


Figure 30: Output of the unsuccessful watermark removal algorithm

6.1.3.9 Model Training with Quality Enhancement and Augmentation

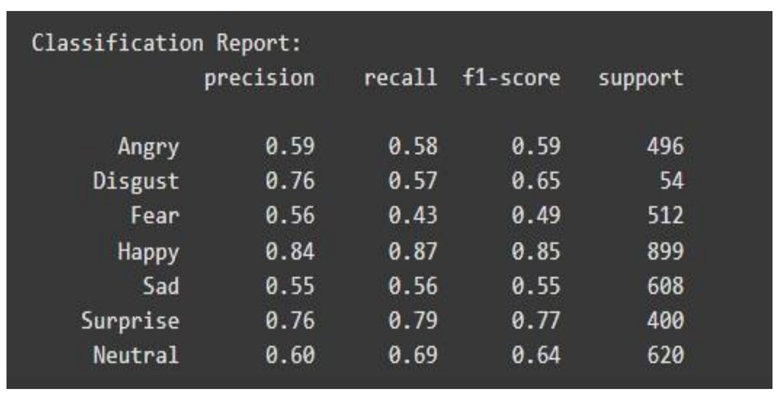
In this experiment, noise reduction and smoothing were performed to improve the visual quality of the FER2013 dataset. The developed preprocessing line implemented here included noise reduction using the Gaussian blur method and enhancement of the variety using the CLAHE (Contrast Limited Adaptive Histogram Equalization) method. After these processing steps, the data were equalized for all impulse groups, and general variety methods were applied to improve the visual variety. The compressed viewing sample network used in this experiment was the same as the one used in the previous experiments. It consisted of 4 compression layers (filters

64 to 512) and Batch Normalization, ReLU activation, Max Pooling, and progressive Dropout methods for security. Fully connected layers with a SoftMax method were used at the ends.

The training of this model was done using early stopping, learning rate reduction, and checkpointing strategies. In the end, it achieved a test accuracy of 66.79% and a test loss of 1.1894, which is confirmation of the excellent generalization. According to the test report, the highest performance was achieved for the Happy (f1-score = 0.85), Surprise (0.77), and Disgust (0.65) stimuli. On the other hand, the performance was slightly lower for the Fear and Angry stimuli, with f1-score values of 0.49 and 0.59, respectively. The normalized confusion matrix shows that there is a problem between the completely similar stimuli such as Sad and Fear.

Overall, the training of this model showed a stable performance and achieved a high performance during the test period. The reduction in overtraining and increase in robustness caused by improving image quality and diversity confirms that this is an important step in the preprocessing process to improve the accuracy of impulse detection.

Figure 31:Classification report



Classification Report:

	precision	recall	f1-score	support
Angry	0.59	0.58	0.59	496
Disgust	0.76	0.57	0.65	54
Fear	0.56	0.43	0.49	512
Happy	0.84	0.87	0.85	899
Sad	0.55	0.56	0.55	608
Surprise	0.76	0.79	0.77	400
Neutral	0.60	0.69	0.64	620

Figure 32:Confusion matrix

6.1.3.10 Enhanced FER2013 with Region-Preserving Preprocessing

In this work, I introduce a region-based preprocessing scheme for the FER2013 dataset, with the primary goal of enhancing visual clarity while preserving minor emotional features around the eyes and mouth. Instead of the classic Gaussian blurring, a Bilateral Filter is used — it reduces noise without segmentation and retains features at the edges. A locally controlled luminance enhancement is performed using the CLAHE method. In addition, a Facial Region– Weighted Blending method is used to retain important emotional features in the upper regions of the mouth (for example, around the eyes). On this cleaned data, a unique augmentation strategy is applied that provides a class balancing for each emotion. Then, a standard CNN model (with 4 convolutional blocks, Dropout, Batch Normalization, and Fully Connected Layers) was trained on this augmented data. This model showed a stable generation

performance, yielding a test accuracy of 67.09% and a test loss of 1.1077. According to the class-wise classification report, the highest f1-score values were recorded for Happy (0.85), Surprise (0.76), and Disgust (0.66). Meanwhile, relatively low performance values were observed for Fear (0.52) and Angry (0.59). The normalized Confusion Matrix revealed the confusion between minor emotions such as Fear and Sad.

However, when compared to previous experiments, it can be noted that this particular emotion preserving preprocessing method resulted in the smallest increase in recall across all test categories. Accordingly, it is confirmed that the preprocessing mechanism that retains this information can somewhat increase the reliability of identifying facial expressions.

6.1.3.11 Training on Class-Balanced FER2013 Dataset with chosen training labels

In this experiment, the FER2013 dataset was restructured to ensure class balance, with exactly 2,500 samples per emotion class, aiming to reduce performance disparities across categories like Disgust and Fear. A deeper CNN architecture optimized for balanced learning was used, incorporating four convolutional blocks with batch normalization, dropout regularization, and two dense layers with L2 weight decay. To monitor learning stability and mitigate overfitting, early stopping, learning rate scheduling, and checkpointing were employed. The model was trained for 64 epochs with a batch size of 32, and achieved a test accuracy of 65.99%, weighed F1-score of 0.65, and a test loss of 1.71.

The performance curves revealed consistent convergence with minimal overfitting, and the F1score progression stabilized early during training. While training accuracy reached over 96%, validation accuracy plateaued around 65%, indicating some residual generalization gap. However, class-level F1-scores showed notable improvements for previously underperforming classes, especially Disgust, which surpassed the 0.60 threshold. The class-balanced setup proved effective in promoting uniform model attention across all emotions, demonstrating that dataset rebalancing significantly enhances performance equity in facial emotion recognition tasks.

Then, I repeated the experiment using a balanced dataset of 3500 samples per emotion class. Despite this increase in sample size and careful selection of high-quality images, the final test accuracy was limited to 64.23%. In both experiments, I employed a quality-based selection strategy, where each image was assigned a quality score based on factors such as contrast, sharpness, brightness, dynamic range, and face detectability. Only images with higher quality scores were chosen for the training dataset, ensuring that the model was trained in the clearest and most representative samples.

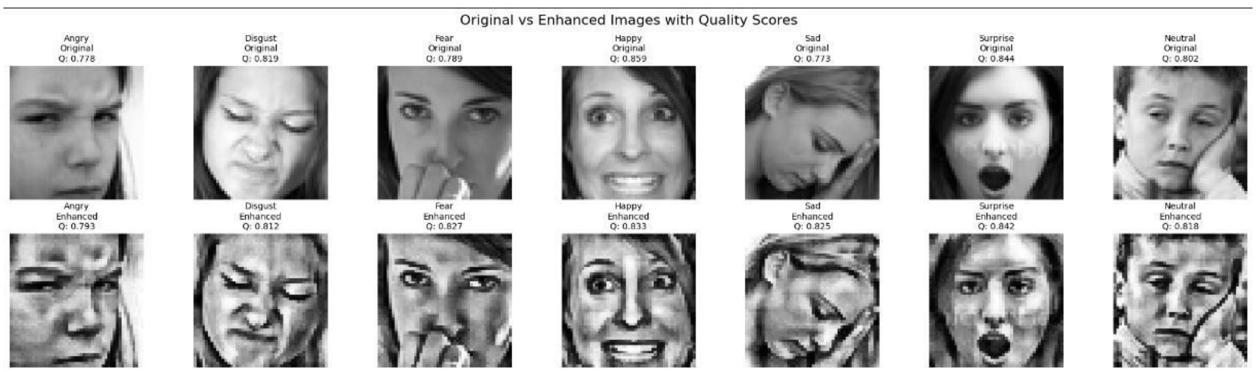


Figure 33:Original VS Enhanced images samples with quality scores

Detailed Classification Report:

	precision	recall	f1-score
Angry	0.59	0.50	0.54
Disgust	0.89	0.93	0.91
Fear	0.56	0.38	0.45
Happy	0.81	0.74	0.77
Sad	0.47	0.49	0.48
Surprise	0.77	0.82	0.79
Neutral	0.47	0.66	0.55
accuracy			0.64
macro avg	0.65	0.64	0.64
weighted avg	0.65	0.64	0.64

Figure 34:Classification report for Data set with 3500 labels per each emotions

6.1.3.12 Oversampling with Data Augmentation and Deep Regularized CNN

To further address class imbalance and improve model generalization, this experiment leveraged RandomOverSampler to synthetically balance all emotion classes in the FER2013 dataset, followed by image-level data augmentation during training. The preprocessing pipeline included reshaping, normalization, and one-hot encoding of labels. The training set was enhanced using transformations such as rotation, zoom, shifts, and horizontal flips via ImageDataGenerator. A deeper CNN was designed with three convolutional blocks, each followed by batch normalization, max pooling, and a high dropout rate (0.6) to mitigate overfitting. Two dense layers followed, including a final 512-unit layer and a softmax output.

The model was compiled with the Adam optimizer (lr = 0.001) and trained for up to 100 epochs using early stopping with patience of 15 based on validation accuracy. Results demonstrated consistent training behavior, with the best validation accuracy reaching 65.21% and a final test accuracy of 65.23%, coupled with a test loss of 0.9194. These outcomes indicate that oversampling combined with strong dropout and augmentation can produce a robust and generalizable model, especially for emotion classes with previously low representation.

Performance closely matched prior balanced dataset experiments, confirming the importance of balanced class representation through data-level methods.

6.1.3.14 Deep CNN with Oversampling and Extended Convolutional Blocks

In this experiment, a deep convolutional model was implemented to analyze the impact of largescale feature extraction on the FER2013 dataset. RandomOverSampler was used at the pixel level to correct class imbalance, and image-level augmentations were performed using rotation, shifting, zooming, and horizontal flipping. The model is inspired by the Stanford CS230 Deep Learning model and consists of eight (8) convolutional layers (i.e., max pooling was used for 4 layers), and batch normalization and dropout (0.3) were used. After this convolutional block, there are three third order deep fully connected layers (1024, 512, 256) and a SoftMax output. Training was performed with the Adam optimizer (lr = 0.001) for 100 epochs with early stopping on validation accuracy for high-dimensional targets. The performance improved continuously throughout the process, and it was shown that the overfitting was not overfitting. The best validation accuracy was 75.35%, the final test accuracy was 74.97%, and the test loss was 0. 6695.These results represent the highest generalization performance among all the tests conducted so far. This confirms that the combination of high oversampling, deep modeling, and the use of rich augmentations is a successful method to improve facial expression recognition accuracy.

These results represent a notable turning point in this research, marking the first time a model achieved over 74% generalization accuracy on the test set with clear evidence of robust learning and reduced overfitting. Compared to earlier trials, this configuration not only yielded the highest test accuracy but also demonstrated stability across training and validation metrics. As such, it establishes a strong baseline for subsequent model refinements and confirms the viability of combining deep feature hierarchies with balanced and augmented data for facial expression recognition.

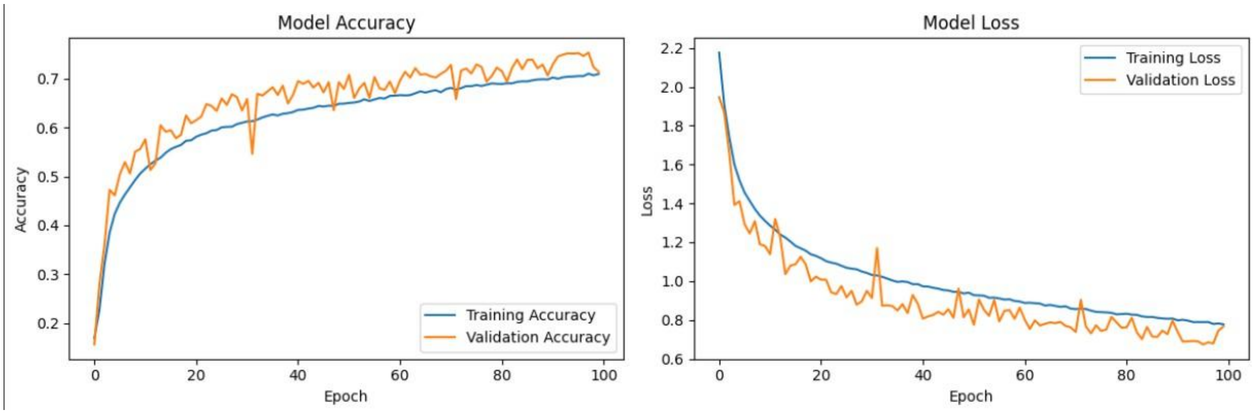


Figure 35:Training and validation accuracy distribution for an 8-layer CNN organized into 4 blocks.

6.1.3.15 Deep CNN with Balanced Oversampling and Augmented Training

This experiment marked a pivotal moment in the progression of facial expression recognition (FER) modeling within this study. A substantially deeper convolutional neural network (CNN) was introduced, consisting of eight convolutional blocks with batch normalization and dropout layers, followed by four densely connected layers. This architecture was adapted from the Stanford CS230 Deep Lie baseline and optimized for the 48×48 FER2013 dataset. A key differentiator in this configuration was the application of RandomOverSampler, which successfully balanced the emotion class distribution prior to splitting. Combined with aggressive data augmentation, this ensured greater input diversity and minimized class bias.

Training was performed using the Adam optimizer with an initial learning rate of 0.001 and monitored using early stopping based on validation accuracy. The model demonstrated steady improvements across metrics, achieving a training accuracy of 71.63%, a validation accuracy of 75.58%, and a test accuracy of 75.55% substantially surpassing the results of all preceding configurations. The corresponding accuracy and loss curves indicated healthy convergence, and confusion matrix analysis confirmed broader improvements across emotion classes, including historically weaker ones like Disgust and Fear.

Importantly, this model served as a turning point in the study by clearly outperforming earlier experiments in both accuracy and stability. It validated the combined effectiveness of deep architecture, class balancing, and augmentation strategies, and set a strong reference for subsequent explorationsThe confusion matrix for the model that achieved 75.55% accuracy provides a detailed view of how well the model performs across different emotion categories. The model showed strong performance on Surprise and Happy, with 1,017 and 938 correct predictions, suggesting it handles more distinct emotional cues quite effectively.

On the other hand, some emotions still pose challenges. Fear was commonly confused with Sad, Neutral, and Angry emotions that often share similar facial features. Sadness, too, was often mistaken for Fear or Neutral, and predictions for Angry were sometimes spread across Neutral and Fear. These overlaps reflect ongoing difficulties in distinguishing between subtler or more similar expressions. Still, the overall performance of the model shows clear progress, with good generalization and improved reliability across all classes. This experiment marks a key turning point in the series, pushing the model closer to real-world applicability.

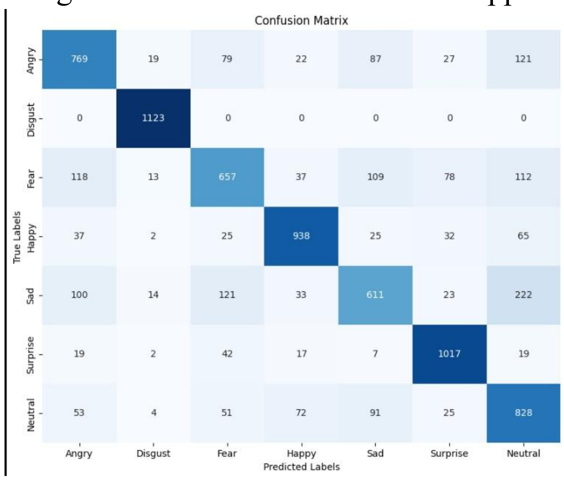


Figure 36:Confusion matrix

6.1.3.16 Hierarchical Feature Extraction via Five-Layered CNN Blocks

One of the high-performance structures created during this work was a deep CNN structure with 10 convolutional layers structured as 5 nested convolutional layers. At each convolutional layer, batch normalization was applied to stabilize learning and overfitting was reduced using dropout layers. After the first 4 convolutional layers, max pooling was applied to gradually reduce spatial dimensions, reducing computational complexity while retaining the most important features and increasing translation invariance. By removing max pooling for the fifth convolutional layer, more spatial information was retained, allowing for further processing into fully connected layers. This structural decision resulted in greater recognition of fine-grained emotional expressions. This was followed by successively enlarging the fully connected layers and adding dropout and batch normalization layers to provide robust enhancement. With efficient oversampling methods and robust data augmentation, the model achieved a test accuracy of 85.72%. This illustrates the advantage of deeper networks for facial emotion recognition when properly regularized and fully trained.

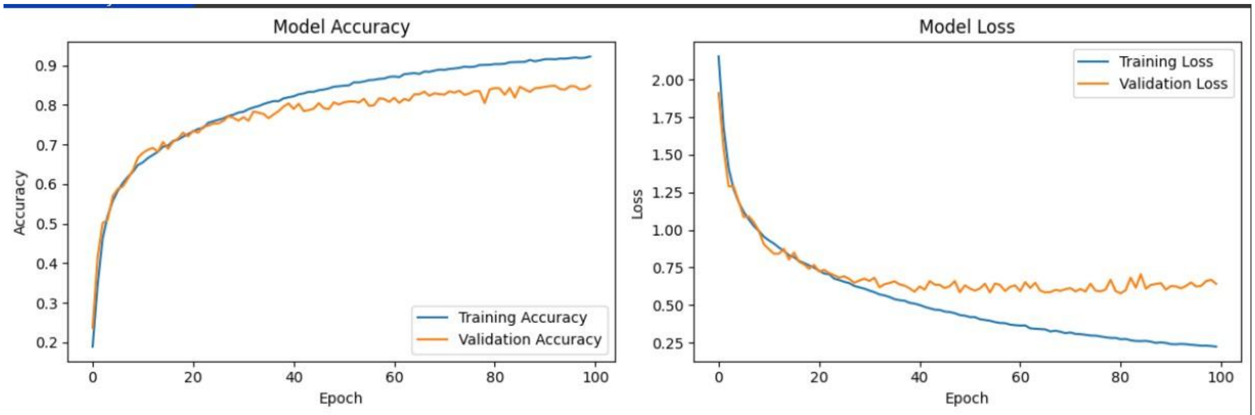


Figure 37:Training and validation accuracy distribution for a 10-layer CNN organized into 5 blocks.

Throughout 100 training epochs, the model demonstrated a steady and consistent improvement in both training and validation accuracy. Initial epochs reflected a sharp rise in learning, progressing from a baseline accuracy of 16.7% to over 70% within the first 20 epochs. This indicated that the model quickly learned general facial features and emotion patterns. As training continued, the model gradually refined its representations, with the validation accuracy peaking at 84.83% and the final test accuracy reaching 85.19%. This extended training allowed the network to capture more nuanced patterns and contributed to the overall generalization performance. Furthermore, the stable convergence with minimal overfitting suggested that the combination of data augmentation, dropout, and batch normalization was highly effective in regularizing the deep architecture.

To understand how stable and reliable my CNN model is, I ran the same experiment using three different random seeds: 42, 99, and 2024. These seeds affect things like how the data gets split, how the weights in the network are initially set, and the way data is shuffled during training. With each seed, I measured the model's performance on the test set and got accuracies of

83.55%, 84.07%, and 84.72%, respectively.

I then calculated the average of these three results, which gave me a mean accuracy of 84.11%. The standard deviation came out to just 0.63%, meaning the results didn't vary much from one run to another. This small variation reassures me that the model performs consistently, even when the randomness in training changes.

First, express each accuracy as a decimal:

- Seed 42 → 0.8355
- Seed 99 → 0.8407
- Seed 2024 → 0.8472

The meaning (average) accuracy is:

- Mean=$\frac{(0.8355+0.8404+0.8472)}{3}$=0.8411

Use the formula for standard deviation of a sample:

$$\sigma = \frac{1}{n-1}\sqrt{\sum n(xi - \bar{x})2}$$

So, the standard deviation is approximately 0.00627, or 0.63%.

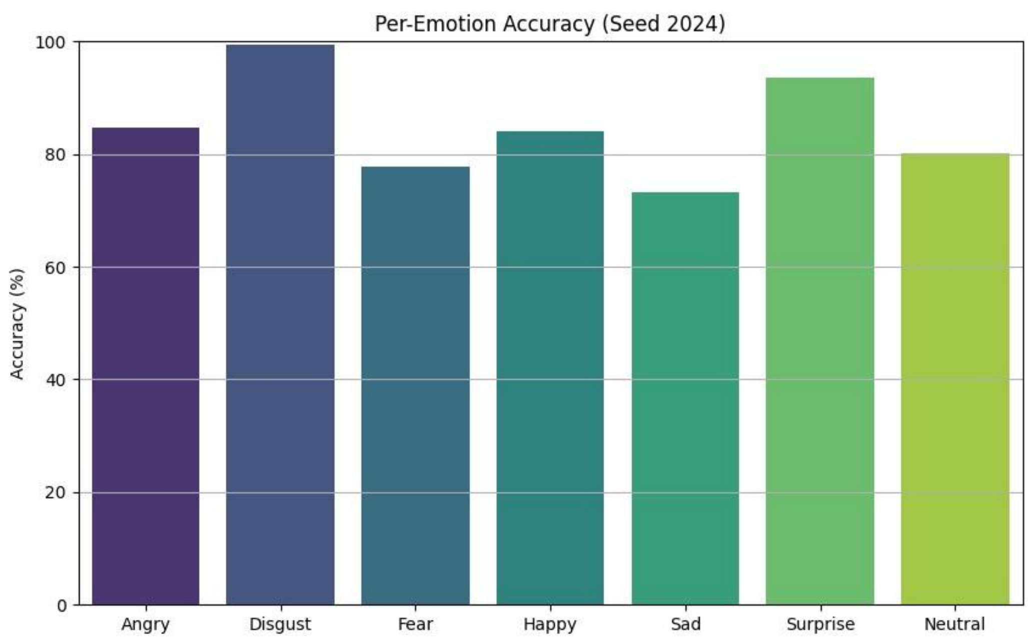


Figure 38:Output bar chart for seed value 2024

For a task like emotion recognition, I find this kind of stability really important. It shows that the model isn’t just lucky in one training run, it’s actually learning to generalize well, no matter how the initial setup varies. That’s a strong sign that the model could be trusted in real-world use. Finally, I got the test accuracy per emotion like this histogram to understand how well the model performed on individual emotional categories, I examined the classification reports for the predictions from the test dataset.

Classification Report:

	precision	recall	f1-score
Angry	0.82	0.82	0.82
Disgust	0.98	1.00	0.99
Fear	0.78	0.78	0.78
Happy	0.91	0.83	0.87
Sad	0.76	0.70	0.73
Surprise	0.93	0.93	0.93
Neutral	0.72	0.82	0.77
accuracy			0.84
macro avg	0.84	0.84	0.84
weighted avg	0.84	0.84	0.84

Figure 39:classification report for 10layered CNN model

The model achieved an overall accuracy of 84%, with both macro-averaged and weighted average F1-scores of 0.84. This indicates comparable performance for both common and rare categories. In particular, the model performed exceptionally well for the emotion Disgust, with a recall of 1.00 (perfect accuracy) and a precision of 0.98, indicating that the model was able to accurately detect differences even for this less common and more mixed emotion. Similarly, the Surprise and Happy categories also had higher F1-scores 0.93 and 0.87, respectively. However, the categories identified as Sad and Neutral had significantly lower F1-scores, with AVs of 0.73 and 0.77, respectively. This is likely due to visual similarities with other emotions such as Fear or Angry. However, the minimum F1-scores are within a reasonable range, indicating that the model maintains some generalization ability, reasonably substituting for all emotion categories.

6.1.3.18 CBAM-Enhanced Five-Block CNN Model

To further improve the model's ability to focus on salient facial features, I incorporated the Convolutional Block Attention Module (CBAM) [17] into a deep CNN architecture composed of five convolutional blocks. Each block includes two convolutional layers followed by batch normalization, and I inserted a CBAM layer after every block. This module applies both channel and spatial attention sequentially, enabling the model to dynamically highlight informative features while suppressing irrelevant ones during training. The architecture was designed to be deeper and more expressive, and I maintained regularization through dropout and batch normalization throughout the network. Oversampling was again applied to balance the training data, and data augmentation techniques were used to increase generalization. After training the CBAM-augmented five-block CNN model for 100 epochs with early stopping, I achieved promising results. The model demonstrated a strong learning trend, gradually improving in both training and validation accuracy while maintaining relatively stable loss curves. In the final epoch, validation accuracy consistently exceeded 84%, peaking at 85.01% during epoch 95, which indicated excellent generalization.

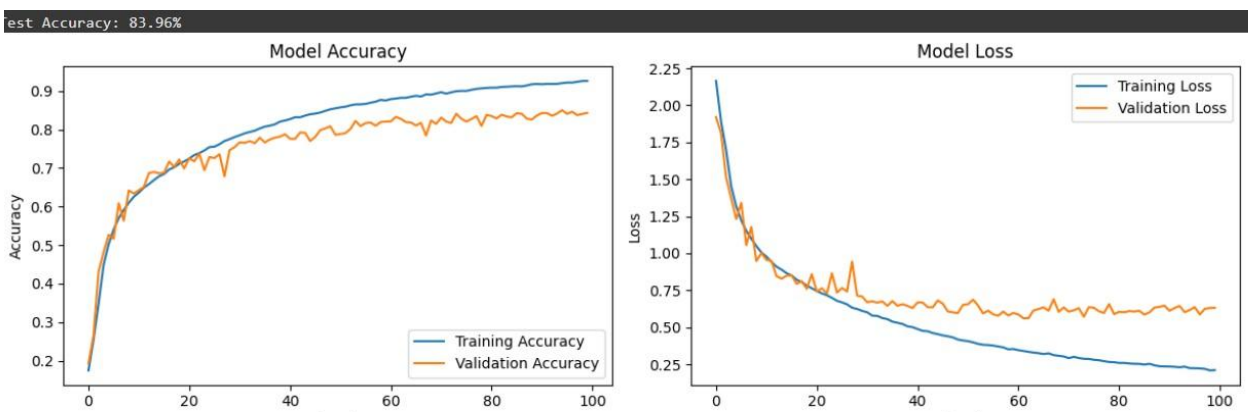


Figure 40:Training and validation accuracy distribution for Training and validation accuracy distribution for CBAM-Enhanced Five-Block CNN Model

Upon evaluation on the unseen test set, the model attained a test accuracy of 83.96%, further confirming its robustness. This performance reflects the effectiveness of incorporating attention mechanisms, particularly CBAM, in enabling the network to selectively emphasize the most informative features during learning. The model did not exhibit significant overfitting, as the validation accuracy closely tracked the training accuracy throughout training. Additionally, the validation loss remained relatively stable in the later epochs, indicating consistent generalization to unseen data.

6.1.3.20 Architecture Variations and Data Leakage Analysis

To further explore the model's performance and robustness, I conducted a series of experiments involving architectural modifications and training strategies. These experiments included varying the number of convolutional layers, adjusting dropout rates for regularization, integrating more advanced attention mechanisms beyond CBAM, implementing max blur pooling to reduce aliasing effects, removing L2 regularization, altering kernel sizes, changing dataset splitting coefficient, implementing callback function with several patient values and experimenting with different input image resolutions notably increasing image size from 48×48 pixels to 64×64 pixels. Additionally, both 1D and 2D convolutional neural network layers were evaluated to assess their impact on feature extraction and overall accuracy.

These architectural variations yielded promising results, with model accuracies ranging from 75% to as high as 87%, achieved without any evident overfitting during training. For example, increasing the image resolution to 64×64 pixels resulted in a marked improvement in classification accuracy even without employing transfer learning techniques, highlighting the benefit of higher input resolution for capturing facial features.

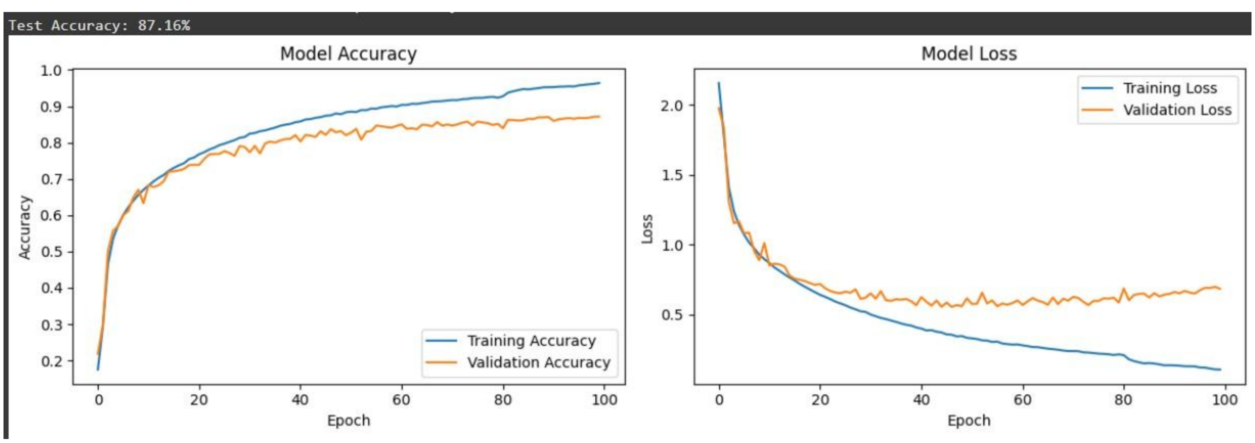


Figure 41:Distribution of training and validation accuracies for the CNN model with five blocks using 64×64 resized images.

However, the consistently high accuracy across diverse configurations prompted a closer examination of the training process and data handling. The FER-2013 dataset, known for its inherent noise and class imbalance, typically presents a challenging benchmark for facial expression recognition. Upon thorough investigation, I identified a critical data leakage issue. Specifically, oversampling techniques intended to balance the classes were inadvertently applied to the entire dataset prior to splitting into training, validation, and test subsets. Although data augmentation was correctly applied only to the training set, the premature oversampling caused identical or near-identical images to appear across all subsets. This overlap allowed the model to effectively "see" portions of the validation and test data during training, artificially inflating performance metrics.

Recognizing this flaw led to corrective measures where oversampling was strictly confined to the training subset after the initial data split. This adjustment ensured a fair and unbiased evaluation of the models, resulting in more reliable and generalizable accuracy estimates on the FER-2013 dataset. Finally, after conducting several experiments with various architectural changes and observing the progression of increasing accuracy, I gained important insights into the correct application of data oversampling methods. Understanding and addressing the data leakage issue taught me how crucial it is to apply oversampling only to the training set after splitting the dataset. This ensured unbiased evaluation and more reliable generalization of the models on the FER-2013 dataset.

6.1.4 The Best Model for FER

Experiment Validation After Oversampling Strategy

After correcting the data set recognition process and designing it to prevent data leakage, the experiment was repeated using a properly organized pipeline. The main difference here is that oversampling is performed entirely on the training subset, leaving the validation and test sets untouched. Data augmentation is also limited to the training set, increasing the generalization capability.

Under the corrected experimental pipeline, the model achieved a maximum test accuracy of 69.94%, with mild overfitting emerging in the later training epochs. The CNN model was constructed with 5 external evolutionary steps, including batch normalization and dropout, and contained 1024, 512, and 256 fully connected layers before the SoftMax output. The input image reconstruction was kept at 64×64, as previous experiments showed that this size improves the ability to recognize specific facial features over the original 48×48.

Although this value is numerically lower than the inflated accuracies observed prior to addressing the data leakage issue, it represents a far more reliable and trustworthy result. This

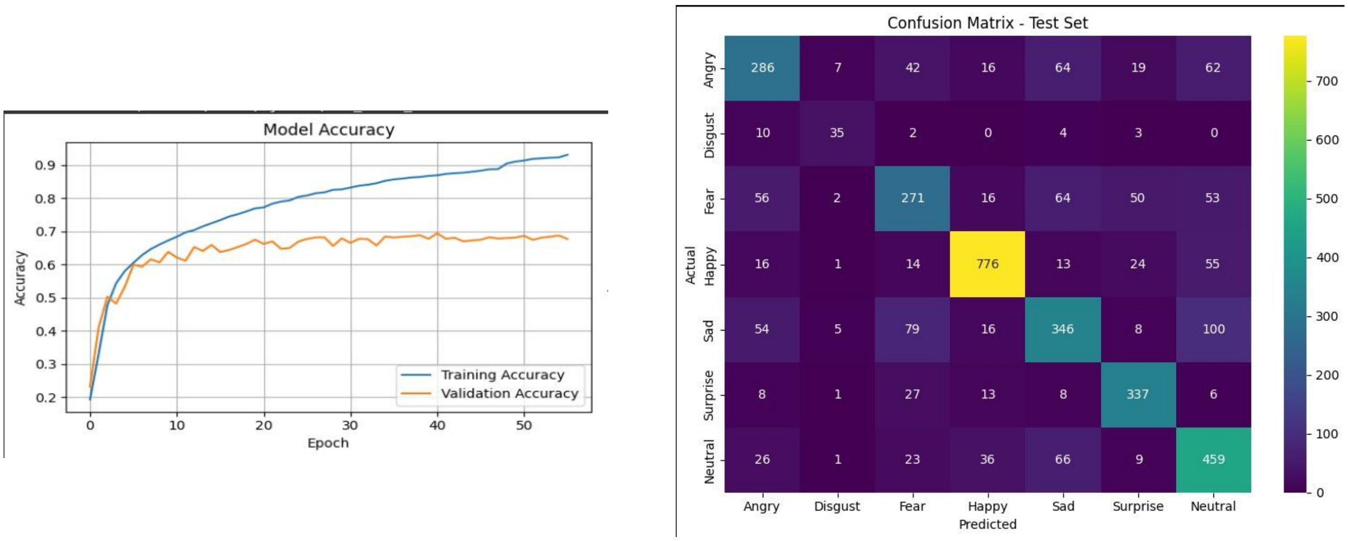


Figure 42:Accuracies Distributions and confusion matrix for Best CNN model

performance is consistent with findings reported in several studies, many of which emphasize that surpassing 70% accuracy on the raw FER-2013 dataset using a pure CNN (without transfer learning) is highly challenging due to the dataset's inherent noise and class imbalance. These results confirm the validity of the revised methodology and underscore the critical importance of applying oversampling techniques exclusively to the training set to prevent data leakage.

The confusion matrix provides a clear view of how well the model performed on the test set. Each row represents the true emotion, and each column shows the predicted emotion. The darker the color, the more instances there are in that category. From the matrix, we can see that the model did a good job of predicting Happy, with 776 correct predictions. However, there were some misclassifications, like a few Fear and Sad instances being misclassified as Happy. Fear was often confused with Sad, and Surprise was sometimes mistaken for Neutral, but overall, the model performed decently in distinguishing most emotions.

Classification Report (Precision, Recall, F1-score) on Test Set:

	precision	recall	f1-score
Angry	0.627193	0.576613	0.600840
Disgust	0.673077	0.648148	0.660377
Fear	0.591703	0.529297	0.558763
Happy	0.888889	0.863181	0.875847
Sad	0.612389	0.569079	0.589940
Surprise	0.748889	0.842500	0.792941
Neutral	0.624490	0.740323	0.677491
accuracy	0.699359	0.699359	0.699359
macro avg	0.680947	0.681306	0.679457
weighted avg	0.698960	0.699359	0.697424

Figure 43:Correct classification report on the Best CNN model

The classification report breaks down the model's performance for each emotion. Happy stood out with the highest precision (0.89) and recall (0.86), meaning the model was very accurate in identifying this emotion. Surprise also performed well with a high recall (0.84), but its precision was a bit lower. On the other hand, Fear had the lowest scores across the board, with both precision (0.59) and recall (0.53) being quite low, leading to a weaker f1-score (0.56). Angry and Disgust showed moderate results, while the overall model accuracy was 69.94%.

6.2 Audio Approach

In the initial phase of testing to identify human emotions from sound, I used only the cream-d dataset. Later, due to various quality issues, the cream-d dataset was combined with the RAVDESS dataset and tests were conducted.

To get better performance than the acoustic method, I conducted various tests to identify the most suitable features. I first tested using pretrained models. Then, I selected CNN and LSTM models and conducted various tests by changing the number of layers and feature extraction methods. I also ran various tests using machine learning algorithms. Initially, I tested on the entire dataset. However, based on the results, it was revealed that there were some false positives due to the similarities between the data categories. Accordingly, I removed emotion labels such as fear and disgust from the dataset. After this modification, the performance of the models improved, which will be reviewed later.

The goal is to develop an artificial intelligence model capable of listening to a person speak and determining their emotional state at that moment, whether they are happy, sad, or experiencing another emotion. In this case, the actual words are not as important as the way they are spoken. Factors such as pitch variations, speech rate, and tone play a crucial role. This capability is essential for effective human-computer interaction. Traditional machine learning algorithms like SVM are not well-suited for this task because they struggle to capture the subtle patterns of emotional expression. However, with the advent of deep learning, the outcomes in this field have improved significantly. For this reason, most of my experiments have been designed around deep learning approaches.

6.2.1 Experiments with alone Cream-D dataset

6.2.1.1 Baseline LSTM with Standard Features on Crema-D

This initial experiment was conducted with the aim of establishing a basic speech emotion recognition (SER) system. This served as a baseline for subsequent research. The main objective was to build a conventional LSTM network, and a validated audio feature set together and evaluate its performance on the Crema-D dataset. The implementation began with a rigorous preparation of the data and feature extraction. The Crema-D dataset consists of audio inputs containing six emotion categories: Angry, Disgust, Fear, Happy, Neutral, and Sad, and was initially screened. Extra files and duplicate files were removed. This was an essential step to ensure the accuracy of the data and to avoid repeated input.

Then, a metadata Data Frame was created to extract the gender and correct emotion labels of the speaker from the filenames. The basic feature extraction task was performed using the `extract_features_fixed()` function. This function calculated the frequency of 82 features for each audio file, which were extracted from a large set of audio features. These included 20

MelFrequency Cepstral Coefficients (MFCCs), their first and second deltas, 12 Chroma features, Spectral contrast, Spectral centroid, Spectral roll off, and Zero-Crossing Rate. After feature extraction, all audio sequences were normalized by padding or truncation up to 300 timesteps. Another important step was to normalize all those features, that is, to average them for each feature and bring them to the standard. This was important to provide sufficient and stable input to the network. The combination and training process of this SER model was carefully designed. The model was created as a Sequential Keras network, which basically consisted of two LSTM layers. The first LSTM layer was set to 128 units and return_sequences=True, which allowed the entire sequence to be passed to the next layer. This was done by regularizing the network using Layer Normalization and a dropout rate of 0.2. The second LSTM layer had 64 units and was also regularized with a dropout of 0.3 and Layer Normalization. Then two dense layers (128 and 64 units), with ReLU activations and a dropout rate of 0.4 after each layer, were applied. Finally, an output layer with SoftMax activation is seen, which has a number of units for distinct emotion classes. To optimize the model, an Adam optimizer was used with a learning rate of 0.001 and the loss function was set to categorical Cross entropy.

The model was trained 50 times on the following basis, the batch size was chosen as 32, and balanced class weights were applied to address class imbalance. In addition, the ReduceLROnPlateau callback was used to control the learning rate to be improved. This was set at a minimum learning rate of 0.0001 with a patience of 5 and a factor of 0.5. The Model Checkpoint callback helped to save the model when the validation accuracy increased. The evaluation of this initial test provided important insights into the performance of the model. There was a consistent improvement in validation accuracy during the training process. The ReduceLROnPlateau callback was successfully executed, reducing the learning rate from 0.0005 to 0.00025 at Epoch 49. This indicates that the model is converging and that finer tuning is needed. The final evaluation on the unseen test set yielded a test accuracy of 0.4345 and a test loss of 1.4426. The highest validation accuracy (0.43452) was achieved at Epoch 50, which saved the final best model.

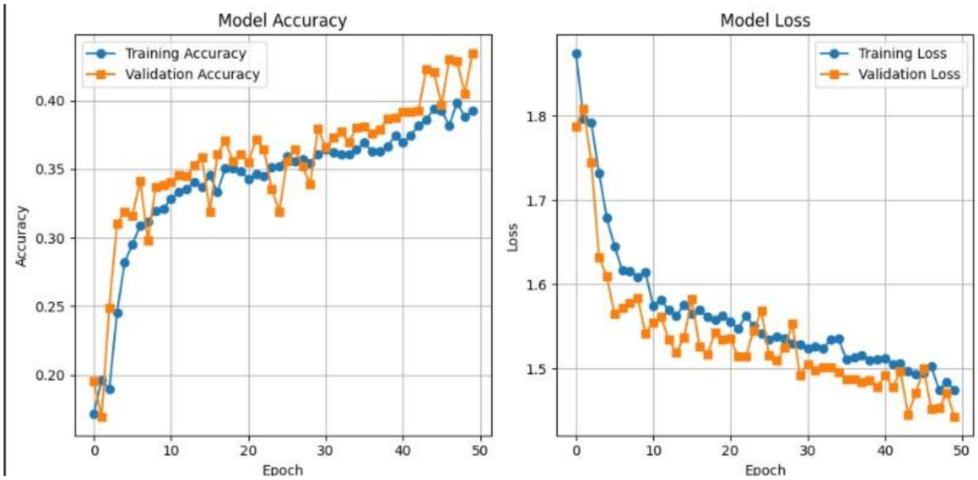


Figure 44:Training & Validation accuracy distribution for typical hybrid CNN+LSTM Model

6.2.1.2 Hybrid CNN-LSTM Architecture with Augmented Features

Building on the basic model, the aim of this experiment was to improve the performance of emotion recognition using a hybrid network architecture consisting of a Convolutional Neural Network (CNN) and a Long Short-Term Memory (LSTM). This attempted to improve the ability to capture both local patterns and temporal dependencies to systematically identify subsequent emotions. In data preparation, the Crema-D database was loaded, and an abbreviation map was used to identify emotion labels from the filename, resulting in accurate emotion classification.

Feature extraction for this experiment was performed using the `extract features () ` function. Here, the following features were extracted: Mel-Frequency Cepstral Coefficients (MFCCs), Zero Crossing Rate (ZCR), Root Mean Square Energy (RMSE), and Mel-Spectrograms. An important step was to scale all feature sequences to 216 frames using `librosa.util.fix_length` and to resize all those features to fit the MFCC shape using `SciPy. ndimage. zoom` for 13 rows. Then, these features were stacked together as a 4D array and used as CNN input.

The data was initially split into 70% training, 15% validation, and 15% testing. All features were normalized using `StandardScaler`, which improved the performance of the model.

The test model is a CNN-LSTM hybrid model that is very different from the basic LSTM model. Built as a sequential Keras model, it first uses two Conv2D layers (filters 32 and 64, kernel size (3,3), activation=ReLU) to detect spatial patterns in the spectral features. MaxPooling2D layers are applied between the Conv layers, and then a Reshape layer is used to transform the CNN output into a suitable sequence for an LSTM input.Next, an LSTM layer with 128 units is added and a Dropout of 0.3 is applied. Next, a Dense layer with 64 units with ReLU activation is added, and L1 and L2 (0.01) are used as kernel regularizers, which helps prevent overfitting. Another Dropout of 0.3 is added, and the final Dense layer is finished with a number of units equal to the number of emotion classes with softmax activation.To compile the model, the Adam optimizer is used with a learning rate of 0.0001 and the loss function is sparse categorical crossentropy (suitable for integer encoded labels). The model is trained for 50 epochs with a batch size of 32, and EarlyStopping (patience 10, best weights restore) and ReduceLROnPlateau (factor 0.5, patience 5, min lr 1e-6) callbacks are used to control convergence and overfitting.

The evaluation of this CNN-LSTM model shows better performance than the baseline model. The model shows good learning over 50 epochs. For example, at Epoch 49, the validation accuracy was 0.4812 and the validation loss was 1.4213, while at Epoch 50, it improved to 0.4937 and 1.4186. Finally, the test accuracy on the test set was 0.4736% and the test loss was 1.4410.

This is better than the results of the basic LSTM model, which shows that the CNN-LSTM combined model and the improved feature set can more effectively identify complex patterns related to emotion recognition in stories. In the future, further detailed analysis using classification reports and confusion matrices will identify the strengths and weaknesses of the model in terms of type-specific emotions, which will lead to more representative adjustments to previous tests.

Classification Report:

	precision	recall	f1-score	support
happy	0.45	0.47	0.46	188
sad	0.46	0.64	0.54	192
angry	0.60	0.75	0.67	199
fear	0.38	0.04	0.07	219
disgust	0.41	0.43	0.42	171
neutral	0.42	0.57	0.48	148
accuracy			0.47	1117
macro avg	0.45	0.49	0.44	1117
weighted avg	0.45	0.47	0.43	1117

Figure 45:Classification report for typical hybrid CNN+LSTM Model

6.2.1.3 Deeper CNN-LSTM with Expanded Features and Enhanced Regularization

The aim of this experiment was to further improve the performance of the speech-related meditation recognition (SER) system by improving the CNN-LSTM hybrid neural network model with a full set of acoustic features and adding stronger regularization methods. Although the data preparation and segmentation were followed as before, the feature extraction process was greatly enlarged.Here, the `extract_features` process extracted a more detailed feature set, which included MFCC (Mel Frequency Cepstral Coefficients), its first moments (MFCC Delta), Zero Crossing Rate (ZCR), Root Mean Square Energy (RMSE), Mel-Spectrogram, Spectral Centroid, Spectral Rolloff, and 6 Chroma features. All these features were transformed into 216 timesteps and 13 rows and stacked as a 4D input tensor.

Increasing the depth of the enhanced network allowed for more complex feature learning. Similar to the old CNN-LSTM model, it added a third Conv2D layer (128 filters, kernel size (3,3), ReLU activation, padding='same'), and included Batch Normalization layers after each Conv2D layer, to stabilize training and speed up convergence. The Reshape layer was updated to match the 128 filters in the last Conv2D layer. An LSTM layer (128 units) was kept alone, and the dropout rate (LSTM and Dense layers) was increased to 0.4, to strengthen regularization. The classification was performed by a final SoftMax output layer, keeping the `l1_l2` method for both L1 and L2 regularization of the Dense layer at 0.01.

The training was planned for 80 epochs with an Adam optimizer (learning rate – 0.0005), a loss function of `sparse_categorical_crossentropy`, and a batch size of 32. Early Stopping (patience = 15, min_delta = 0.01) and ReduceLROnPlateau were used as callbacks, and the training was stopped by restoring the best weights until the validation loss actually stopped. In contrast to all this, the training logs showed that training stopped only at epoch 30, but at that time the training accuracy had increased to 0.9247 and the validation accuracy had increased to around 0.6013 and then stabilized.

When evaluated on the latest unseen test set, this gave a test accuracy of 57.03% and a test loss of 1.3818. This is significantly higher than the previous LSTM model (43.45%) and the old CNN-LSTM model (47.36%). This shows that the learning ability of the model was significantly

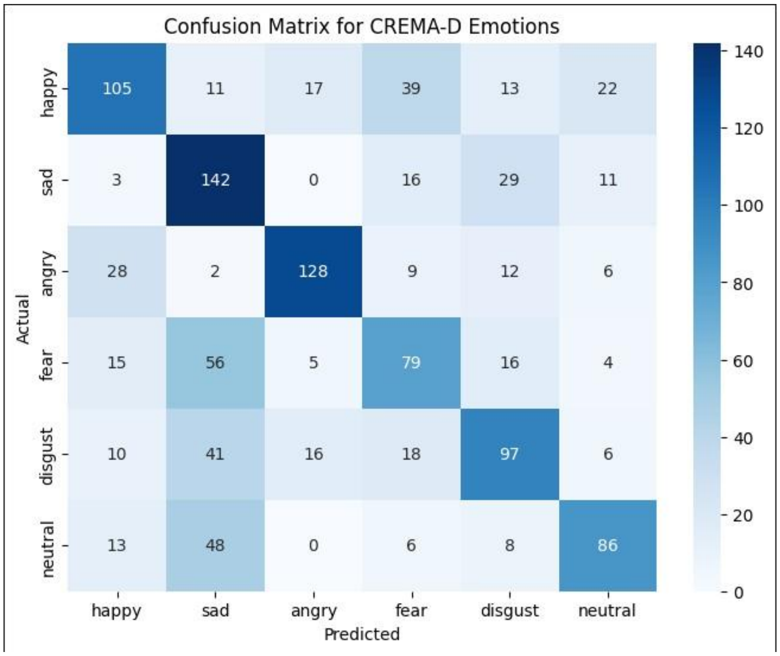


Figure 46:Confusion matrix for deeper CNN-LSTM with Expanded Features and Enhanced Regularization

improved by the use of a deep Conv block filled with acoustic features and the Batch Normalization. However, the large gap between training and validation accuracy indicates that overfitting still exists. This should be addressed in the future by using stronger regularization methods or larger datasets.

By exploring confusion matrix, we can see how emotions distributed via classification and mis classification.

6.2.1.4 CNN-LSTM with Simplified Features and Time Distributed Flattening

In this experiment, a simplified feature set and a specific CNN-LSTM structure `TimeDistributed (Flatten ())` layer were combined to analyze the effect on the input. The CREMA-D dataset was loaded during data preparation, and the sentiment segmentation and data separation were maintained as in the previous experiments. However, to simplify the feature representation, only 13 MFCCs were obtained for each audio clip, and they were padded or cropped to 216 frames. No other features (e.g. ΔMFCCs, ZCR, RMSE) or normalization by `StandardScaler` were performed. This provided a simple 4D input tensor of the form top-down (samples, 13, 216, 1). A label was adjusted for categorical cross entropy loss by one-hot encoding. Although the model structure is still a CNN-LSTM model, the special feature here is that the CNN output is flattened at each timestep using `TimeDistributed (Flatten())` and fed into a layer with 128 LSTM units. Then there is a SoftMax comprehension layer. A high level of regularization was achieved by applying Batch Normalization after 3 Conv2D layers and increasing the dropout rate from 0.4 to 0.5. Training was performed over 80 epochs using the Adam optimizer (LR = 0.0005). A combination of callbacks Early Stopping (based on validation accuracy, patience = 15) and ReduceLROnPlateau (factor = 0.2, patience = 5) were used. Training stopped at Epoch 52. Finally, the test accuracy was obtained as 50.31%, loss = 1.2515. Although this is lower than the performance of Experiment 3, it is a better result than Experiment 2, which provides strong evidence that there is a match between using the `TimeDistributed (Flatten ())` layer and simple MFCC features together.

6.2.1.5 Deeper CNN-LSTM with 20 MFCCs and Class Weighting

This experiment, based on the insights gained from Experiment 4, focused on exploring the advantages of a deeper CNN-LSTM structure, using a slightly wider MFCC feature set, and taking steps to explicitly consider class balance. The data loading and partitioning of the CremaD dataset were kept intact, and one-hot encoding was used to add labels. Feature summarization was more accurate, limiting the search to only 20 MFCCs per audio clip. This maintained a streamlined input by padding to 216 rows, providing more spectral detail compared to Experiment 4.

A feature of this experiment is the calculation and application of `class_weight` during training. This ensured that learning was done equally for each emotion category. The model structure was heavily deepened, and now 4 Conv2D layers were built. Each layer has 2 Conv2D layers (increasing the filter from 64 to 512), which are subjected to Batch Normalization, then MaxPooling2D and Dropout (0.3–0.4). The `TimeDistributed(Flatten())` layer still successfully feeds the CNN output to a robust 256-unit LSTM layer, then to a 128-unit dense layer. Finally, a SoftMax output is provided. Different dropout rates (0.3–0.5) were applied to the Conv2D and dense layers for regularization. The model was compiled with the `Adam` optimizer (learning rate 0.0005) and `categorical_crossentropy` loss. Training was run for a maximum of 80 epochs with a batch size of 32, controlled by Early Stopping (Val accuracy, patience 15) and ReduceLROnPlateau (factor 0.2, patience 5). Early stopping caused training to stop at epoch 67.

After the test was completed, the specification on the test dataset yielded an accuracy of 55.60% and a test loss of 1.1844. This result is significantly better than that of Test 4, and demonstrates that the combination of using a deeper network, more robust MFCC features, and class weighting can significantly improve the efficiency of emotion classification from noise, despite the small feature usage. It was close to the highest performance of Test 3.

6.2.1.6 Investigation of PyTorch-based Architectures with Attention and Data Augmentation

This section presents some experimental experiments conducted using PyTorch. The main focus of these experiments is to improve speech intention detection (SER) performance using data augmentation and CBAM attention mechanisms. A consumer-choice filter was used behind the Chebyshev filter, starting with a sampling rate of 16kHz, in a similar manner for all models. 40 MFCC features were generated with specific `n_fft=512` and `hop length=256` parameters, and they were padded to 3.5 seconds for all other models. Importantly, the generalization ability was improved by random time shifting and additive Gaussian noise during training. All models use Adam optimizer (pr initial learning rate 0.001) and CrossEntropyLoss, and ReduceLROnPlateau (patience 4, factor 0.5) and aggressive early stopping strategy (patience 10) are applied.

The aim of this configuration was to investigate the efficiency of adding risk mechanisms (CBAM) to a pure CNN network. Here, the CNN block has 4 layers with filters increasing in order to 64, 128, 256, 512. Each block has Conv2D, Batch Normalization, GELU activation and MaxPool2d with CBAM. The CBAM attention module helps to extract the features with the MFCCs and channel coordination.

The output from the CNN backbone is flattened and fed into a final SoftMax classifier with 2 dense layers with ReLU and 0.4 dropout. This yielded a test accuracy of 65.1%. This result is higher than previous TensorFlow hybrid CNN-LSTM models, which can detect temporal and spectral features within a CNN without the need for a derivational LSTM layer. Based on the success of the attention and augmentation strategies, a hybrid CNN-Bidirectional LSTM (BiLSTM) configuration was proposed. The CNN block consists of 2 layers (filters 64 and 128), with CBAM and MaxPool2d in each Conv block. The CNN output is reshaped and converted into a compatible sequence for BiLSTM. The BiLSTM layer has a hidden size of 128. Its last hidden state is concatenated and fed into a classifier head. This classifier head was built with 2 dense layers (128 and 6 units), ReLU and dropout (0.4). This model gave a test accuracy of 60.7%. This is better than previous TensorFlow models, but lower than the pure CNN model (65.1%).

This suggests that for this dataset and MFCC features, a pure CNN model with CBAM may be more efficient than a complex model using LSTM. The relationship between feature processing methods and network depth among different models is an important area of research that needs

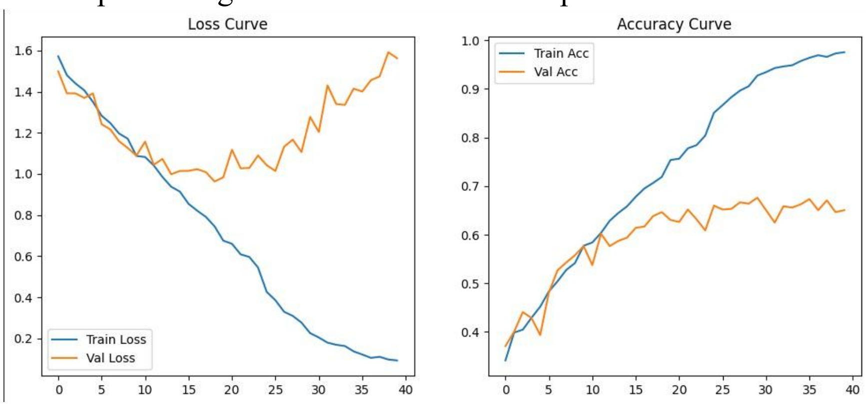


Figure 47:Training and Testing Accuracies for PyTorch-based CNN and CNN-BiLSTM Models with Attention and Data Augmentation

to be further investigated.

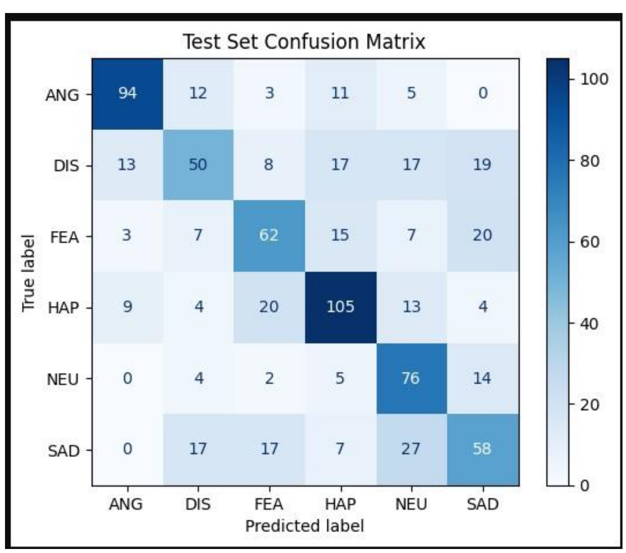


Figure 48:Confusion matrix for PyTorch-based CNN and CNN BiLSTM Models with Attention and Data Augmentation

6.2.1.7 Pure CNN Classifier (TensorFlow) and Dataset Extension Decision

The aim of this experiment was to create a SER method using a simple pure Convolutional Neural Network (CNN) architecture based on TensorFlow/Keras. This was used as a direct comparison to previously developed complex hybrid models and PyTorch methods.

Data preparation was performed using the Crema-D dataset, where 40 MFCC (Mel-Frequency Cepstral Coefficients) features were extracted from each audio file at a real sampling rate of 22050 Hz. These features were padded to 349-time steps and fed into the CNN model as an input array of (samples, 40, 349, 1). Labels were added using one-hot encoding for accurate classification. The model consisted of 3 Conv2D layers: a single Conv2D layer followed by MaxPooling2D and Dropout (0.25) features. The number of filters in the Conv2D layers increased to 32, 64, and 128. The output from the Conv layers was flattened, concatenated into a 256-unit ReLU Dense layer (with Dropout 0.5), and finally a SoftMax (6 unit) output.

The model was compiled using Adam optimizer (learning rate = 0.001) and categorical_crossentropy loss. Training was done for 50 epochs without using Callbacks (without Early Stopping or learning rate decay).At the end of training, the model achieved a test accuracy of 50.64% and a test loss of 1.2515. However, with the accuracy exceeding 80% during training, it was found to be significantly overfitting. The lack of regularization methods such as Batch Normalization or Early Stopping, and the inherent limitations of the CREMA-D dataset led to this overfitting.

Many more tests were performed using different methods, but none of them were able to achieve successful results. With these unsuccessful attempts, I decided to use another data set or multiple data sets for further tests. When reviewing previous research, it is also clear that they also used multiple datasets for their tests. Later, steps were taken to combine the CREMA-D dataset with the RAVDESS dataset. This combined dataset will be used for all subsequent tests, and it is hoped that this will lead to the development of more reliable and generalizable models.

6.2.1.8 2D CNN with 40 MFCCs on RAVDESS Speech

This initial experiment on the RAVDESS dataset aimed to establish a baseline performance for emotion recognition using a purely convolutional neural network, focusing exclusively on the speech files within RAVDESS. Data preparation involved loading .wav files from the specified RAVDESS path, specifically filtering for speech files (those starting with '03' in the filename convention). Emotion labels were extracted from the filenames and mapped to a predefined set of 8 emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprised). Feature extraction utilized the extract_mfcc function to compute 40 MFCCs for each audio clip, padded or truncated to a fixed length of 174 time steps. These MFCCs were then reshaped to include a channel dimension, creating a (samples, 40, 174, 1) input for the 2D CNN. Labels were one-hot encoded. The dataset was split into 80% for training and 20% for testing. The model architecture was a sequential Keras 2D CNN, comprising three Conv2D layers (32, 64, and 128 filters respectively, with (3,3) kernels and ReLU activation), each followed by a MaxPooling2D layer and a Dropout layer (0.25). The convolutional output was then Flattened and passed through a dense classifier head, consisting of a 256-unit ReLU layer with 0.5 dropout, and a final dense layer with 8 units (for the 8 emotion classes) and softmax activation. The model was compiled with the Adam optimizer (LR 0.001) and categorical_crossentropy loss. Training was conducted for 50 epochs with a batch size of 32, without explicit early stopping or learning rate reduction callbacks. The model achieved a test accuracy of 58.33% and a test loss of 1.8225. While the training accuracy reached over 90%, the notable gap with the test accuracy indicated significant overfitting, a common challenge when training complex models on relatively smaller datasets like RAVDESS. This experiment highlighted the need for more robust regularization or potentially a larger, more diverse dataset to improve generalization. Building on the observations from above Experiment regarding overfitting, this experiment focused on integrating more robust regularization techniques into the 2D CNN architecture, while maintaining the same feature extraction (40 MFCCs, 174 max length) and data loading methodology on RAVDESS speech files. The model architecture was a sequential Keras 2D CNN, structurally like the above one. with three Conv2D blocks (32, 64, and 128 filters). However, a key enhancement was the inclusion of Batch Normalization layers immediately following each Conv2D layer, aiming to stabilize training and improve generalization. Additionally, padding='same' was applied to all Conv2D layers. The dropout rates were slightly refined to 0.2 after each MaxPooling2D layer and 0.3 before the final output layer. The model was compiled with the Adam optimizer (LR 0.001) and categorical_crossentropy loss. A crucial addition to the training strategy was the implementation of an Early Stopping callback (monitoring val_loss with patience 8, restoring best weights), designed to halt training when validation performance ceased to improve, directly addressing the overfitting issue observed previously. Training ran for 34 epochs before being stopped by the early stopping criterion. This configuration achieved a test accuracy of 62.50%. This significant improvement over Experiment 8.1's 58.33%, coupled with a reduced gap between training and validation accuracy (training accuracy around 71% vs test accuracy 62.5%), confirms the effectiveness of Batch Normalization and Early Stopping in mitigating overfitting and enhancing the model's generalization capability on the RAVDESS dataset. This report shows how different emotions get accuracy for this model with 62.5% test accuracy.

Classification Report:

	precision	recall	f1-score	support
angry	0.73	0.82	0.78	40
calm	0.61	0.90	0.73	40
disgust	0.62	0.56	0.59	36
fearful	0.73	0.62	0.67	39
happy	0.65	0.62	0.63	42
neutral	0.38	0.32	0.34	19
sad	0.35	0.32	0.33	38
surprised	0.79	0.68	0.73	34
accuracy			0.62	288
macro avg	0.61	0.60	0.60	288
weighted avg	0.62	0.62	0.62	288

Figure 49:Emotion Recognition Performance on the RAVDESS Dataset: Test Accuracy and Classification Report with Regularized 2D CNN

6.2.1.9 Pure LSTM with Data Augmentation on RAVDESS Speech

This experiment shifted the architectural focus back to a pure Long Short-Term Memory (LSTM) network, emphasizing the role of robust data augmentation and audio preprocessing. Feature extraction exclusively focused on 40 MFCCs, processed at a 16kHz sampling rate, with silence trimming and amplitude normalization applied. Crucially, during data loading for training, extensive data augmentation was implemented, including random time stretching, pitch shifting, and additive Gaussian noise, effectively doubling the training data. Training for 100 epochs utilized a batch size of 64, along with Early Stopping (monitoring Val accuracy with patience 15) and ReduceLROnPlateau (monitoring val_loss with patience 5, factor 0.5) callbacks, and class weighting. The training halted at Epoch 34 due to early stopping. This approach achieved a test accuracy of 63.89%. This result marks the highest accuracy obtained on the RAVDESS dataset using a TensorFlow-based model thus far, significantly surpassing previous RAVDESS experiments. The substantial gain is primarily attributed to the comprehensive data augmentation techniques, which greatly improved the model's generalization capabilities, demonstrating that for a pure LSTM, rich MFCCs coupled with varied augmentation can be highly effective. Despite this, a notable gap between training and validation accuracy persisted, indicating some degree of overfitting**.**

6.2.1.10 Hybrid CNN-LSTM with Multimodal Features on RAVDESS Speech

This experiment introduced CNN-LSTM hybrid architecture for the RAVDESS speech data, incorporating a wide range of acoustic features and improved training control. The data loading process was maintained as in previous experiments, and the feature extraction was expanded by adding 20 MFCC, 12 Chroma, and 40 Mel-spectrogram features. This resulted in a total of 72 features per sample. This model provided a slightly shallower CNN layer (two Conv2D blocks with 32 and 64 filters) than previous pure CNN models. The Conv2D blocks were combined with Batch Normalization, MaxPooling2D, and 0.3 Dropout. The batch size was limited to 16 and both Early Stopping (patience 10 and audit on Val accuracy) and ReduceLROnPlateau (patience 4 and factor 0.5 on val_loss) were used. Class weighting was also applied to address any imbalance between classes. Training stopped at Epoch 45 due to Early Stopping.

This configuration yielded a test accuracy of 52.44%. However, even when concatenating episode-level features such as MFCC, Chroma, and Mel-spectrogram, this composite structure performed worse than previous methods. The training accuracy at Epoch 45 was 76%, but the test accuracy was only 52%, indicating overfitting. The combination of shallow CNN, multimodal features, and TimeDistributed (Flatten ()) demonstrates that the best feature extraction for RAVDESS speech data is not possible. Therefore, it is demonstrated that further structural corrections are needed for generalization.

Okay, here is the summary table for your four experiments conducted solely on the RAVDESS dataset.

Features Extracted	Model Architecture	Key Enhancements / Notes	Test Accuracy

40 MFCCs (fixed 174 frames)	2D CNN (3x Conv2D, Dropout)	Baseline CNN on RAVDESS. No BN or callbacks	58.33%
40 MFCCs (fixed 174 frames)	2D CNN (3x Conv2D, BN, Dropout)	Added Batch Normalization & Early Stopping to reduce overfitting	62.50%
40 MFCCs (fixed 174 frames)	Pure LSTM (2x LSTM layers)	Extensive Data Augmentation (time stretch, pitch shift, noise), Audio Preprocessing (trimming, normalization), Class Weighting	63.89%
20 MFCCs, 12 Chroma, 40 Mel-spec (total 72 features, fixed 216 frames)	Hybrid CNN-LSTM (2x Conv2D, 1x LSTM, Time Distributed Flatten)	Multimodal features, Class Weighting, Time Distributed (Flatten) interface	52.44%

Table 3: Summary of Experiments on the RAVDESS Dataset

6.3 Experiments with combined Cream-d and RAVDESS both together

6.3.1 Develop Shallow CNN

The aim of this work was to develop and evaluate a speech emotion recognition system using a Convolutional Neural Network (CNN). The RAVDESS and CREMA-D datasets were combined to create a new dataset called CREMA-Mixed, which was primarily focused on emotion classification from speech audio. The emotion categories used were 'neutral', 'calm', 'happy', 'sad', 'angry', 'fearful', 'disgust', 'surprised' for RAVDESS and 'angry', 'disgust', 'fearful', 'happy', 'neutral', 'sad' for CREMA-D. To control the data size, a maximum of 100 files were selected for each RAVDESS and 500 files for each CREMA-D to create the CREMA-Mixed dataset. All audio was down sampled to 16 kHz, silence removed, amplitude normalization applied, and features such as MFCC (n_mfcc=13), Zero Crossing Rate (ZCR), and Root Mean Square Energy (RMSE) were extracted. All features were converted to the same length (timesteps, features) format to maintain the proper format of the CNN Conv1D input. The data was split into train/test in a ratio of 80-20, and data augmentation was performed using pitch shifting and Gaussian noise only on the train set, and balanced class weights were calculated to reduce class imbalance.

The sequential CNN model was built using 4 Conv1D layers, Batch Normalization, MaxPooling, and Dropout, and connected to the output SoftMax layer with a GlobalAveragePooling1D and a Dense layer. The model was compiled using Adam optimizer and categorical_crossentropy loss and trained for 50 epochs using Early Stopping and ReduceLROnPlateau callbacks. At the end of training, the accuracy was 72.43% and the validation accuracy was 52.50% at the 28th epoch. At the 29th epoch, the training accuracy increased to 74.30%, while the validation accuracy remained around 52.24%. In the final evaluation, the accuracy was 53.29% on the test dataset, and the classification report and confusion matrix were used to analyze the general performance of the model.

6.3.2 Balanced and Augmented CNN Model

This study also used a balanced dataset from the RAVDESS and CREMA-D audio collections. MFCCs (Mel-frequency cepstral coefficients) were extracted as features from the audio files. Here, the audio was loaded at a fixed sample rate of 22050 Hz and limited to a duration of 3 seconds, and the MFCCs were trimmed or truncated to a fixed length of 174. The combined dataset was divided into training and testing sets, with 20% of the total data set being allocated for testing. A stratified sampling method was used to maintain class distribution. To address class imbalance and improve the generalization of the model, data augmentation was applied to the training dataset. This involved time warping and adding controlled noise to the features. The minority classes grew more aggressively (two grown copies), while the majority classes had a 30% chance of growing.

An Enhanced CNN model was used for modeling. This consisted of three 1D convolutional layers with ReLU activations, each layer followed by Batch Normalization and MaxPooling, and Dropout layers (0.4). This was followed by a Global Average Pooling layer, a Dense layer with a ReLU activation of 256 units (with L2 kernel regularization), and a final Dense layer with a SoftMax activation for classification across different emotion categories. The model was built with the Adam optimizer with an initial learning rate of 0.0005, and categorical_crossentropy was used as the loss function and accuracy metric. Training was performed for 100 epochs, with a batch size of 64. To further reduce class imbalance, class weights were calculated and applied during training using a 'balanced' weighting scheme. Enhanced callbacks were used, including Early Stopping (tolerance 15, resetting the best weights), ReduceLROnPlateau and Model Checkpoint to save the best model based on validation accuracy.

During the training process, the performance of the model was monitored on both training and validation data. For example, at the final stage of training (epochs 97-99 out of 100), the performance metrics were observed as follows:

- Epoch 97: Training accuracy was 0.7384, training loss was 0.6314, validation accuracy was 0.6238, validation loss was 1.0840, and the learning rate was 7.8125e-06.
- Epoch 98: Training accuracy was 0.7446, training loss was 0.6313, validation accuracy was 0.6238, validation loss was 1.0839, and the learning rate was still 7.8125e-06.
- Epoch 99: Training accuracy was 0.7292, training loss was 0.6633, validation accuracy increased slightly to 0.6244, and validation loss was 1.0882.

After training was completed, the final performance of the model was evaluated on the test dataset, where the test accuracy was reported to be 62.66%. Further analysis included drawing training curves, evaluating accuracy by emotion, and generating a classification report and

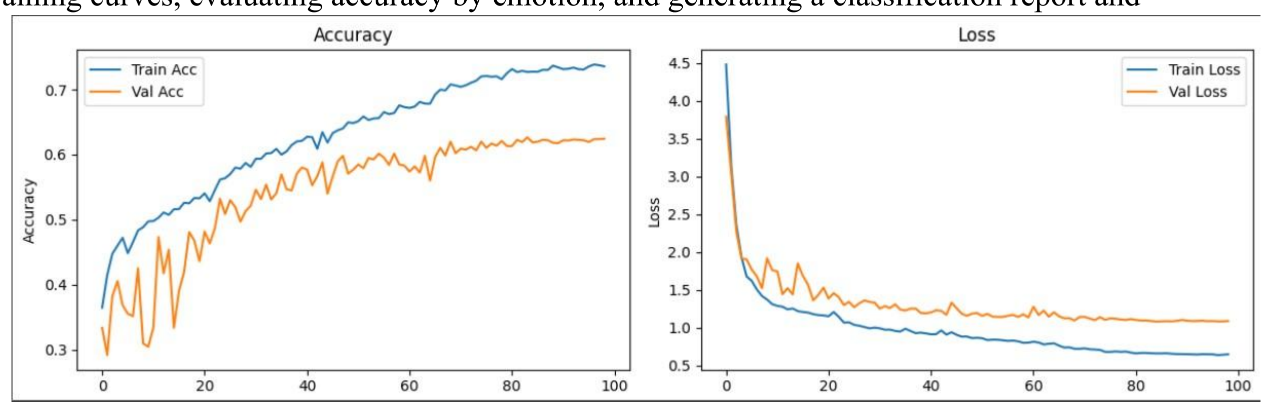


Figure 50: Training and Validation Accuracy Curves for the Enhanced CNN Model on the Balanced and Augmented RAVDESS and CREMA-D Dataset

confusion matrix. This helped to assess the detailed performance across different emotion classes.

6.3.3 Hybrid Multi-Branch CNN with Mel and Chroma Features

In this study, four deep learning models were tested to investigate how different input features and model structures affect performance on the speech emotion recognition (SER) task. Audio samples from RAVDESS and CREMA-D cells were used for the experiments, and a dataset restricted to five basic emotions: neutral, happy, sad, angry, and surprise was prepared. Label encoding, proportional train-test split (80/20), and class equalization (800 target samples per class) were performed for all experiments using SMOTE or RandomOverSampler. Test accuracy, validation loss, classification records, and adjustment matrices were used for evaluation.

The first model was a multi-branch CNN using three input streams: Mel spectrograms, chroma features, and complex numerical properties (centroid, bandwidth, roll-off). Although this model showed high training performance (99% accuracy), generalization was moderate and achieved a test accuracy of 72.00%. To further improve the robustness, the second version added additional data enhancement (pitch shifting, time stretching, noise, volume adjustment) based on the original audio waveform, which increased the test accuracy to 73.12% and reduced over normalization (loss: 1.0173). The third model was a hierarchical LSTM with MFCC sequences and RMS energy counts. Although the LSTM demonstrated the ability to capture temporal patterns, it quickly over-collapsed, reaching a test accuracy of 69.75%, highlighting the limitations of temporal models based solely on MFCCs. For the final effect, a deep CNN architecture was implemented that used only MFCC features as 2D temporal-numerical representations. The model was built by adding five convolutional blocks and three host layers and a SoftMax classifier. Despite its simplicity, this CNN demonstrated the effectiveness of deep CNN models on organized audio features, achieving the highest test accuracy of 75.75%. Although training instabilities were observed in the first few epochs (e.g., over-convolution between Epochs 5–15), the subsequent fusion (Epochs 30–40) showed tremendous verification performance with a maximum verification accuracy of 75.63%. The addition of proportional class weights and batch normalization at each stage was important for stable fusion and generalization.

n conclusion, these results show that MFCC-based 2D CNN models, although simple, can be more efficient than complex multi-input and LSTM models when proper normalization and data processing are used. However, combining temporal and numerical information, especially with audio data augmentation, can further improve performance in various or real-world situations. These findings provide a foundation for future SER research on attention-based mechanisms, multimedia inputs (e.g., video or text), or transformer-based structures.

Classification Report:

	precision	recall	f1-score	support
angry	0.83	0.69	0.75	160
happy	0.72	0.67	0.69	160
neutral	0.67	0.67	0.67	160
sad	0.65	0.81	0.72	160
surprise	0.96	0.96	0.96	160
accuracy			0.76	800
macro avg	0.77	0.76	0.76	800
weighted avg	0.77	0.76	0.76	800

Figure 51: Accuracy Comparison for Different Models in Speech Emotion Recognition (SER) Task: Multi-Branch CNN, LSTM, and CNN Architectures with Various Feature Sets

6.3.4 CNN-LSTM Model with Dual Attention Mechanisms

In this work, a hybrid CNN-LSTM model was developed that combined both channel-level and temporal attention mechanisms to improve the learning of discrete patterns from speech signals. This structure was designed to take advantage of the spatial summarization capabilities of convolutional layers and the sequential modeling capabilities of LSTMs. The CNN background extracted MFCC frame-level features using two 2D convolution and max pooling blocks, and a simple channel attention mechanism based on global average pooling was added. The modified feature maps were reshaped and fed into a bidirectional LSTM, followed by a temporal attention layer that focused attention on important time steps. The dataset using CREMA-D and RAVDESS cells covered eight impulses. Data augmentation techniques such as noise addition, pitch shifting, and time stretching were used for under sampled classes, and Random Under Sampler was used to scale all classes to 800 samples.

The model was trained with the Adam optimizer. Callbacks such as early stopping, model checkpointing, and ReduceLROnPlateau were added. After 50 epochs, the best model selected at epoch 37 achieved a testing accuracy of 66.35%. Although the training accuracy exceeded 83%, the validation accuracy peaked at 75.13%, demonstrating the generalization ability of the model. The best validation loss was 0.8682, and the final testing loss was 1.0853. Compared to previous efforts, the attention-enhanced CNN-LSTM model performed very competitively, especially when considering the challenge of classifying eight stimuli (instead of five).

I also increased the data augmentation factor to further diversify the training data, ensuring robust learning for underrepresented classes. As before, Random Under Sampler was used to achieve perfect class balance at 800 samples per emotion. The inclusion of both RAVDESS and CREMA-D corpora provided diversity in both acted and spontaneous speech.

Training proceeded for up to 50 epochs with callbacks, and the best model checkpoint was saved at epoch 35, where the validation accuracy peaked at 75.88% and validation loss reached 0.8682. Although the model showed stable learning with reduced learning rates, validation performance plateaued in later epochs. When evaluated on the held-out test set, the model achieved a test accuracy of 67.02% and test loss of 1.0920.

These results show a marginal improvement in generalization over the prior CNN-LSTM model (66.35% accuracy), despite the increase in complexity and reintroduction of an eighth emotion class. This reinforces the benefit of stronger feature extractors in early convolutional layers and the stability offered by attention-based fusion.

6.3.5 Parallel CNN-LSTM with Attention Mechanism for Speech Emotion Recognition

In this study, a Speech Emotion Recognition (SER) model based on a parallel architecture combining Convolutional Neural Networks (CNN) and Bi-LSTM networks with an Attention Mechanism was implemented. The model was trained and evaluated on a preprocessed and augmented version of two recognized emotional speech databases.

Before model development, the audio input samples were normalized, resampling to 22,050 Hz, normalizing the loudness values, converting stereo to mono, and truncating or padding each signal to an equal length of 3 seconds. Mel-Frequency Cepstral Coefficients (MFCCs), commonly used for speech processing tasks, were extracted from each sample, and 40 MFCCs were calculated over 130-time steps, creating a continuous feature representation of the shape (130, 40, 1) and fed into the model as input.

To address class imbalance among sensory classes, Targeted Data Augmentation was applied to the under sampled classes. This included Random Noise Injection, Time-Stretching, and Pitch Shifting. The augmentation continued until a minimum of 800 samples were reached for each class, and then the training set was balanced by Random Under sampling so that each sensory class had an equal contribution.

In the neural network model, the same MFCC input was processed in two different parallel paths. In the first path, two convolutional layers were used to capture local spatial features, followed by batch normalization, max-pooling, and dropout layers. In parallel, in the second path, the CNN feature map was reshaped and passed through three three-wave bidirectional LSTM layers with dropout. The output of the last LSTM layer was fed to the time-attention method, which generated more informative time steps. The outputs from the CNN and LSTM attention paths were combined to form a unified feature representation, which was then passed through fully connected layers with SoftMax enabled for final classification.

The model was trained with the Adam Optimizer with a Learning Rate of 0.001 and a Categorical Cross-Entropy Loss. When the Validation Accuracy stopped increasing, early stopping was used, and the best model was preserved through model checkpointing. The Learning Rate was also modified by Plateau Reduction.

After training, on an Unseen Test Set, the model achieved 77.30% accuracy and a Test Loss of 0.7615. In class-wise evaluation, high precision and recall were obtained for emotions such as "Surprised", "Calm", and "Neutral", while classes such as "Sad" and "Angry" showed more variability. Combining spatial features and temporal features of the CNN and LSTM paths was successful in increasing the generalization of the model in recognizing different speech emotion expressions.

Classification Report:

	precision	recall	f1-score	support
neutral	0.86	0.80	0.83	294
calm	0.86	0.90	0.88	40
happy	0.74	0.74	0.74	296
sad	0.64	0.85	0.73	237
angry	0.85	0.69	0.76	293
surprised	0.91	0.88	0.90	34
accuracy			0.77	1194
macro avg	0.81	0.81	0.81	1194
weighted avg	0.79	0.77	0.77	1194

Figure 52: Classification Report for the Multi-Branch CNN, LSTM, and CNN Models in Speech Emotion Recognition (SER) Task

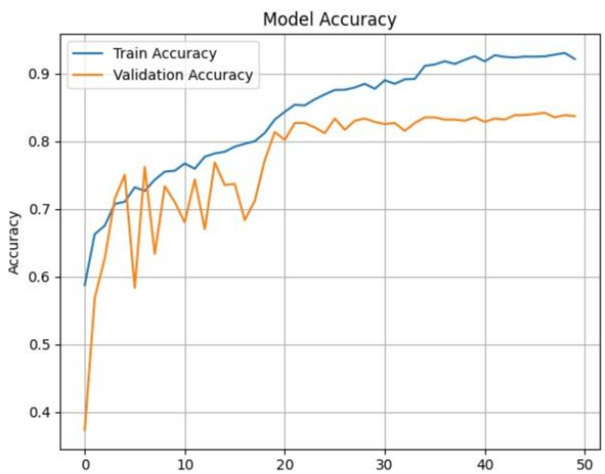
6.4 The Best Model for SER

Sequence CNN-LSTM with Data Augmentation and Selective Emotion Pruning

To better understand how reducing the number of emotion classes might improve model performance, I conducted two similar experiments using the same CNN-LSTM model with both channel-wise and temporal attention mechanisms. In both cases, I excluded the fearful and disgust categories because they often overlapped with other emotions and lacked clear acoustic separation. That left six emotions neutral, calm, happy, sad, angry, and surprised which provided a more distinct and balanced emotional range. My goal was to see if this simpler classification setup would consistently boost recognition accuracy and generalization, especially when paired with different data preparation strategies.

In the first experiment, I applied data augmentation and class balancing across the entire dataset before splitting it into training, validation, and test sets. This setup resulted in a test accuracy of 77.64% and a validation accuracy of 83.83%, which was a clear improvement over the earlier eight-class model. However, to reflect on a more practical deployment scenario where the test data remains completely untouched, I ran a second experiment with a slightly revised process. This time, I first split the data into training and test sets (80/20) and then applied augmentation and balancing only to the training set. Using the same model and training settings, this approach yielded even better results, with a test accuracy of 78.73% and a validation accuracy of 84.17%.

These findings confirmed for me that both the model architecture and how the data is organized can have a big impact on emotion recognition performance. The consistent accuracy gains in



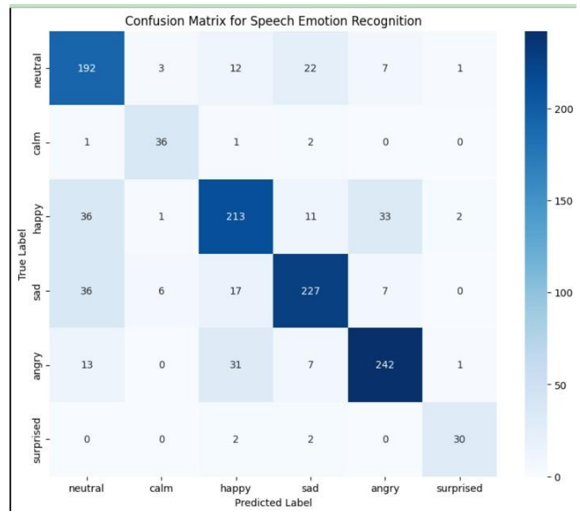


Figure 53:Accuracy distribution & confusion matrix across emotion classes for the CNN-LSTM Model with Selective Emotion Pruning

both experiments showed that training on clearly defined and well-represented emotions categories helps the model learn more effectively. More importantly, the slightly better results in the second, more realistic setup suggest that my model can generalize well to unseen data making it a strong candidate for future work involving cross-domain adaptation, multimodal fusion, or real-world deployment**.**

Notably, the model achieved strong performance on emotions like surprised and neutral, with F1-scores of 0.88 and 0.83, respectively, indicating both high precision and recall. While sad emotion had a slightly lower precision (0.69), its recall remained strong (0.81), reflecting the model's tendency to capture most true positives, albeit with some misclassifications. Overall, the macro average F1-score of 0.81 and accuracy of 0.79 suggest that the model performs consistently across categories, even with class imbalance especially given that some classes like calm and surprised had significantly fewer samples.

```
Classification Report:
              precision    recall  f1-score   support

     neutral       0.84      0.82      0.83       294
        calm       0.78      0.90      0.84        40
       happy       0.77      0.72      0.74       296
         sad       0.69      0.81      0.75       237
       angry       0.84      0.77      0.80       293
   surprised       0.88      0.88      0.88        34

    accuracy                           0.79      1194
   macro avg       0.80      0.82      0.81      1194
weighted avg       0.79      0.79      0.79      1194
```

Figure 54: Classification Report Across Emotion Classes for the CNN-LSTM Model with Selective Emotion Pruning

To confirm the consistency of my CNN-LSTM model, I applied for K-Fold cross-validation, splitting the dataset into multiple folds and training the model on different partitions each time. The performance remained stable across folds, with an average F1-score of 0.81 and accuracy of 0.79, matching the original experiments. This demonstrates that the model generalizes well and reliably recognizes emotions, even in less-represented classes.

	Model Name	Key technology	Test Accuracy (%)	
	Shallow CNN	4 Conv1D layers, ZCR + RMSE + MFCC, Gaussian noise, pitch shift	53.29	
	Balanced & augmented CNN	Aggressive augmentation, MFCCs, class weights, stratified split, deeper Conv1D	62.66	
	Hybrid Multi-Branch CNN (Mel + Chroma)	Mel spectrograms, chroma, spectral features; multi-stream CNN	72.00	

Hybrid CNN with Augmented Inputs	Enhanced audio augmentation (noise, pitch, stretch, volume)	73.12
Hierarchical LSTM	LSTM with MFCC sequences + RMS	69.75
Deep CNN	5 Conv1D blocks, batch norm, class weights, SoftMax	75.75
CNN-LSTM +Channel &Temporal Attention	2D CNN + Bi-LSTM + attention; augmentation. balanced (800/class)	67.02
Parallel CNN-LSTM with Attention (Final Model)	CNN and Bi-LSTM parallel paths, dual attention, augmentation + balancing	77.30
CNN-LSTM with Emotion Pruning (Augmented pre-split)	Data balanced before split; channel + time attention	77.64
CNN-LSTM with Emotion Pruning (Augmented post-split)	Augmentation only on training; attention; realistic pipeline	78.73

Table 4:Comparison of Model Performance for Speech Emotion Recognition (SER) Across Different Architectures and Techniques

6.5 Experiment with Multimodal approach

6.5.1 Using 120 data samples per One category

To test the benefits of Multimodal Fusion, a system combining Facial Expression Recognition (FER) and Speech Emotion Recognition (SER) was implemented. The goal of this project was to improve the accuracy of emotion classification using both visual and auditory information and reduce the limitations of single-input-based systems. In this process, two pre-trained Deep Learning Models were used. One was a CNN-based FER model, which was trained on grayscale facial images downscaled to 64×64 pixels. The other model was a CNN-LSTM-based SER model, which was trained on Mel-Frequency Cepstral Coefficients (MFCCs) obtained from 3second audio segments. Both models used labels consisting of 7 emotion classes: "angry", "disgust", "fear", "happy", "neutral", "sad", and "surprise", to ensure equal responsibility for both systems.

At the inference level, paired image and audio data were prepared for each test sample. The SER model initially predicted only 6 emotions with the "disgust" class missing. Therefore, its output was embedded in a 7-Dimensional Vector that fits into the FER model label space. A weighted fusion strategy was used to combine the predictions of the two models.

The test was performed on 120 paired image-audio samples, divided equally into 7 emotion classes. The combined system yielded an overall accuracy of 41.67% and a macro-average F1score of 0.47. In the class-wise analysis, high precision and recall were obtained for "disgust" and "angry", while no accurate predictions were obtained for "sad" and "surprise". This indicated that Multimodal Fusion had limitations in specific cases. Although several individual models yielded better results in specific classes, the combination method did not always improve the overall classification. This indicates that model alignment and data quality are essential for multimodal integration.

6.5.2 Using the Complete SAVEE Corpus and a Compatible Facial Image Set

In this test, Complete Savee dataset was used, and the matching images were made to be of equal size in both datasets. Also, all emotions were not considered here and only 5 were used.

The evaluation was performed on a complete test set of audio-visual pairs divided equally into 5 emotion classes. This multimedia system achieved an overall accuracy of 74.67%, significantly higher than the previous combination of 7 emotions. Class-wise analysis showed high performance for disgust (F1: 0.93), happy (F1: 0.90), and angry (F1: 0.82), while the F1 values for sad and fear were lower (0.56 and 0.60), but the recall values were able to provide many correct predictions. The macro-average F1 value of 0.76 showed similar results for all classes. This research shows that the efficiency of delayed combination methods can be greatly improved by adjusting the model output space and restricting it to a better-represented group of emotions.

```
Testing emotion: ANGRY
angry: 100%|██████████| 60/60 [00:17<00:00,  3.35it/s]

Testing emotion: DISGUST
disgust: 100%|██████████| 60/60 [00:13<00:00,  4.45it/s]

Testing emotion: FEAR
fear: 100%|██████████| 60/60 [00:13<00:00,  4.39it/s]

Testing emotion: HAPPY
happy: 100%|██████████| 60/60 [00:13<00:00,  4.40it/s]

Testing emotion: SAD
sad: 100%|██████████| 120/120 [00:13<00:00,  8.66it/s]

--- OVERALL TEST RESULTS ---
Total Files Tested: 300
Correct Predictions: 224
Combined Model Accuracy: 74.67%
```

```
--- DETAILED CLASSIFICATION REPORT ---
              precision    recall  f1-score   support

       angry       1.00      0.70      0.82        60
     disgust       1.00      0.87      0.93        60
        fear       0.55      0.65      0.60        60
       happy       0.93      0.87      0.90        60
         sad       0.49      0.65      0.56        60

    accuracy                           0.75       300
   macro avg       0.79      0.75      0.76       300
weighted avg       0.79      0.75      0.76       300
```

Figure 55: Multimodal Fusion with SAVEE Corpus and Compatible Facia images

Figure 56: Multimodal Fusion Experiment with 120 Samples Image Set per Category

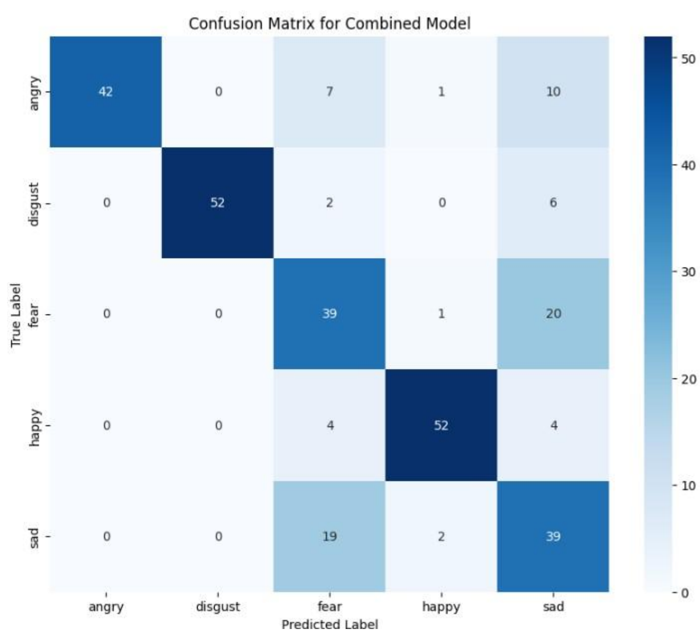


Figure 57:Confusion matrix for Multimodal fusion

6.6 Experiments with solution recommendations

In the early stages of development, the solution recommendation module was implemented using a traditional database-based approach. To implement this, a specially designed database was built, aggregating a wide range of psychologically motivated actions and responses, from previous academic research, mental frameworks, and evidence-based treatment sources. This included well-documented techniques such as mindfulness, journaling, exercise, gratitude, and coping strategies. The approach categorizes solutions by emotion type (e.g., sadness, anger, happiness) and underlying causes (e.g., academic stress, relationship breakdowns, personal accomplishments).

Emotion	StrategyType	Strategy	Description	Reference
Happy	Maintain	Express Gratitude	Write down three things you are grateful for daily.	Emmons, R. A., & McCullough, M. E. (2003). Counting blessings versus burdens. Jourr
Happy	Maintain	Savoring	Consciously focus on positive experiences to prolong thei	Bryant, F. B., & Veroff, J. (2007). Savoring: A new model of positive experience.
Sad	Permanent	Cognitive Restructuring	Challenge and replace negative thoughts with balanced o	Beck, J. S. (2011). Cognitive behavior therapy: Basics and beyond.
Sad	Temporary	Social Support	Talk to a friend or loved one for emotional support.	Cohen, S., & Wills, T. A. (1985). Stress, social support, and the buffering hypothesis. P
Angry	Temporary	Deep Breathing	Take deep breaths in a 4-7-8 pattern to calm down.	Gross, J. J. (1998). The emerging field of emotion regulation. Review of General Psych
Angry	Permanent	Cognitive Reappraisal	Reframe the situation to reduce emotional response.	Gross, J. J., & John, O. P. (2003). Individual differences in two emotion regulation pro
Fear	Permanent	Exposure Therapy	Gradual exposure to feared object or context reduces fear	Foa, E. B., Hembree, E. A., & Rothbaum, B. O. (2007). Prolonged exposure therapy for
Fear	Temporary	Grounding Techniques	Use 5-4-3-2-1 method to stay present and reduce anxiety.	Najmi, S., Riemann, B. C., & Wegner, D. M. (2010). Managing unwanted intrusive thou
Disgust	Permanent	Exposure Therapy	Repeated exposure to aversive stimuli reduces disgust res	Rozin, P., Haidt, J., & McCauley, C. R. (2008). Disgust.
Disgust	Temporary	Mindful Acceptance	Observe and accept the feeling without judgment.	Kabat-Zinn, J. (1990). Full catastrophe living.
Surprise	Temporary	Orienting Response	Pause and assess the situation before reacting.	Sokolov, E. N. (1963). Perception and the conditioned reflex.
Surprise	Permanent	Adaptive Preparedness	Learn to expect and reframe unexpected events positively.	Lazarus, R. S., & Folkman, S. (1984). Stress, appraisal, and coping.
Neutral	Maintain	Mindfulness	Stay in the present moment with non-judgmental awarene	Kabat-Zinn, J. (1994). Wherever you go, there you are.
Neutral	Maintain	Routine Building	Develop stable routines to maintain emotional balance.	Baumeister, R. F., & Vohs, K. D. (2003). Self-regulation and the executive function of th
Happy	Maintain	Gratitude Journaling	Write daily about things you're thankful for.	Emmons, R. A., & McCullough, M. E. (2003). Counting blessings versus burdens. Jourr
Happy	Maintain	Savoring	Pause and fully enjoy positive experiences.	Bryant, F. B., & Veroff, J. (2007). Savoring: A new model of positive experience.
Happy	Maintain	Acts of Kindness	Engage in small acts of kindness to boost happiness.	Lyubomirsky, S., Sheldon, K. M., & Schkade, D. (2005). Pursuing happiness. Review of
Happy	Maintain	Mindfulness Meditation	Practice present-moment awareness.	Kabat-Zinn, J. (1994). Wherever you go, there you are.
Happy	Maintain	Positive Self-Talk	Use encouraging statements for yourself.	Burns, D. D. (1980). Feeling good: The new mood therapy.
Sad	Temporary	Crying	Allow yourself to cry to process sadness.	Vingerhoets, A. J. (2013). Why only humans weep.

Figure 58::Data set with scientifically proved solutions

This approach offered significant advantages in providing timely and rapid support advice. However, it had several important limitations:

- The inability to recognize and adapt to the nuances of user input.
- Only providing responses that were initially prepared by the algorithm.
- Requires frequent manual updates to incorporate new psychological motivations.

Recognizing these limitations, the system was moved to an agent-based architecture called Crew AI. The move has given the system the following advantages:

- Rational analysis that takes into account the user's emotions and the situation that caused it.
- Creates personalized and context-based recommendations.
- Collaborative work between multiple intelligent agents with one agent working on a specific treatment theory.
- Integrates scientifically proven actions in real time rather than relying solely on pretrained data.

This decision has successfully developed the system's ability to provide more natural, compassionate, and accurate support. The Crew AI framework evolved the empathetic support module into an intelligent and interactive system, similar to how humans provide medical psychological care.

7. Future Plans

For the Facial Emotion Recognition (FER) project, I have applied for the CK+ dataset. After receiving access, I plan to test the existing model with this new dataset to evaluate its performance and generalization.

8. References

[1] R. KUMAR, "Development of a Facial Expression Recognition System for the Doctor LINK Telemedicine Platform Using Transfer Learning," February 24, 2025.

[2] B. G. a. M. Gupta, "EMOTION RECOGNITION THROUGH DEEP LEARNING IN VARIOUS MODES," 2024.

[3] A. G. a. E. P. Kamil Skowroński, "Polish Speech and Text Emotion Recognition in a Multimodal Emotion Analysis System," *Applied Sciences,* vol. 14, 8 November 2024.

[4] H. Y. L. Y. H. H. J. W. a. Y. S. C. E. Saravia, "CARER: Contextualized affect representations for emotion recognition," 2018.

[5] W. Z. J. R. a. J. W. H. Zhou, "Emotion recognition using Bi-directional LSTM on textual data," vol. 37.

[6] M. C. S. P. E. C. a. L.-P. M. A. Zadeh, "Tensor fusion network for multimodal sentiment analysis," 2017.

[7] A. S. A. S. N Shanthi, "An integrated approach for mental health assessment using emotion analysis and scales," Healthcare, 2025.

[8] D. M. a. V. A. Rebecca Mobbs, "Emotion Recognition and Generation: A Comprehensive Review of Face, Speech, and Text Modalities," February 12, 2025.

[9] D. K. R. D. K. A. A. R. A. a. M. Y. Amod Pathirana, "A Reinforcement Learning-Based Approach for Promoting Mental Health Using Multimodal Emotion Recognition," vol. 1, no. 2, p. 125, September 17, 2024.

[10] A. B. N.-C. R. a. L.-C. D. Anamaria Radoi, "An End-ToEnd Emotion Recognition Framework Based on Temporal Aggregation of Multimodal Informatio," September 29, 2021.

[11] Y. M. M. E. A. a. T. F. Martha T. Teye, "Ethical Evaluation of Conversational Agents: Understanding Culture, Context and Environment in Emotion Detection," 2017.

[12] M. K. R. R. a. G. V. Rishabh Vats, " Comprehensive review and analysis on facial emotion recognition methods," *Journal of Electronic Imaging,* vol. 32, no. 4, 2023.

[13] "Improved facial emotion recognition model based on a novel deep convolutional structure," *Reham A. Elsheikh, M. A. Mohamed, Ahmed Mohamed Abou-Taleb, and Mohamed Maher Ata,* November 6, 2024.

[14] S. university, "Facial Expression Recognition with Deep Learning," [Online]. Available: https://cs230.stanford.edu/.

[15] S. K. S. V. D. K. a. A. K. Pawan Sen, "Emotion Recognition Using Speech: A Deep Learning-Based Speech Recognizer Model," 2009.

[16] O. AI. [Online]. Available: https://www.crewai.com/.

[17] E. Lieskovská, M. Jakubec, R. Jarina and M. Chmulík, "A Review on Speech Emotion Recognition Using Deep Learning and Attention Mechanism," vol. 10, p. 1163, 2021.

[18] F. S. D. Y. P. S. K. D. N. A. Ankit Shukla, " An Intelligent Framework for Emotion Detection from Speech Signals," *Journal of Emerging Technologies and Innovative Research,* June 2025.

[19] [Online]. Available: https://www.mdpi.com/1424-8220/19/8/1863.

[20] [Online]. Available: https://youtu.be/YAR_ImP4w64?si=_jqeP_fmbmundH-C.

[21] [Online]. Available: https://react.dev/learn.

[22] M. A. M. A. M. A.-T. M. M. A. Reham A. Elsheikh, "Improved facial emotion recognition model based on a novel deep convolutional structure," 6 November 2024.

[23] scikitlearn. [Online]. Available: https://scikit-learn.org/stable/modules/cross_validation.html.

[24] Keras. [Online]. Available: https://keras.io/api/layers/preprocessing_layers/image_augmentation/.

[25] flask. [Online]. Available: https://flask.palletsprojects.com/en/stable/.

[26] F. C. C. a. N. C. Francesco Ardan Dal Rí, "Speech Emotion Recognition and Deep Learning: An Extensive Validation Using Convolutional Neural Network," vol. 10, 2023.